Healthcare NLP: where are we and what is next?

Tutorial @ LREC, May 12, 2026
Palau de Congressos de Palma, Spain

Lifeng Han, Paul Rayson, Andrew Moore, Goran Nenadic, Suzan Verberne

PICTURE
https://4dpicture.eu/

Universiteit Leiden

LUMC Leids Universitair Medisch Centrum

MANCHESTER 1824 The University of Manchester

Lancaster University

Quick survey on attendee background

https://www.menti.com/alyixj47x2s1

Join at menti.com with code 1563 1233

open source websites

https://github.com/4dpicture/HealthNLP

https://arxiv.org/abs/2512.08617

May 12th, morning session
Room 7
Tutorial 13: HealthcareNLP
https://lrec2026.info/wp-content/uploads/2026/02/LREC2026-programme-Day2.pdf

logistics

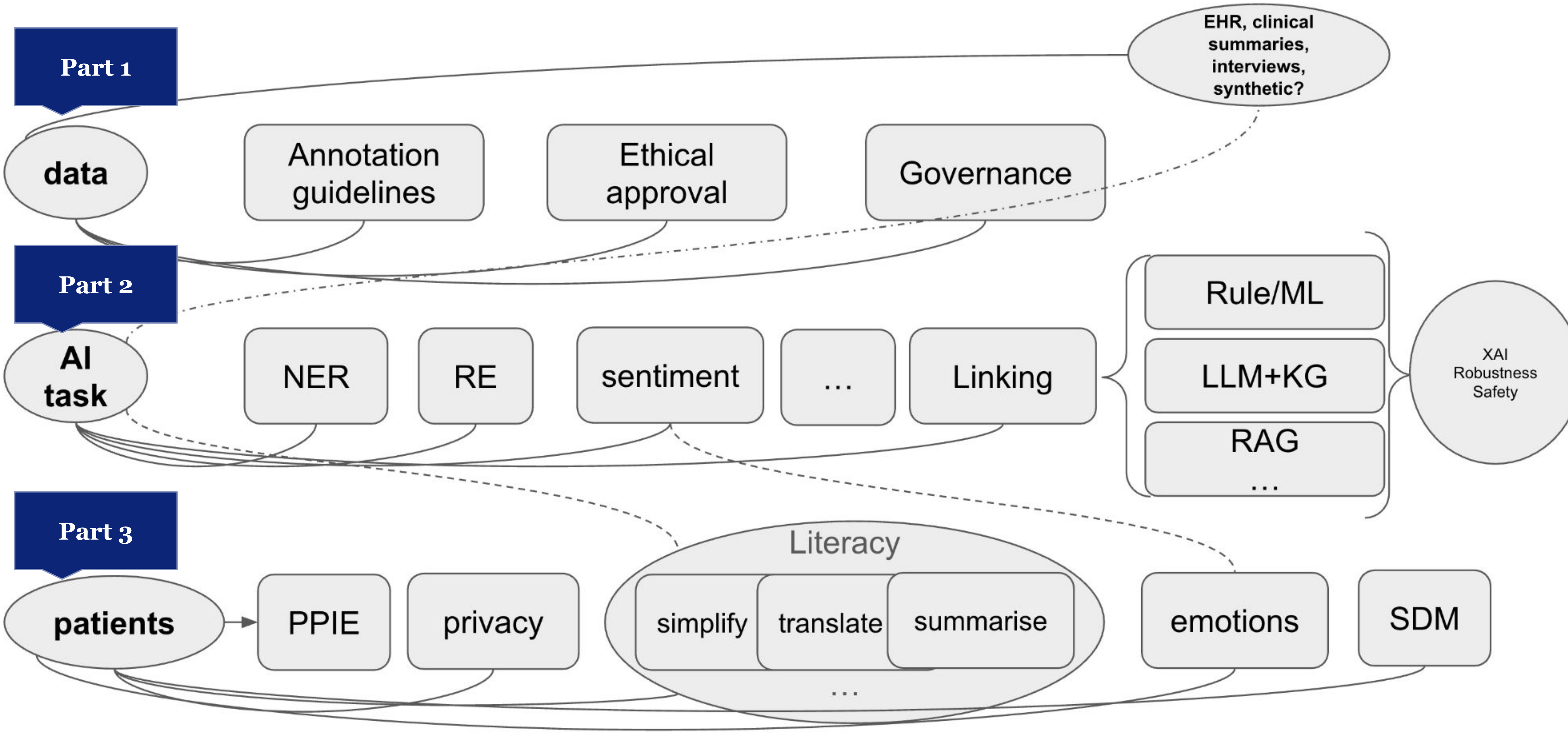
Morning session: 9:00 to 13:00
Morning Coffee break: 10.30-11.00

LREC 2026

Schedule
Tuesday, 12 May 2026

	Morning session	Afternoon session
Room 1	W2 The Third Workshop on Patient-oriented Language Processing (CL4Health)	
Room 2	W36 SIGUL 2026 Joint Workshop with ELE, EURALI, and DCLRL. Towards Inclusivity and Equality: Language Resources and Technologies for Under-Resourced and Endangered Languages	W31 The Seventh Workshop on Resources for African Indigenous Languages (RAIL)
Room 3	W44 Joint Workshop on Legal and Ethical Issues in Human Language Technologies (LEGAL2026) & Computational Approaches to Language Data Pseudonymization, Anonymization, De-identification, and Data Privacy (CALD-pseudo 2026)	
Room 4	W26 NLPerspectives: The 5th Workshop on Perspectivist Approaches to Natural Language Processing	
Room 5	T15 Hallucination in Large Foundation Models *Tutorial*	W45 The Information Disorder Workshop
Room 6	W15 3rd International Workshop on Natural Scientific Language Processing	
Room 7	T13 HealthcareNLP: where we are and what's next? *Tutorial*	W13 Resources and Processing of linguistic, para-linguistic and extralinguistic Data from people with various forms of cognitive/psychiatric/ developmental impairments (RaPID-6@MENTAL.ai@LREC)

structure of the tutorial



1. Data

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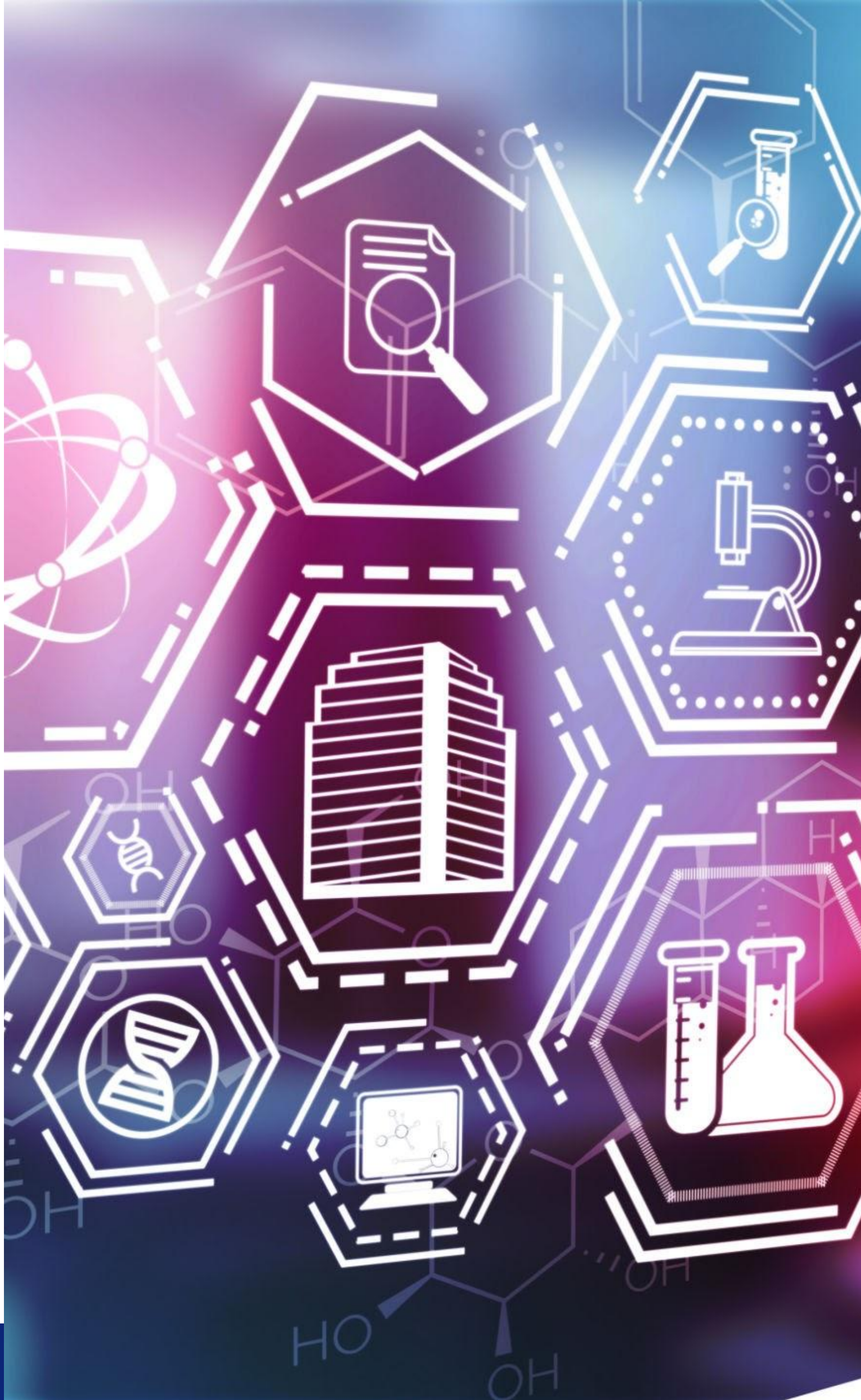
Data

Large amounts of data in the biomedical & health domain:

- Scientific literature
- Experimental data
- Patents
- **Electronic Health Records**
- **Patient surveys**
- **Health social media (e.g. patient support groups)**

We focus on

1. Text data
2. Patient related data (no scientific articles)
3. Data Ethics



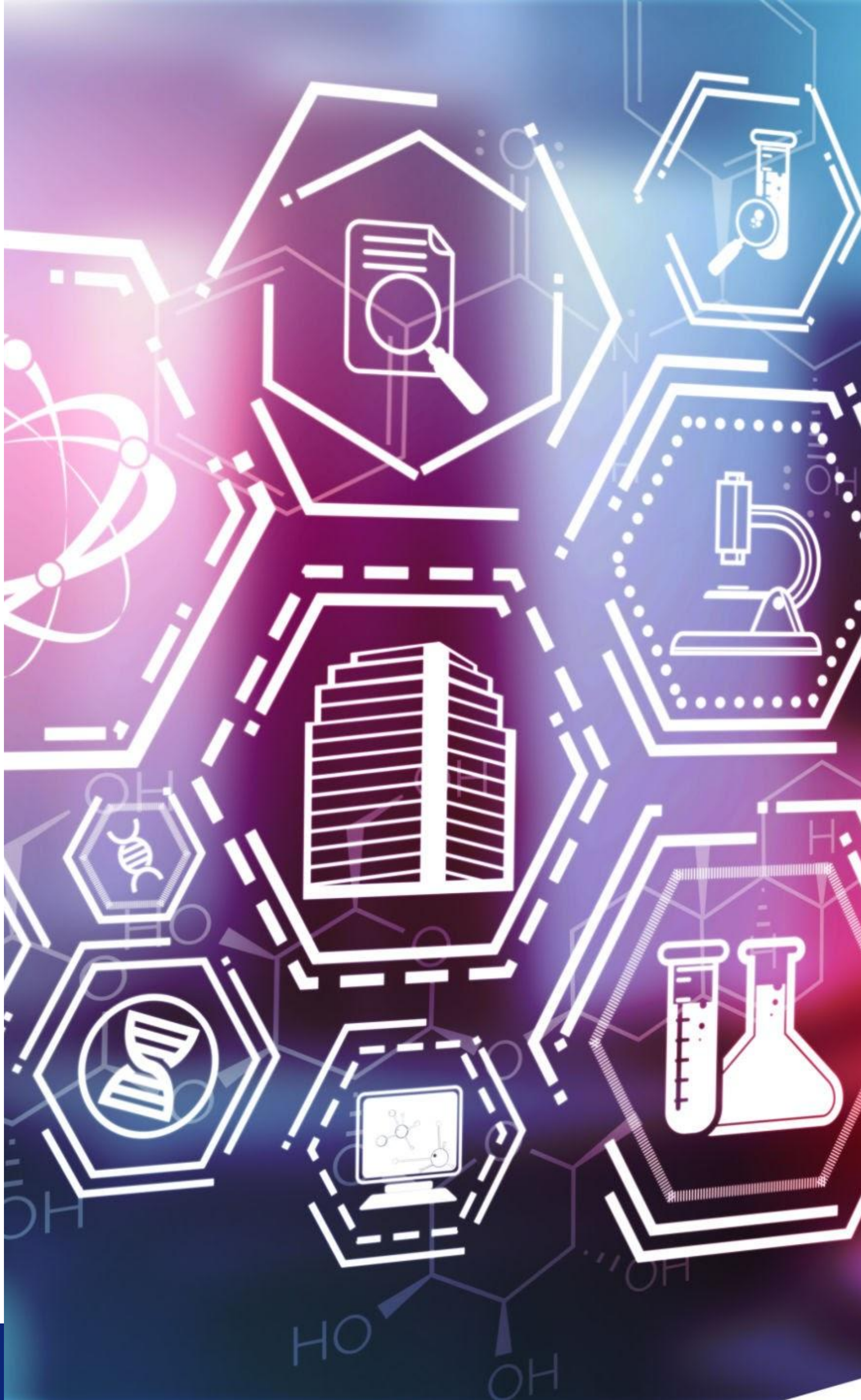
Data

Large amounts of data in the health domain:

- Electronic Health Records
- Patient surveys
- Health social media (e.g. patient support groups)

But the majority of this data is not curated for machine learning purposes:

- Noise in the text
- No labels or noisy labels
- Data is not always accessible or available for research due to privacy reasons
  - Text is more difficult to anonymize than numerical data



Data: annotation

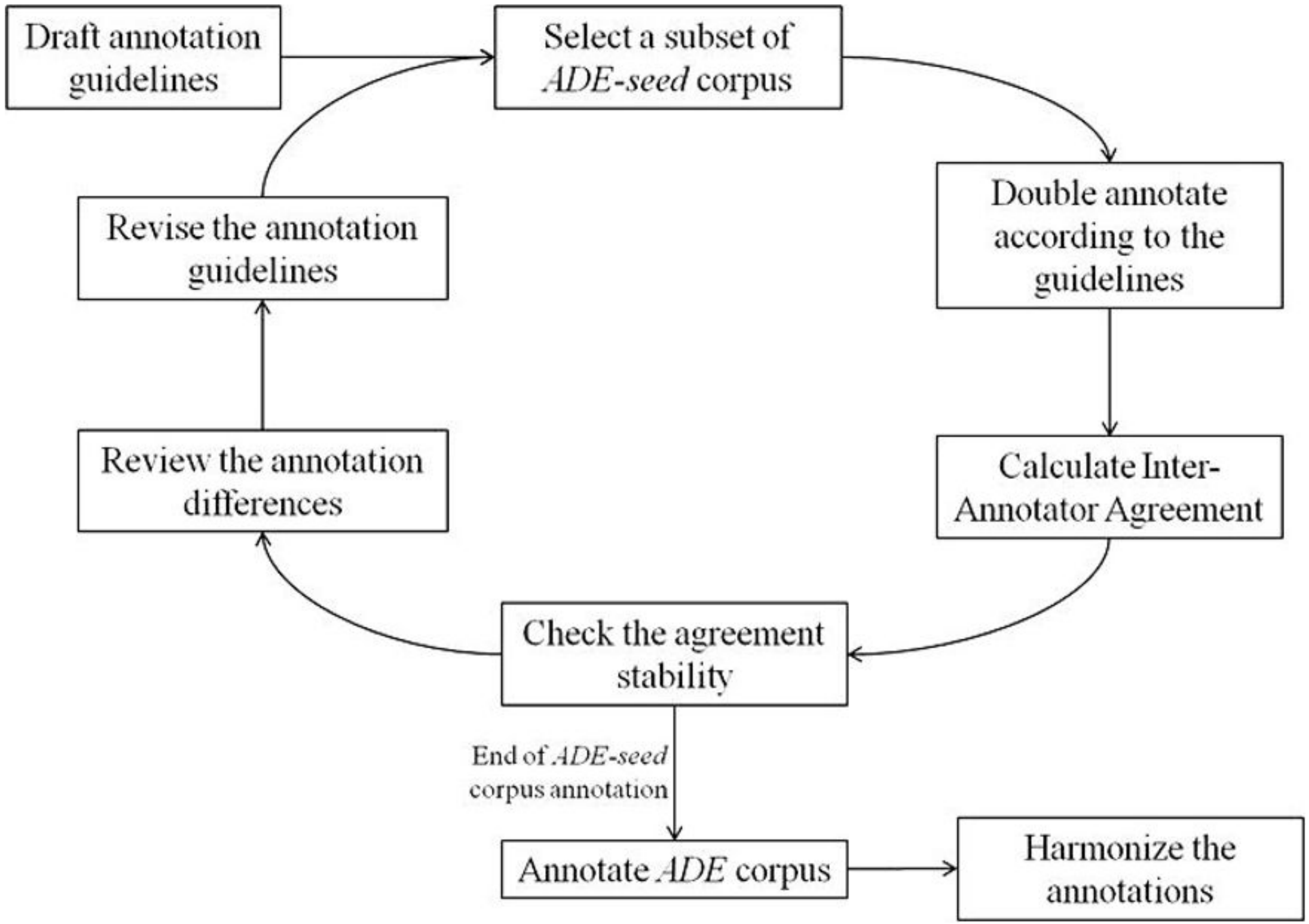
The Adverse Drug Effect (ADE) benchmark corpus is a good example of an annotation effort in the medical domain

Adverse event = side effect or other negative event of treatment

- a set of nearly 3000 case reports
- manually annotated with 5063 drugs and 5776 conditions
- + ontology of adverse events
- "With these methods, a number of drug label changes for the drugs rituximab, efalizumab, and natalizumab could successfully be predicted"

H. Gurulingappa et al., Development of a benchmark corpus to support the automatic extraction of drug-related adverse effects from medical case reports J. Biomed. Inf. 45 (5) (2012) 885–892.

Data: annotation



H. Gurulingappa et al., Development of a benchmark corpus to support the automatic extraction of drug-related adverse effects from medical case reports J. Biomed. Inf. 45 (5) (2012) 885–892.

Example task: what do we need

Task: **find side effects for medications in online cancer patient forum discussions**

How would you approach this? What do you need?

Discuss with your neighbour for a few minutes

yes, bad chills at the beginning of my treatment that I end up at emergency, this happened about three times . After two years taken Gleevec still very cold, but not too bad.

The water retention is a pain in the ass . Can you do sports? It's what helps me most.

Example task: what do we need

Steps to take

1. Filter the potentially relevant messages
2. Get/create training data for NER
3. Train an NER model to identify drug names and side effects in the messages
4. Normalize the side effects (map to ontology)
5. Relation extraction: cooccurrences of drug names and side effects in one message
6. (Match the found relations to an existing knowledge base to identify which relations are new)

Example task: what do we need

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Needed:

- Lists/ontologies of drug names and known side effects (e.g. SNOMED CT)
- Pre-processing tools
- Pre-trained BERT models for NER and ontology linking
- Labelled data for supervised NER finetuning and evaluation

Example task: what do we need

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Exercise: annotation for NER

Task: mark in the following messages all strings (single- or multi-words) that refer to:

- A disease
- A medication
- A side effect

Use 3 different colors for your markup

You can find the PDF on our git repository

After your annotation, discuss with your neighbour

#	post
1	I was only on Gleevec for 1 year so I guess about 3 years later.
2	I'm not sure what you mean by severe but while taking gleevec I was prone to eye bleeds, more than just blood shot but bright red . It was only in my left eye and at the same time I was being seen by the eye hospital about something else . They new about my medication and gave my eyes a good examination but could find no reason for the bleeding and I had no cause to worry . It has stopped since I stopped taking gleevec . I'm not saying this is the same in your case so get it checked out just in case.
3	Most important: Consult an oncologist who is knowledgeable and experienced with GIST, specifically . Ask lots of questions, and keep asking until you get the answers you understand.
4	yes, bad chills at the beginning of my treatment that I end up at emergency, this happened about three times . After two years taken Gleevec still very cold, but not too bad.
5	Someone asked me yesterday how cancer has changed my life . You said it much better than I did! Thanks.
6	Those cysts are quite common in just about everyone . However us GIST warriors it makes us worry . Always good to check and ask for an opinion.
7	The water retention is a pain in the ass . Can you do sports? It's what helps me most.
8	Votrient is another cancer drug that has been used off label for GIST with mixed results . I was on Stivarga also, it made my BP skyrocket . Fornunately, my primary care physican is a good friend and has taken over the managing of BP meds for me . It is great to have a doctor who gives you their personal cell phone number and will take your call or text even on the weekend! I am blessed.

Exercise: annotation for NER

- What challenges did you encounter?
  - What is the boundary of a side effect mention?
  - Spelling variations
  - Lack of domain knowledge

- But: Even without having heard of GIST and Gleevec before, we are able to identify these as a disease and a medication

- We do this based on context

- Similarly, sequence labelling models also rely on context and word order

#	post
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2	I'm not sure what you mean by severe but while taking gleevec I was prone to eye bleeds, more than just blood shot but bright red . It was only in my left eye and at the same time I was being seen by the eye hospital about something else . They new about my medication and gave my eyes a good examination but could find no reason for the bleeding and I had no cause to worry . It has stopped since I stopped taking gleevec . I'm not saying this is the same in your case so get it checked out just in case.
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Note that our annotation is not the only truth - human annotation is always partly subjective

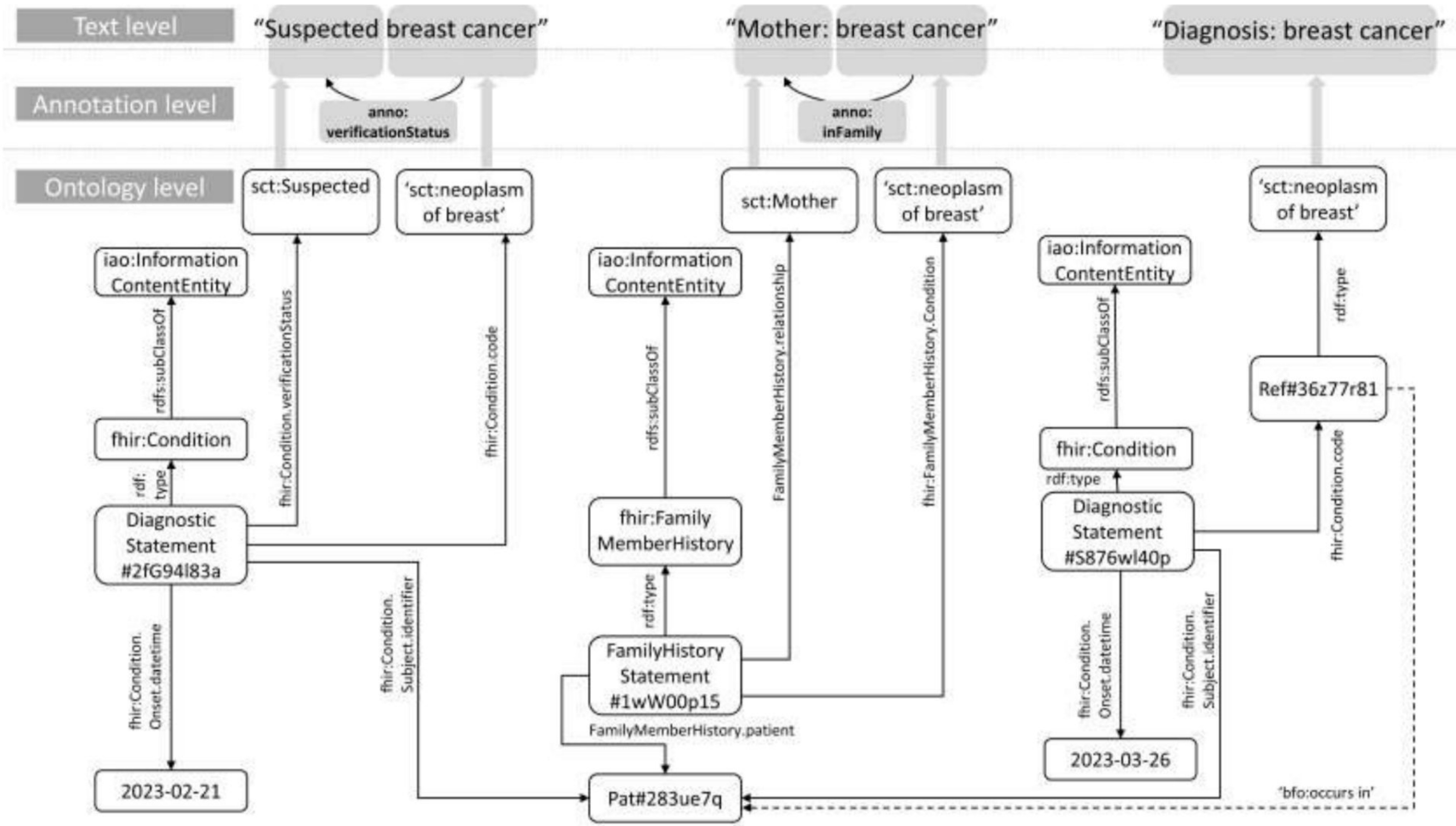


Figure 2: From text to ontology-based knowledge graphs. The text level shows three text snippets (quoted) belonging to clinical texts about one patient. The human annotations (grey background) use SNOMED CT and "anno:" as annotation vocabularies. The lower, most complex level shows the instantiations of FHIR resources with references to SNOMED concepts, instances thereof, and literals. Nodes with random IDs represent individuals, linked to the concept whose code appears in the annotation. An example of an inferred predication is shown by a dashed arrow.

Examples for mappings between HL7 FHIR values and SNOMED CT concepts

HL7 FHIR	SNOMED CT (from the *Qualifier value* hierarchy)
Unconfirmed	*Suspected* OR *Probably not present*
Provisional	*Probably present* OR *Suspected*
Confirmed	*Confirmed present*
Refuted	*Known absent*

Schulz, S., Del-Pinto, W., Han, L., Kreuzthaler, M., Aghaei, S., & Nenadic, G. (2023). Towards principles of ontology-based annotation of clinical narratives. In *14th International Conference on Biomedical Ontologies* (pp. 36-47). CEUR Workshop Proceedings. https://ceur-ws.org/Vol-3603/icbo2023_proceedings.pdf#page=51

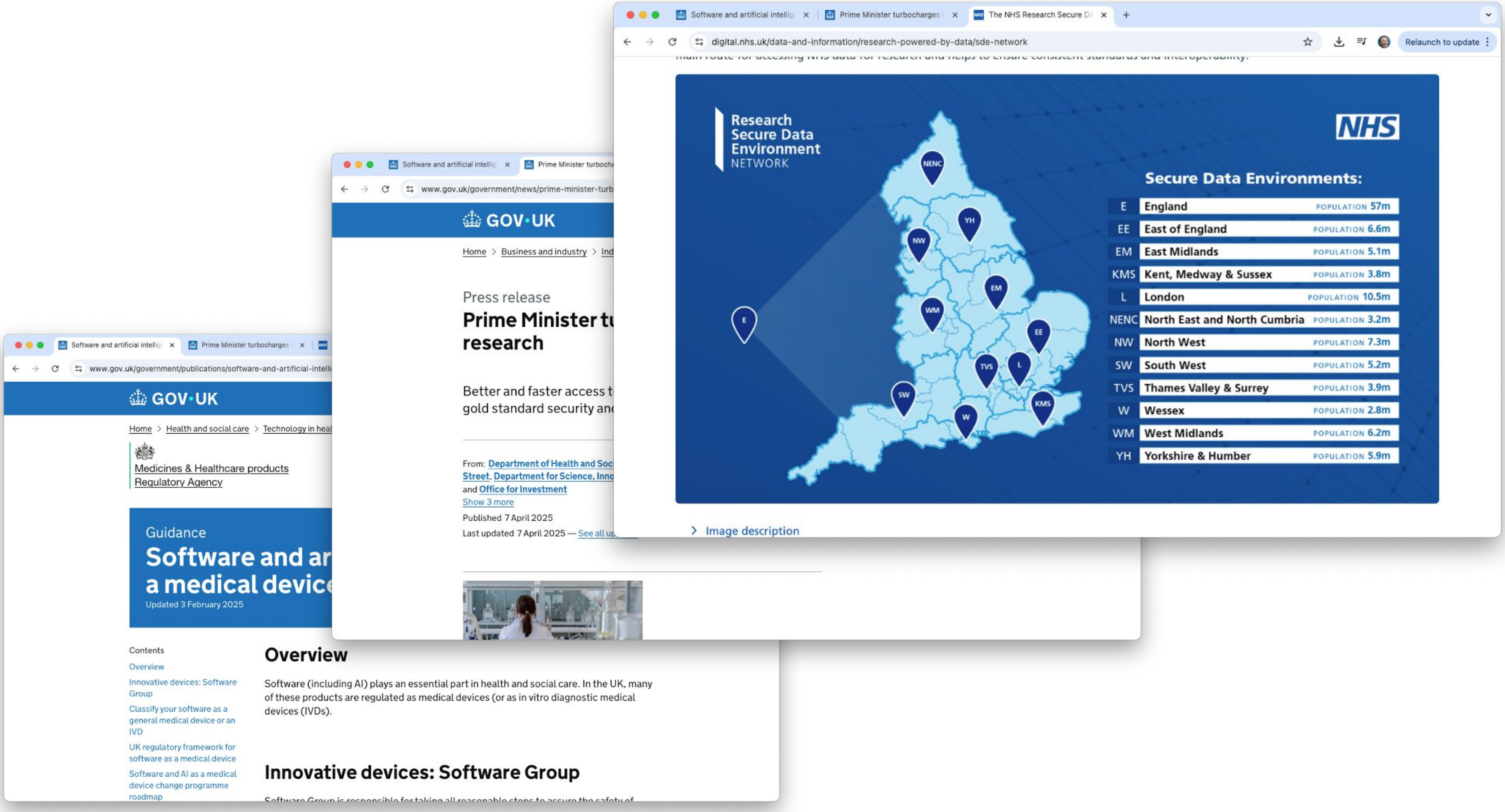
Data: ethical issues

What is your approach when dealing with textual data from healthcare and medical settings …

1. What ethics processes, guidelines and requirements are imposed by your organisation?
2. What are the expectations in your country?
3. If you're running a focus group to discuss a particular health issue with people with lived experience, do you need ethical approval?
4. If you collect health related social media data, do you need ethical approval?
5. What about if you want to release this data, what would you need to do?

Discuss these questions with your neighbour(s)
Take 5 minutes starting from now

Data: ethical issues



Data and patients

Ethical considerations

- Eysenbach & Till (2001) Ethical issues in qualitative research on internet communities. BMJ.
  - https://doi.org/10.1136/bmj.323.7321.1103
- Seale C, Charteris-Black J, MacFarlane A, et al. Interviews and internet forums: a comparison of two sources of data for qualitative research. Qual Health Res 2010;20:595–606.
  - https://doi.org/10.1177/1049732309354094

Summary points

Internet communities (such as mailing lists, chat rooms, newsgroups, or discussion boards on websites) are rich sources of qualitative data for health researchers

Qualitative analysis of internet postings may help to systematise and codify needs, values, and preferences of consumers and professionals relevant to health and health care

Internet based research raises several ethical questions, especially pertaining to privacy and informed consent

Researchers and institutional review boards must primarily consider whether research is intrusive and has potential for harm, whether the venue is perceived as "private" or "public" space, how confidentiality can be protected, and whether and how informed consent should be obtained improving Peer Online Forums

Lancaster University

MANCHESTER 1824

NHS Berkshire Healthcare NHS Foundation Trust

iPOF: Improving Peer Online Forums

"Realist evaluation of online mental health communities to improve policy and practice"

https://fundingawards.nihr.ac.uk/award/NIHR134035

https://www.lancaster.ac.uk/ipof/

March 2022 – December 2024

FUNDED BY NIHR | National Institute for Health and Care Research

Peer Support: Mental Health

People look online for help. COVID has only increased this.

25% increase in depression and anxiety (WHO)[1].

Online forums can be a source of comfort and support to people suffering from mental health problems.

[1] https://www.who.int/news/item/02-03-2022-covid-19-pandemic-triggers-25-increase-in-prevalence-of-anxiety-and-depression-worldwide

iPOF project research questions

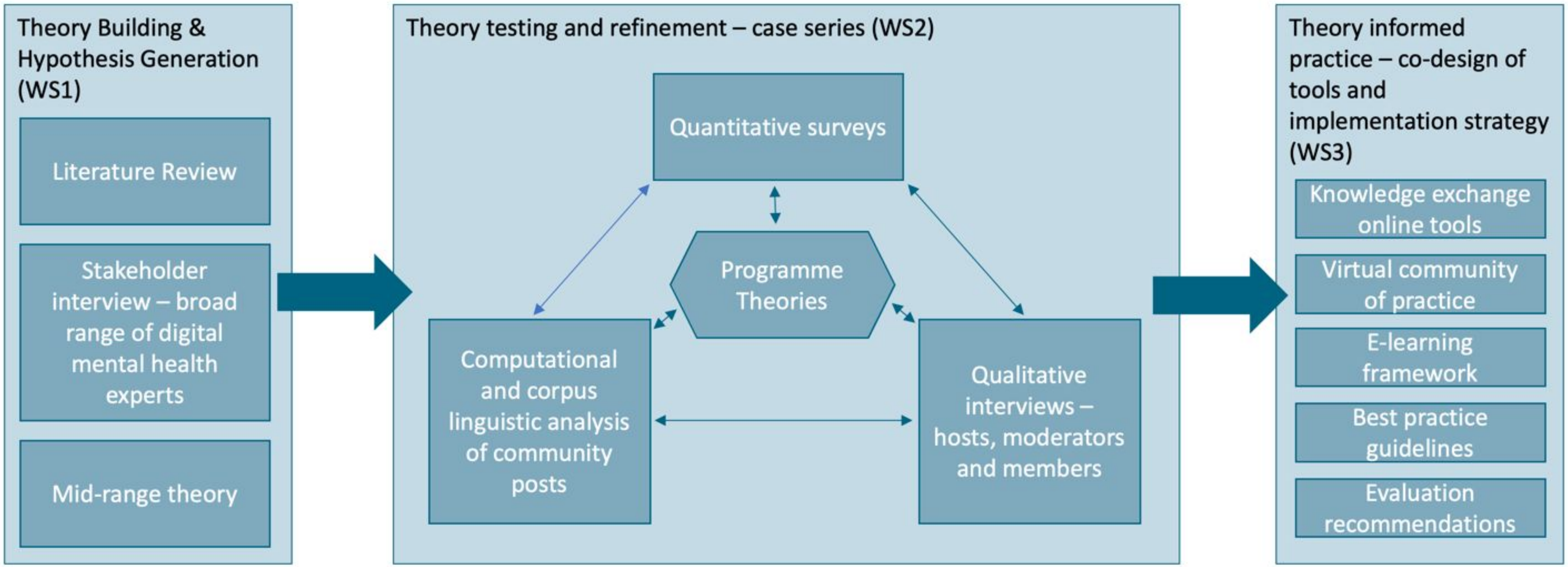
How do online mental health forums work?

Why do some work better than others?

Why do some people find them helpful and others don't?

iPOF project overview

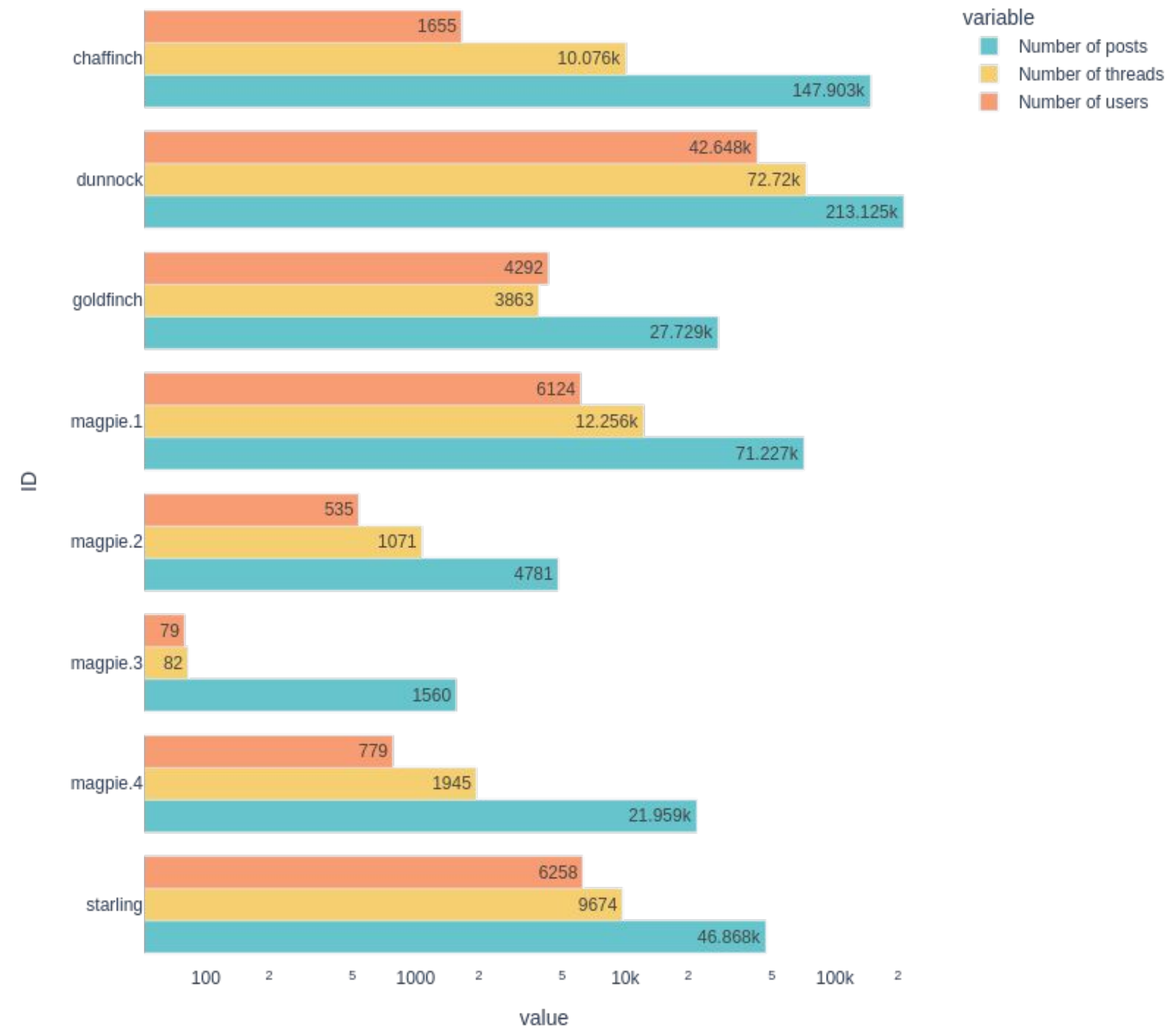
Lancaster University



10

iPOF Forum Analysis (WS2)

- How can NLP and corpus linguistics be leveraged in realist evaluation?
  - Semantic annotation, emotion and sentiment analysis
  - Topic modelling, keywords
- Can data driven analysis inform development of realist programme theories?
- What characterises peer support forum data compared to other forums?
- iPOF case summaries: https://doi.org/10.17635/lancaster/researchdata/716



iPOF Ethics Framework

- Very important ethical issues in analysing forum posts
  - People often share details about the things that are causing them distress, in the hope that other people who have faced similar problems can help them
  - It is vital that the forum feels a safe space in which to do this. We do not want this research to jeopardise this feeling of safety in any way.
- Developed a comprehensive ethical framework for this study with input from legal, clinical, academic and lived expertise, and approved by the Health Research Authority (IRAS 314029).

iPOF Ethics Framework

- Drew on existing guidelines and frameworks including
  - UK Government Data Ethics framework
  - British Psychological Society Ethical Guidelines for Internet-Mediated Research
  - Ethical Guidelines for the Association of Internet Research
  - Worked closely with our NHS host Trust Information Governance Lead to gain Trust Research and Development (R&D) approval and Lancaster University Information Officer to write a Data Protection Impact Assessment (DPIA)
- https://www.lancaster.ac.uk/health-and-medicine/research/spectrum/research/ipof/ethics-framework/
-
- *Lobban et al. (2025) Evaluating peer online forums to support health: ethical and practical challenges. Journal of Medical Internet Research. https://doi.org/10.2196/73427*

Data use: GDPR guidelines

Any text containing medical information qualifies as **personal data**, even when pseudonomized, and even when someone posted it publicly (e.g. on social media)

Health data may be processed when there is an appropriate legal basis (legal consent or public interest) and "**collected for specified, explicit and legitimate purposes**".

Measures you should take:

1. Only collect and store data that you actually need for your goal (e.g. no birth dates if age does not matter)
2. Anonymize the data
3. Do not re-share the data with others
4. Remove the data at the end of the project
5. Get ethical approval from your faculty's ethics review committee

https://gdpr-info.eu/art-5-gdpr/

Pseudonimization:
replacing each user name by a unique identifier. Needed for e.g. longitudinal analysis of people's data

Anonymization:
removing all user ID information, no longer linking multiple texts of the same user to each other

Data: use of synthetic data

Synthetic data can be promising as a **training data source** that does not have ethical issues (mostly; but also depending on how it is generated)

History of tabular data generation in the health domain.

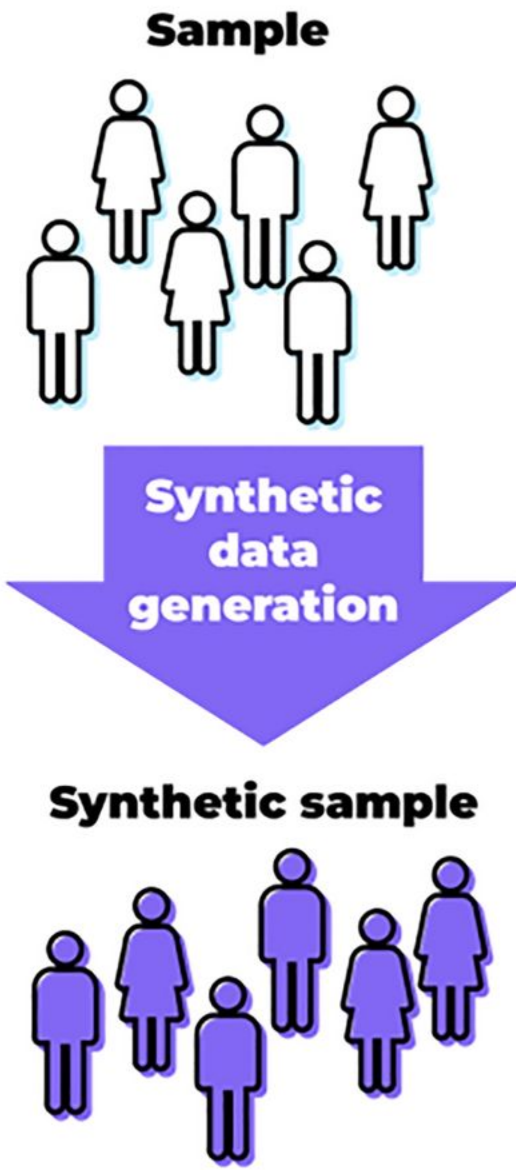


Image: https://www.nature.com/articles/s41598-024-57207-7

Data: use of synthetic data

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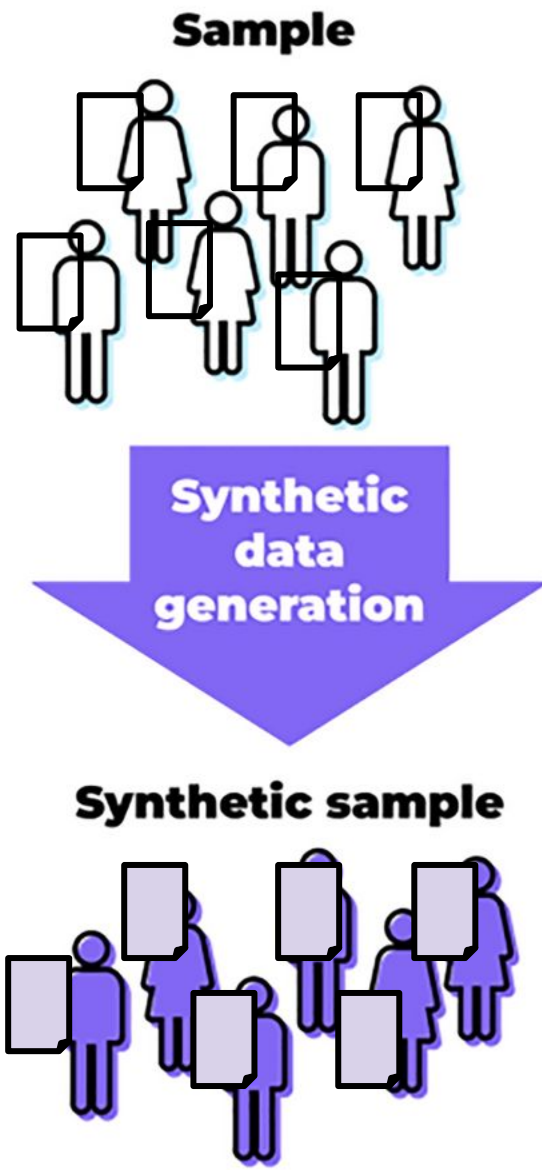
History of tabular data generation in the health domain.

How about text data?

- Text data in health records is diverse and noisy:

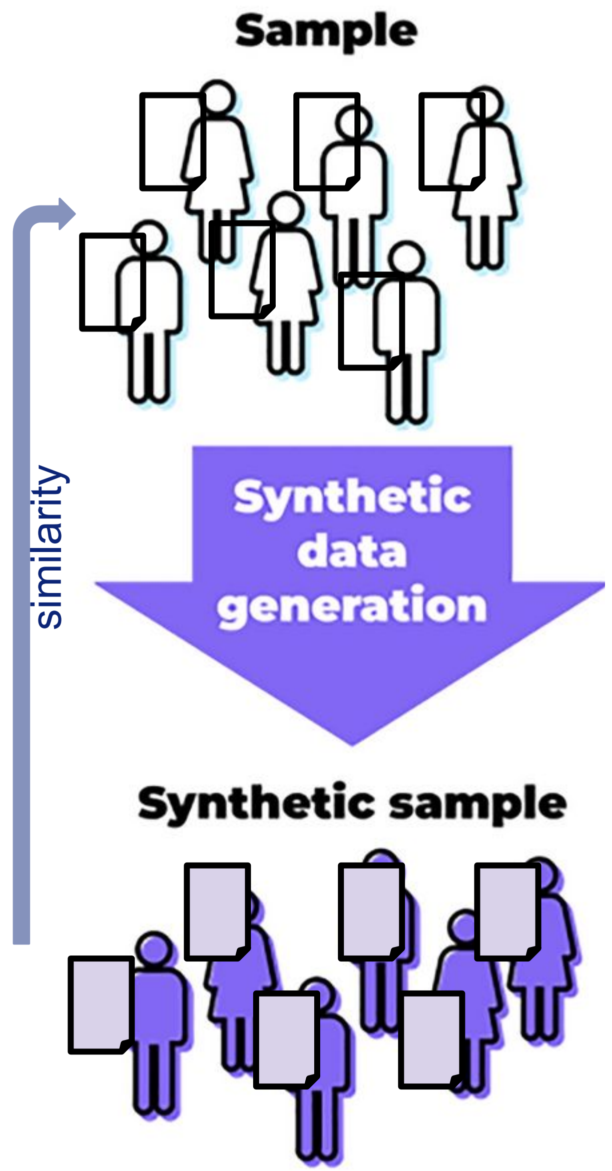
  Spelling errors, typos, abbreviations (e.g. ‘pt’ or even ‘p’ for patient)
- Text data is language-dependent
- Anonymization is more tricky for text data than for numeric data.
- Labels tend to be incomplete because of time pressure (if any)



Data: use of synthetic data

Evaluation aspects of synthetic data:

- **Similarity** to real data
    - lexical similarity (ROUGE)
    - semantic similarity (embeddings)
    - distributions of lengths and frequencies
    - can a classifier distinguish real from synthetic?
- **Utility**: how useful is the data to be used as a replacement for real data in a machine learning task?
    - Train on synthetic data, test on real data
- **Diversity**: how much variation is in the generated data?

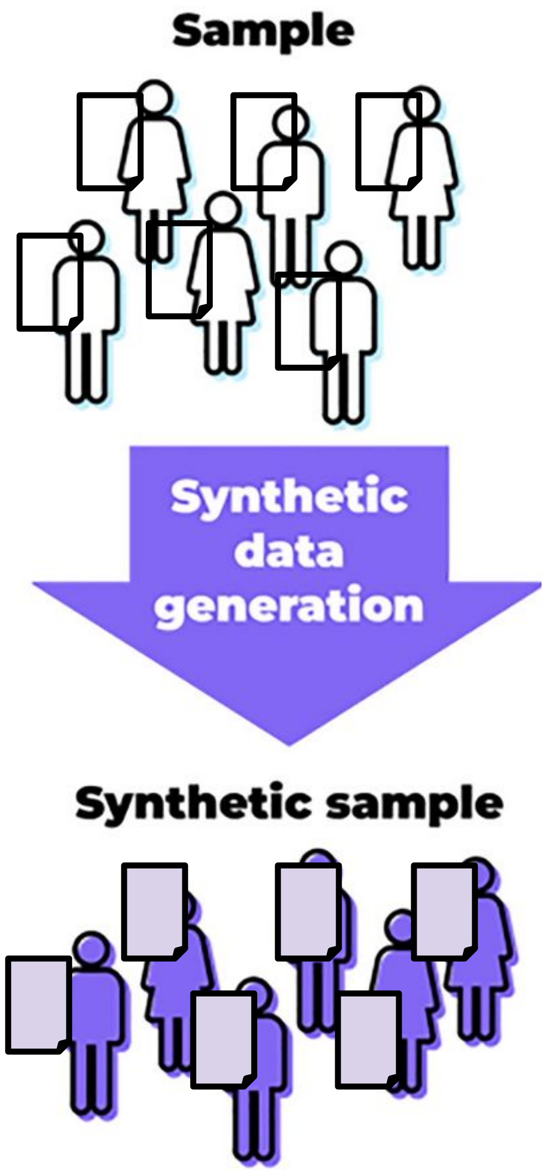


Other example work:
Marchesi, Raffaele, et al. "Mitigating Health Data Poverty: Generative Approaches versus Resampling for Time-series Clinical Data." *NeurIPS 2022 Workshop on Synthetic Data for Empowering ML Research*.

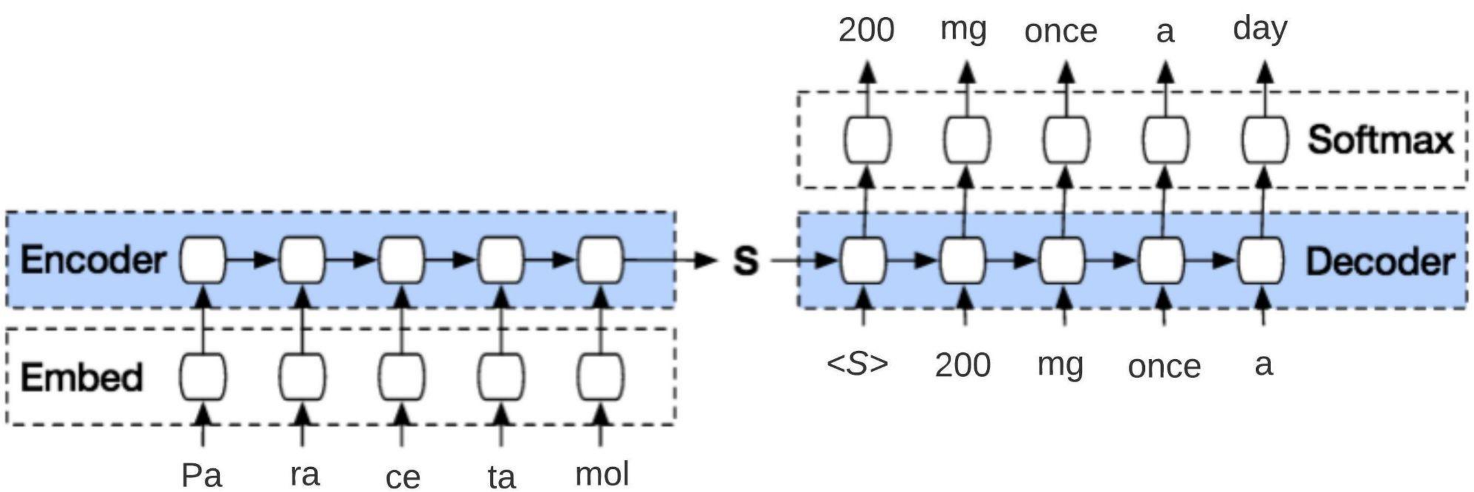
Synthetic data generation

What clinical data can we synthesize?

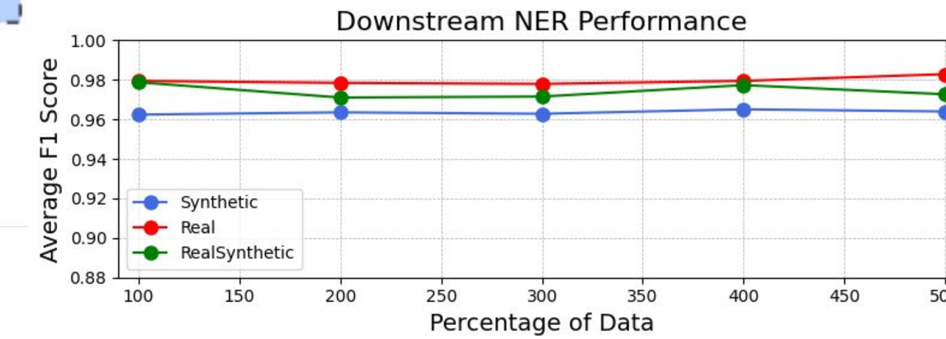
- Medication prescriptions
- Clinical letters
- Episode descriptions (short visit reports)
- Medical records

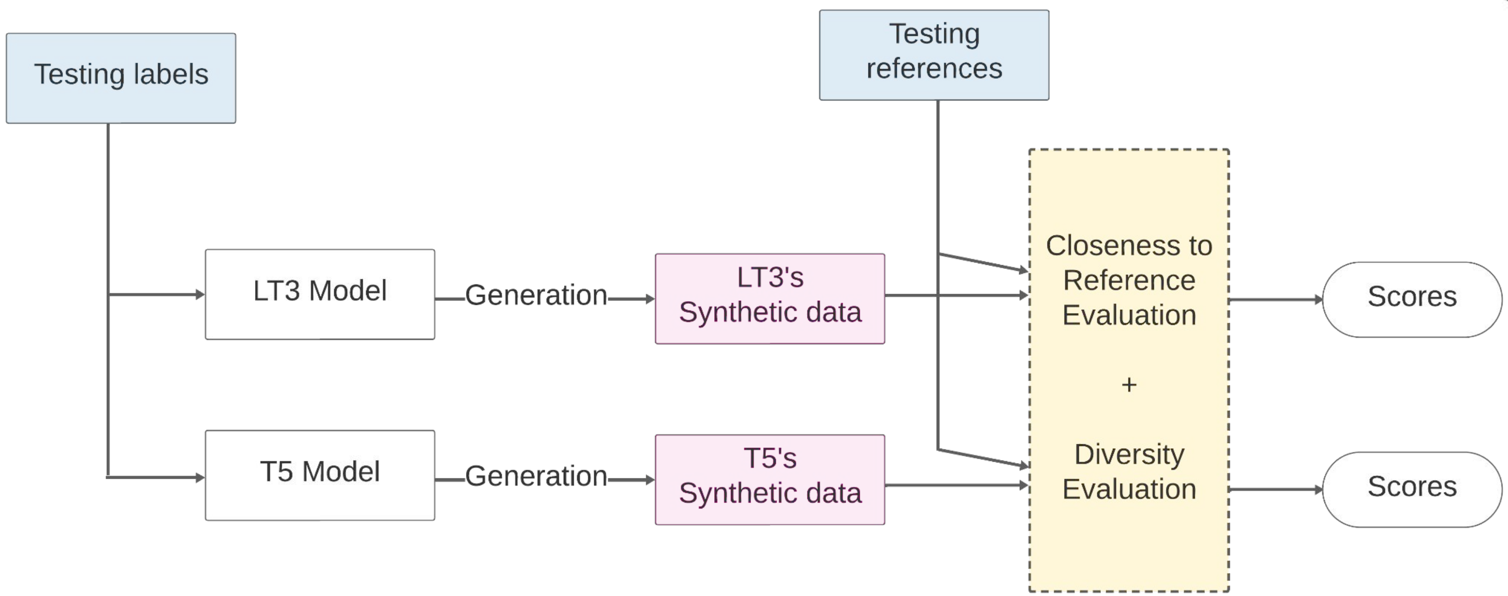


Synthetic data generation: prescriptions



Example of prescription generation

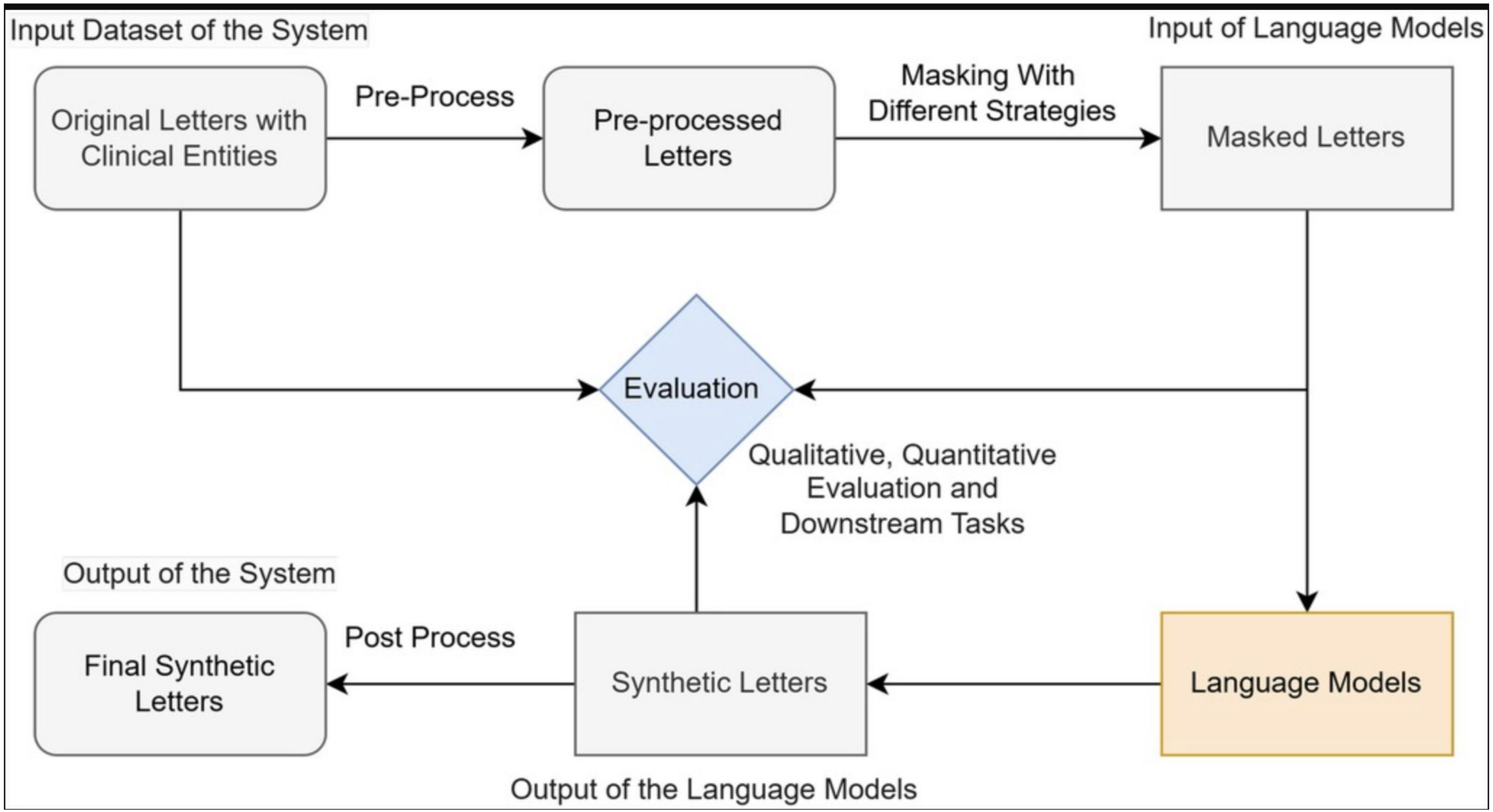




(Belkadi, Micheletti, Han, et al. 2023)
LT3: label-to-text Transformer
Prescription generation for ClinicalNER
Direct text eval vs application-driven eval
https://openreview.net/forum?id=yYIL3uhQ8i

Synthetic data generation: clinical letters

Workflow of Synthetic4health

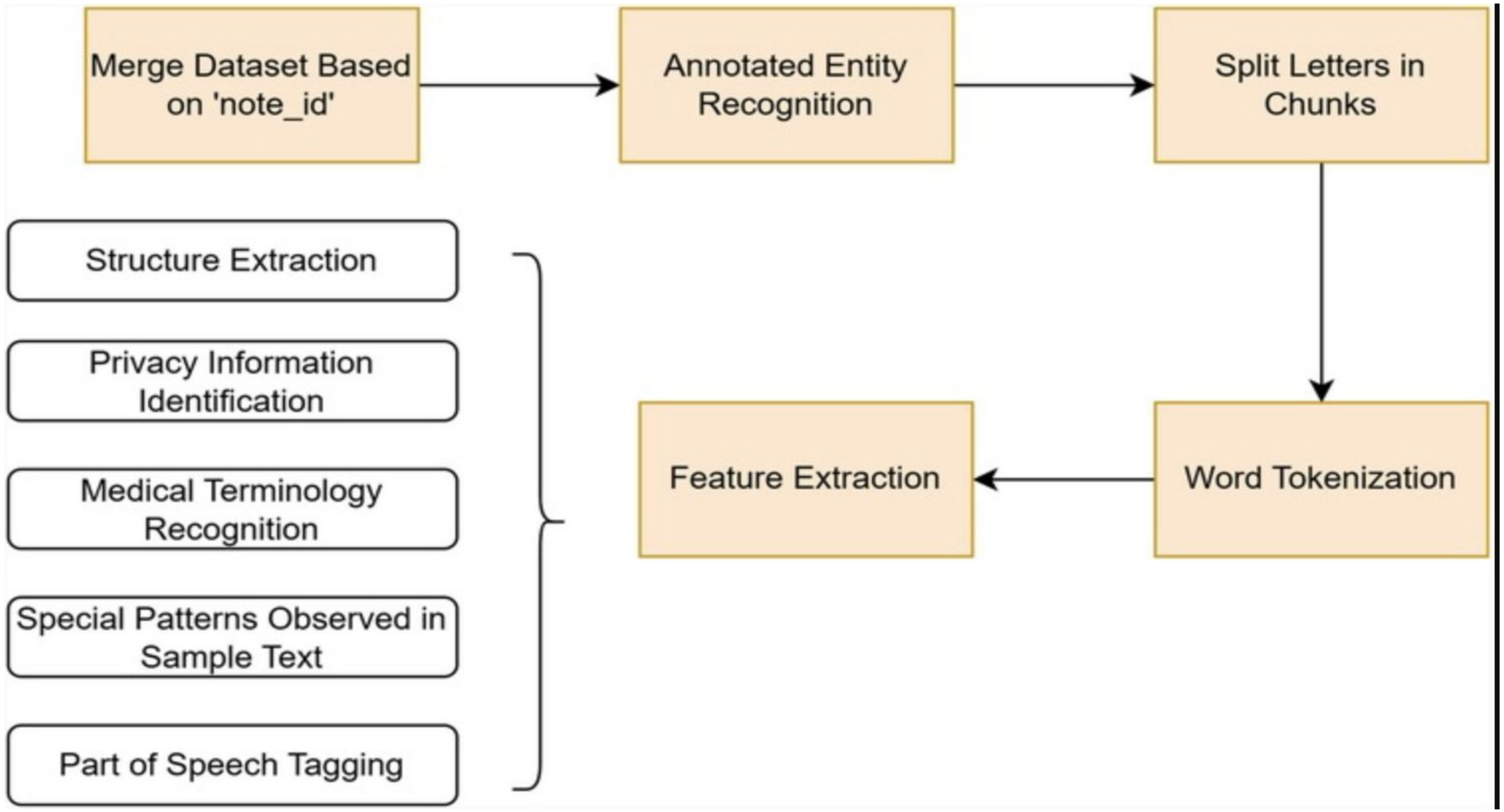


Synthetic4Health: generating annotated synthetic clinical letters

L Ren, S Belkadi, L Han, W Del-Pinto, G Nenadic, Frontiers in Digital Health 7, 1497130

Synthetic data generation: clinical letters

Preprocessing steps



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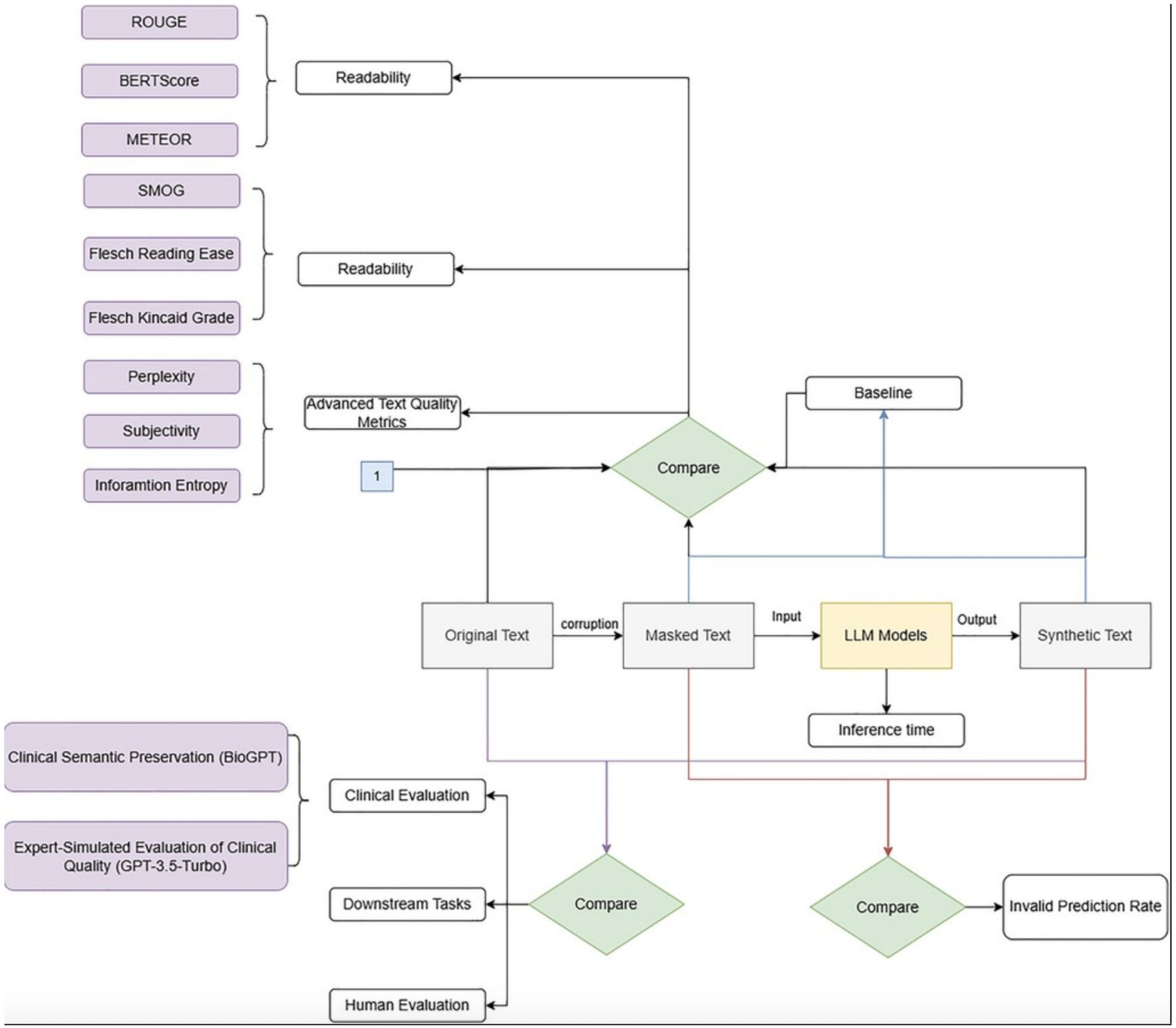
Synthetic data generation: clinical letters

Evaluation



SYNTHETIC4HEALTH: generating annotated synthetic clinical letters

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Synthetic data generation: episodes

Original data: 2 Million episodes from LUMC

- Very short descriptions: avg 2.3 words
- Many labels: ~1,300 ICPC classes
  (International classification of primary care)

Can we replace the real data by synthetic data, based on the distribution of class labels?

- Fine-tune a GPT-2 model

ICPC code → GPT → Episode description

- Measure the utility of the synthetic data by training a classifier
  - Accuracy reasonable
  - But the synthetic data is less **diverse**

Codes	True	Generated
K86	hypertensie	hypertensie
K86	hypertensie CVRM PP hoog chol	hypertensie
K86	Hypertensie hypertensie 170/105; ECG: normaal;R/Capozide 1 dd 1	hypertensie
K86	hypertensie	hypertensie

	Accuracy	F1
Real training data	91.8%	89.8%
Synthetic training data	81.4%	72.2%

Kattenberg, T. (Tristan), Generating a Synthetic Dutch Medical Data Set, Thesis Master Computer Science, LIACS, Leiden University, 2025.

Synthetic data generation: medical records

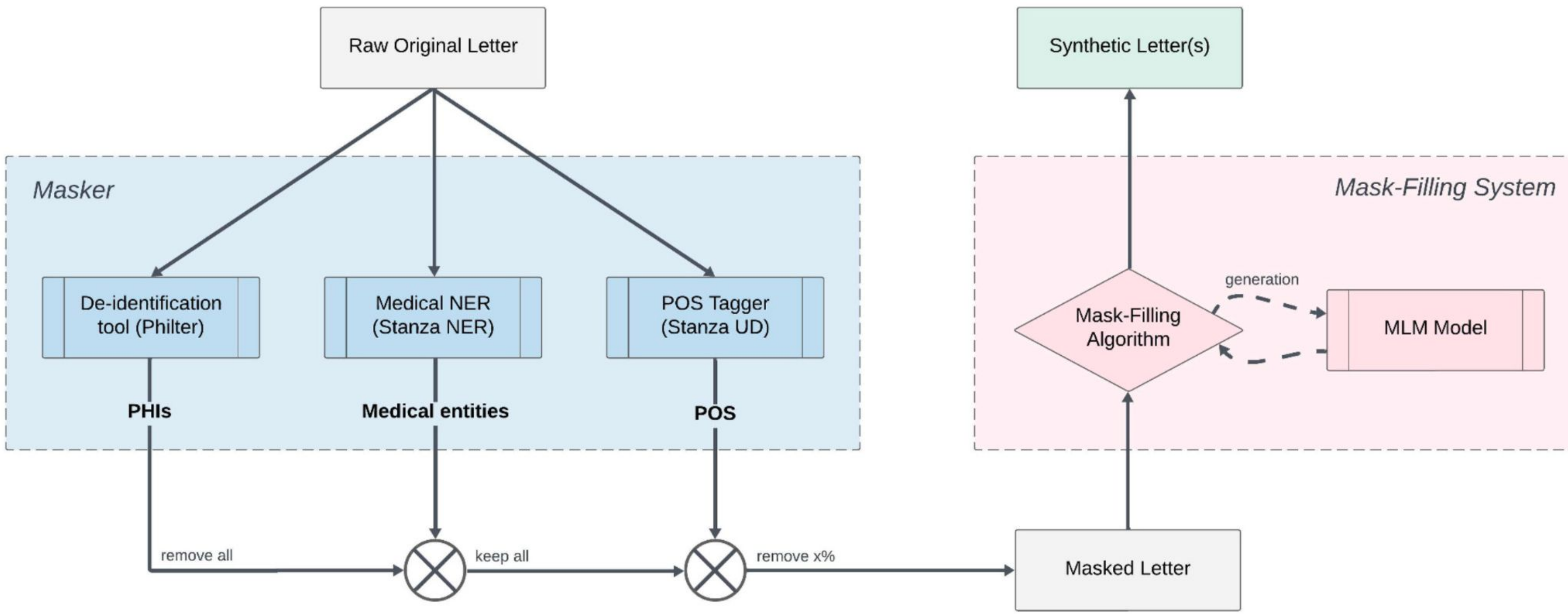

Fig. 1. MLM4SynMed: Design of the entire System with Masker and Mask-Filling components.

MLM4SynMed: Masked Language Modelling for Synthetic Free-text Medical Records Generation
S Belkadi, L Ren, N Micheletti, L Han, G Nenadic
2025 IEEE 13th International Conference on Healthcare Informatics (ICHI)

Synthetic data generation: medical records

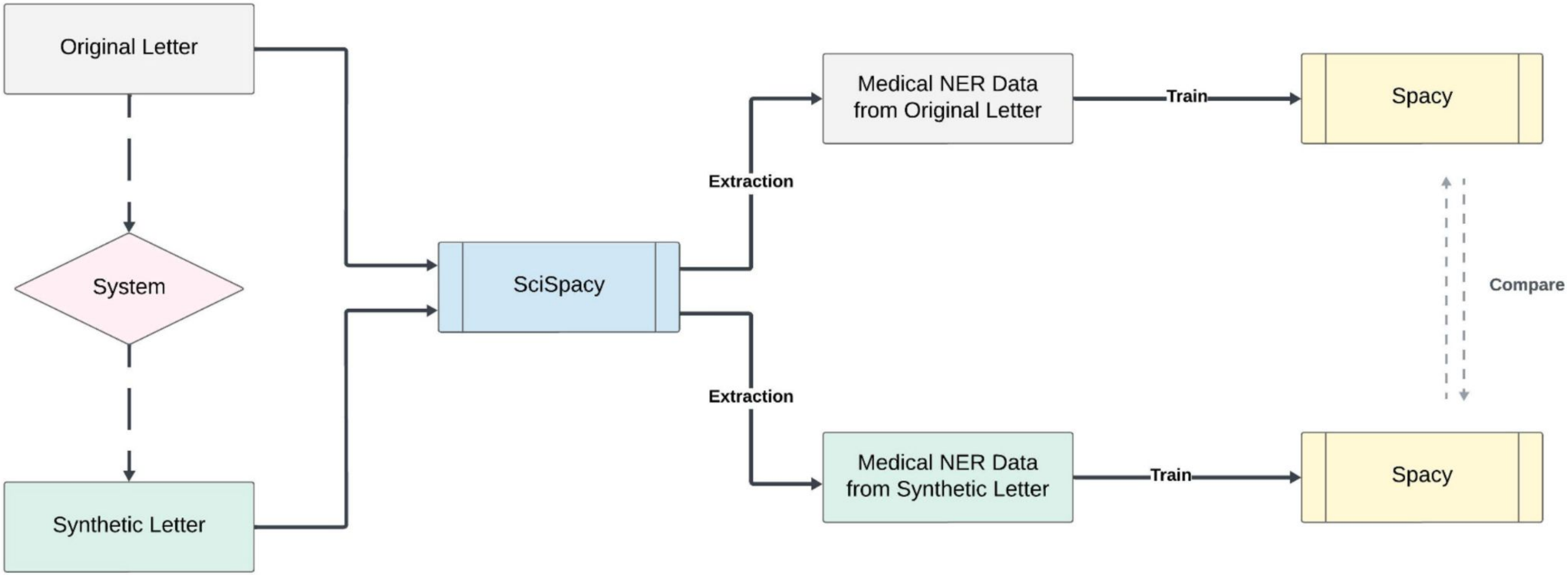

Fig. 2. Design of the Downstream Task Pipeline for Data Utility Evaluation.

MLM4SynMed: Masked Language Modelling for Synthetic Free-text Medical Records Generation

S Belkadi, L Ren, N Micheletti, L Han, G Nenadic

2025 IEEE 13th International Conference on Healthcare Informatics (ICHI)

key points of synthetic data generation

From clinical KB / dictionary to dense data
=> can train NER

From clinical KB / dictionary to synthetic letters
=> clinically soundness (challenging, RE)

=> have you met such challenge? How would you address?

Synthetic data references

Generating Medical Instructions with Conditional Transformer

S Belkadi, N Micheletti, L Han, W Del-Pinto, etc. 2023 NeurIPS WS on SyntheticData4ML - openreview.net

MLM4SynMed: Masked Language Modelling for Synthetic Free-text Medical Records Generation

S Belkadi, L Ren, N Micheletti, L Han, G Nenadic

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Synthetic4Health: generating annotated synthetic clinical letters

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Frontiers in Digital Health 7, 1497130

2. AI/NLP tasks

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Relevant NLP tasks for health data

- Classification and Regression
  - Thematic: ICD-10 coding, ICPC coding
  - Filtering, e.g. relevant posts on a patient forum
  - Predictive: patient re-admission, aggressive behaviour
  - Time to event (e.g. time to death)
- Sentiment
  - Patient experiences with treatments
  - Patient evaluation of hospitals
- Extraction (and De-identification)
  - Entities
  - Relations
- Linking (normalisation / relation)
  - From entities to a knowledge base
  - from entities to entities
- Appendix:
  - Summarization & Simplification
  - Translation

Classification

Motivation for automatic classification of clinical data: saving time for doctors

- Administration costs time
- The typical classification systems are complex and have many different codes
  - International Classification of Diseases (ICD-10, ICD-11)
  - International Classification of Primary Care (ICPC)
  - Diagnostic and Statistical Manual of Mental Disorders (DSM)

- ICPC:
  - ~1,300 classes
  - Assigned on the level of an 'episode' (visit)

L01	Neck Symptoms/Complaints
L08	Shoulder Symptoms/Complaints
L09	Arm Symptoms/Complaints
L10	Elbow Symptoms/Complaints
L11	Wrist Symptoms/Complaints
L12	Hand/Finger Symptoms/Complaints
L13	Hip Symptoms/Complaints
L14	Leg/Thigh Symptoms/Complaints
L15	Knee Symptoms/Complaints
L16	Ankle Symptoms/Complaints
L17	Foot/Toe Symptoms/Complaints
L18	Muscle Pain/Fibrositis
L19	Other Symptoms, Multiple/Unspecified Muscle
L20	Symptoms Multiple/Unspecified Joints
L28	Disability/Impairment
L29	Other & Multiple Musculoskeletal Symptoms
L77	Sprain Of Ankle/Foot
L78	Sprains/Strains of Knees
L79	Sprains/Strains Other Joints
L80	Dislocations
L81	Other Injury Musculoskeletal
L83	Syndrome Of Cervical Spine
L84	Osteoarthritis Spine
L85	Acquired Deformities Of Spine
L87	Ganglion Joint/Tendon
L88	Rheumatoid Arthritis
L89	Osteoarthritis Hip
L90	Osteoarthritis Knee
L91	Other Osteoarthritis
L92	Shoulder Syndrome
L93	Tennis Elbow
L94	Osgood-Schlatter, Osteochondritis
L95	Osteoporosis
L96	Acute Meniscus/Ligament Knee
L97	Chronic Internal Knee Derangement
L98	Acquired Deformities Limbs
L99	Other Musculoskeletal/Connective Disorder
N93	Carpal Tunnel Syndrome

Classification: example project

Automated classification of ICPC codes using episode descriptions

- Data: 2 Million episodes from LUMC
- Very short descriptions: avg 2.3 words
- Many labels: ~1,300 classes
- Anonymization needed

Results with a 80-20 split:

- Precision 74.6%
- Recall 64.7%

Most difficult class: A97 ('no illness')

	A04	A97	F91	H81	K86	L04	L17	L81	L98	L99	R05	R74	R96	S74	S87	S99	T90	T91	T93	U71
A04	4,043	326	0	0	1	3	1	0	0	0	2	7	0	0	0	0	1	7	0	0
A97	5	11,418	2	0	14	1	4	2	1	0	1	1	2	1	0	0	3	9	5	9
F91	1	561	2,448	0	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
H81	0	68	0	3,190	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
K86	0	121	1	0	8,258	2	0	0	0	0	2	1	0	0	0	0	2	0	6	0
L04	6	519	0	0	2	2,744	0	20	0	22	5	3	0	0	0	0	0	0	0	0
L17	1	515	0	0	0	0	2,874	93	19	36	0	0	0	8	4	0	0	0	0	0
L81	0	362	0	0	0	65	198	1,839	0	17	1	0	0	1	0	0	0	0	0	0
L98	0	664	0	0	0	0	105	2	2,375	16	0	0	0	0	0	0	0	1	0	0
L99	1	1056	0	0	0	7	64	48	3	2,395	0	1	0	0	0	0	0	0	[illegible]	[illegible]
R05	4	108	0	0	0	5	0	0	0	0	5,699	64	36	0	0	0	0	0	[illegible]	[illegible]
R74	2	328	0	2	0	1	0	0	0	0	59	5,654	39	1	0	0	0	0	[illegible]	[illegible]
R96	0	379	0	0	1	2	0	0	0	0	40	87	5,954	0	2	0	0	[illegible]	[illegible]	[illegible]
S74	0	392	0	0	0	1	22	0	0	0	0	0	0	3,168	61	2	0	[illegible]	[illegible]	[illegible]
S87	0	233	0	0	0	0	1	0	0	0	0	2	1	29	7,819	21	0	[illegible]	[illegible]	[illegible]
S99	0	678	0	0	0	2	7	0	0	0	0	0	0	3	9	2,308	0	[illegible]	[illegible]	[illegible]
T90	1	412	0	0	9	0	4	0	0	0	0	0	0	0	0	0	3,866	[illegible]	[illegible]	[illegible]
T91	6	296	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2,868	[illegible]	[illegible]
T93	0	547	0	0	8	0	0	0	0	0	1	0	0	0	0	0	1	0	3,472	0
U71	0	115	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3,906

Table 8: Confusion matrix of predictions for the twenty most occurring codes.

Long tail distribution issues in data for NLP

Ebing, W. (Wouter), Analysing Classification Of Episodes From Electronic Health Records With The Use Of Text Mining, Thesis Bachelor Informatica, LIACS, Leiden University, 2021.

Predictive modelling: example project

- Prognostication of life expectancy is difficult for doctors
- We approached the task of predicting life expectancy as a supervised machine learning task (at the time still with an LSTM)

- We compared
  - a model which does not use text features (baseline model)
  - a model which uses features extracted from the free texts of the medical records (keyword model)
  - doctors' performance on a similar task as described in scientific literature.

GP to know life expectancy of their patients - for end of life conversations

Beeksma, Merijn, Suzan Verberne, Antal van den Bosch, Enny Das, Iris Hendrickx, and Stef Groenewoud. "Predicting life expectancy with a long short-term memory recurrent neural network using electronic medical records." *BMC medical informatics and decision making* 19, no. 1 (2019): 36.

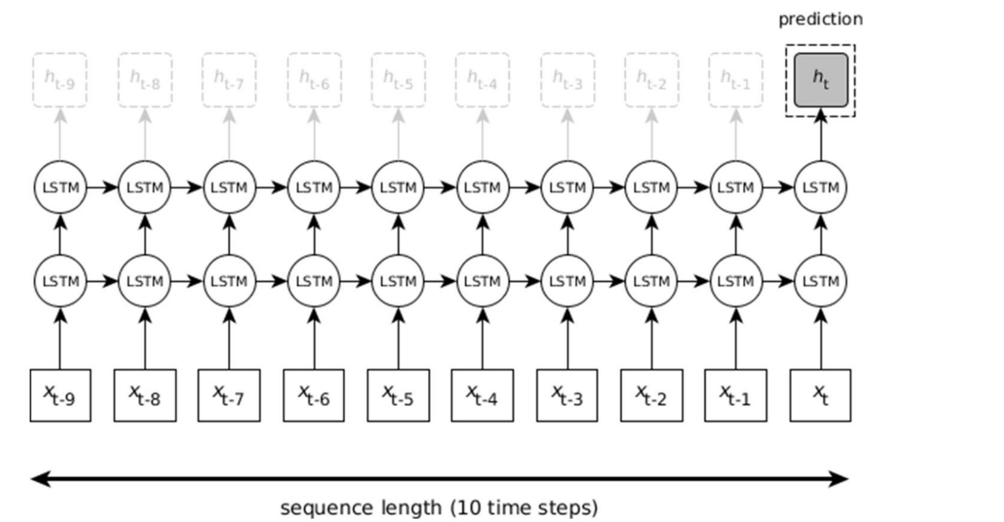


Fig. 1 Simplified LSTM architecture. At final time step t, x_t represents the feature vector used as input to the hidden LSTM units, which activate output h_t. In each preceding time step, output h functions as an intermediate prediction of life expectancy. We are interested in final prediction h_t: a probability distribution for the next 50 months

Predictive modelling: example project

- Results:
  - Doctors and the baseline model were correct in 20% of the cases, (margin 33% around the actual life expectancy)
  - The text-based model had an accuracy of 29% (win)
  - **Doctors overestimated life expectancy** in 63% of the incorrect prognoses, which harms anticipation to appropriate end-of-life care, our model overestimated life expectancy in only 31% of the incorrect prognoses.

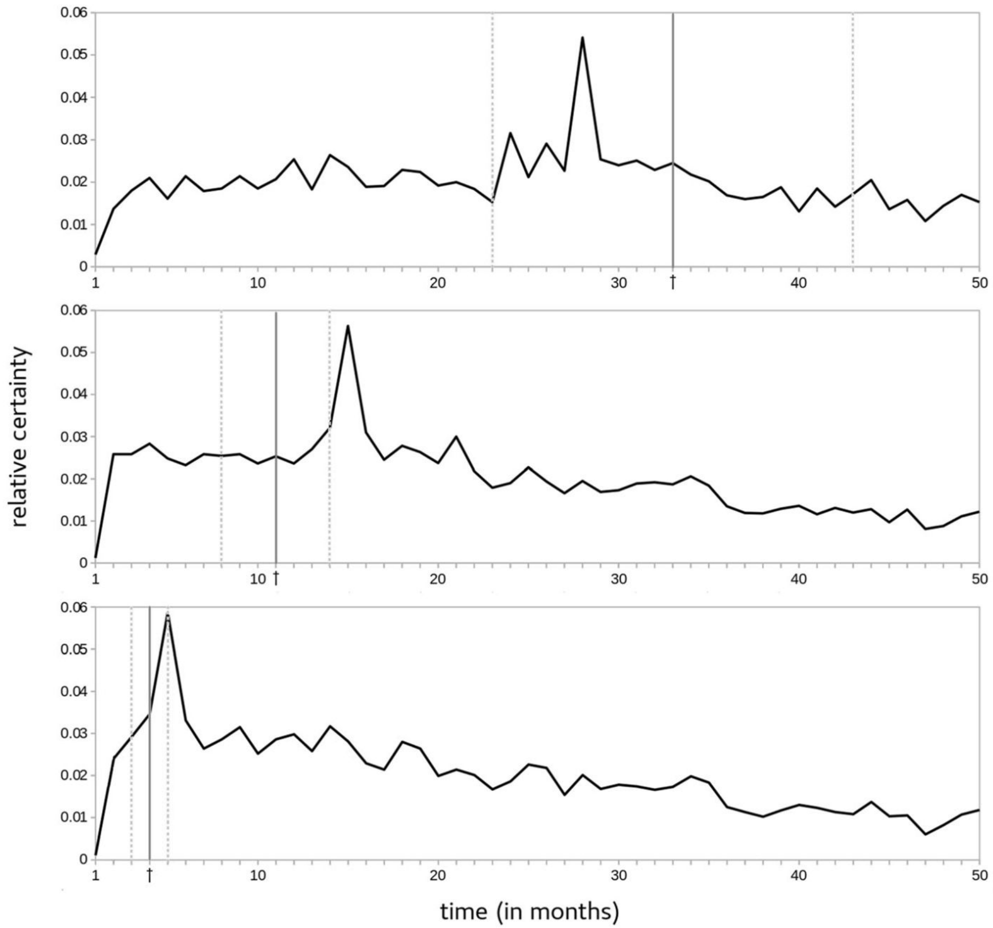
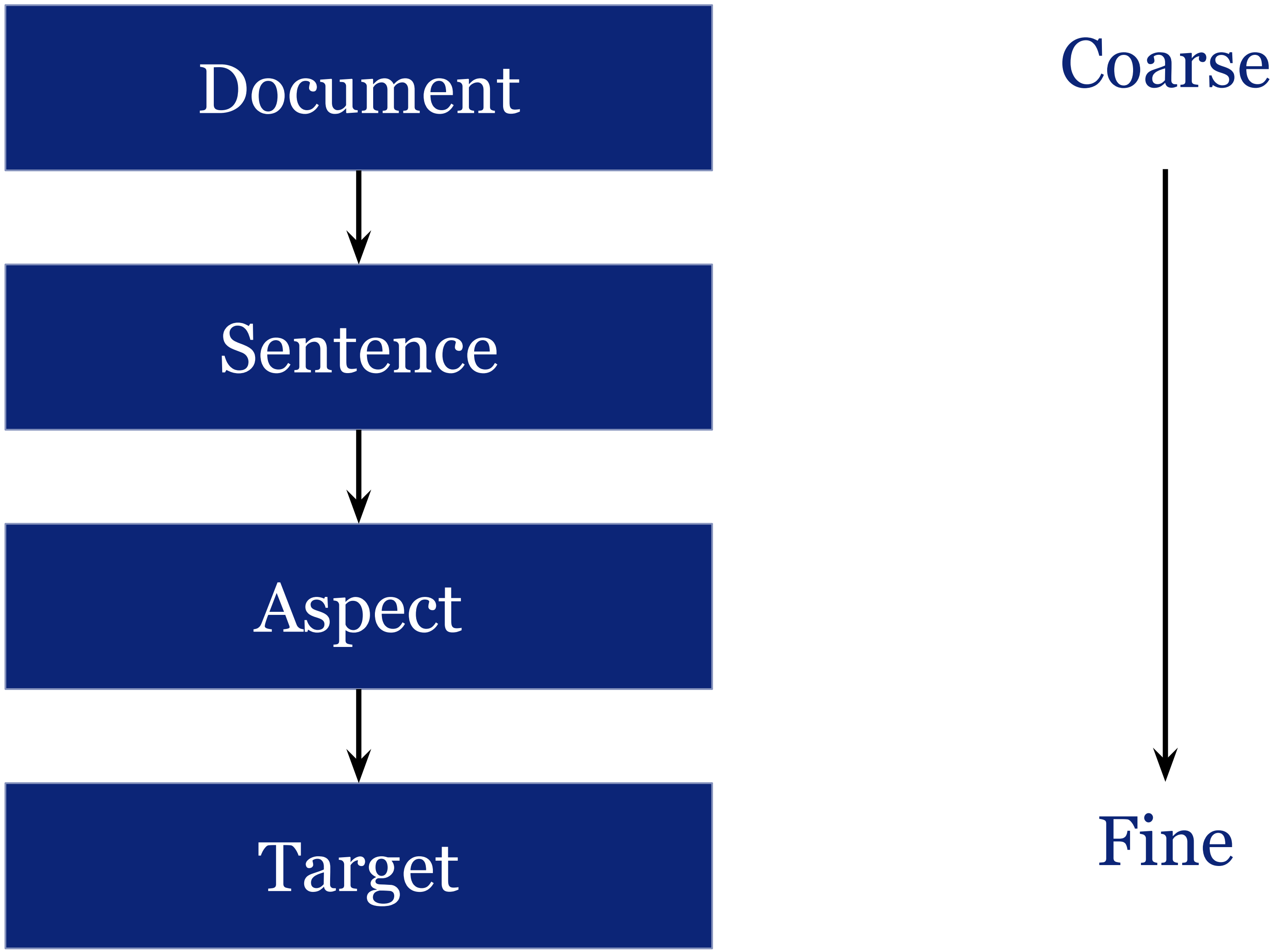


Fig. 2 Probability distributions produced by the baseline model for one patient *at different moments in time*. From top to bottom, the corresponding actual number of months to death are 33 months, 11 months, and 3 months, respectively

Beeksma, Merijn, Suzan Verberne, Antal van den Bosch, Enny Das, Iris Hendrickx, and Stef Groenewoud. "Predicting life expectancy with a long short-term memory recurrent neural network using electronic medical records." *BMC medical informatics and decision making* 19, no. 1 (2019): 36.

Sentiment and its various levels



More in depth review of the different sentiment levels see chapter 2 of: Moore, A. (2021). *Empirical Evaluation Methodology for Target Dependent Sentiment Analysis*. [Doctoral Thesis, Lancaster University]. Lancaster University. https://doi.org/10.17635/lancaster/thesis/1408

Document and Sentence Sentiment Analysis

Objective: To find the sentiment of the text

Assumptions:

- That the text is only discussing one topic
- The sentiment within the text has come from the author of the document.

Methods:

- Zero shot with LLMs
- Pre-Trained Transformers fine tuned.

Document Example (patient survey)

My **Doctor** is **great** and all of the **staff** are **friendly**!! They are always on **time** **except** for the **specialist** but when they do arrive the **treatment** always **fixes the problem**. When my **test results** are ready the **hospital** calls me. However, my **neighbor** is **never happy** with their **treatment** nor the **staff** I have never understood how that can be the case.

Document is based off examples given in the annotation guidelines of It's Difficult to Be Neutral – Human and LLM-based Sentiment Annotation of Patient Comments (Mæhlum et al., CL4Health 2024) which can be found at: https://github.com/ltgoslo/Sentiment-Annotation-of-Patient-Comments/blob/main/documentation/Retningslinjer_for_FHI_SA-1.pdf

Sentiment and Rationales

Objective: Given a transcript (audio or text) predict the sentiment and generate the reason.

Dataset: Open, multimodal, multilingual (Vietnamese, English, Chinese, German, and French) doctor patient conversation.

Methods:

- Fined tuned small BERT like models performed competitively to larger fine tuned LLMs.
- Domain specific BERT like models performed better than non-domain specific.
- The rationale helped predict the sentiment.
- Generated reason were semantically similar to the human ground truth.
- Audio only models performed worse than text.

Sentiment Reasoning for Healthcare (Nguyen et al., ACL 2025)

Example from the dataset

Text	Label	Reason
The patient will suffer from emotional disorder and sometimes depression	Negative	Emotional disorder

Fine Grained Sentiment Analysis

Objective: Identify (H, T, A, E, S, O) Holder, Target, Aspect, Entity, Sentiment, Opinion term.

Methods: Most concentrate on (T, A/E, S, O) also known as Aspect Category Opinion Sentiment (ACOS), and Aspect Sentiment Quad Prediction (ASQP).

- Zero and few shot LLMs
- Pre-Trained Transformers fine tuned.

Fine Grained Sentiment Analysis Example (patient survey)

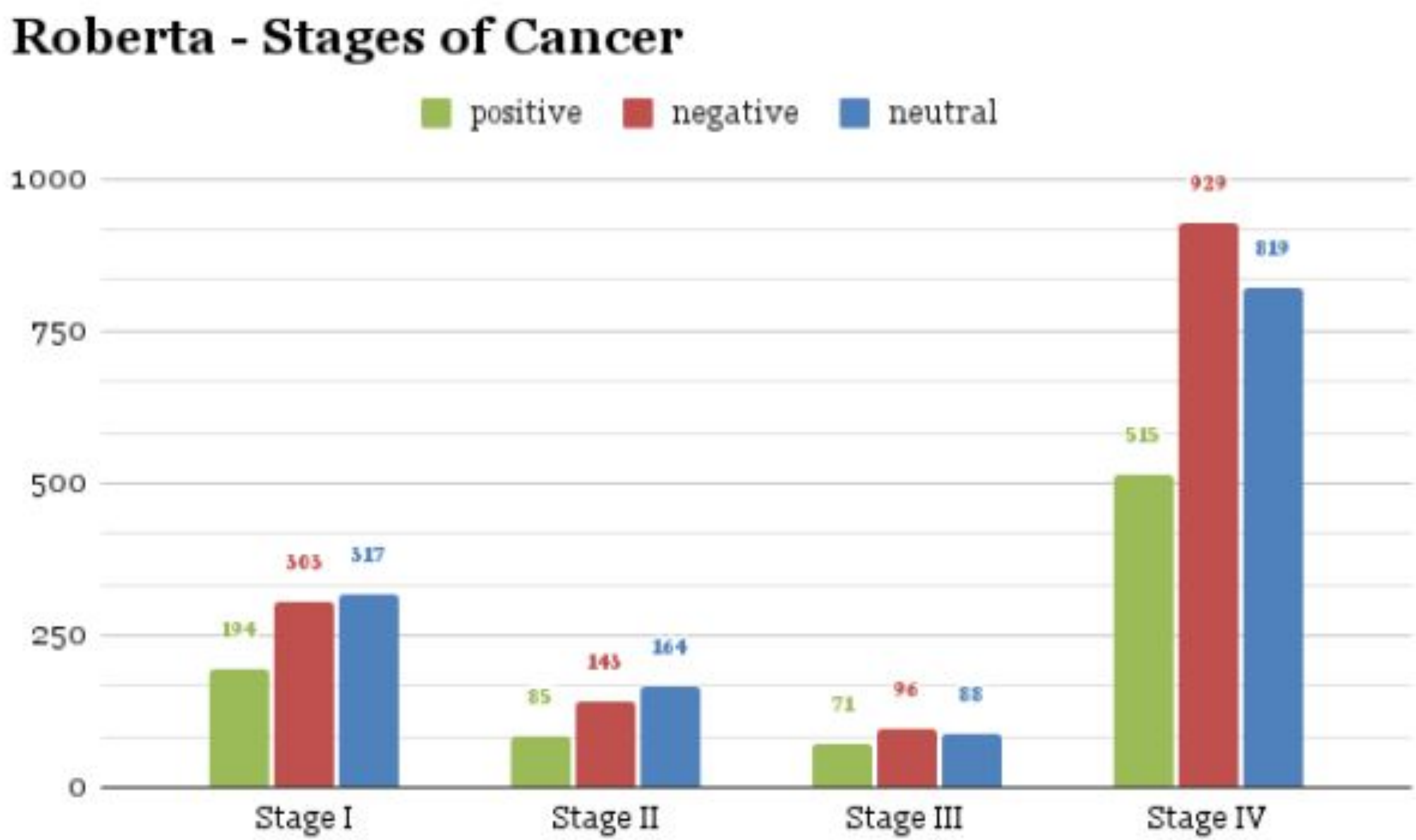
Text = My **Doctor** is **great** and all of the **staff** are **friendly**!! They are always on **time except** for the **specialist** but when they do arrive the **treatment** always **fixes the problem**. When my **test results** are ready the **hospital** calls me. However, my **neighbor** is **never happy** with their **treatment** nor the **staff** I have never understood how that can be the case.

Holder = Author of the text
Target = Doctor
Aspect = Competence of providers
Entity = Healthcare providers and staff
Sentiment = Positive
Opinion Term = great

Example Aspect and Entity categories taken from [From Pain to Praise: Aspect-Based Sentiment Analysis for Norwegian Patient Feedback]() (Storset et al., HeaLing 2026)

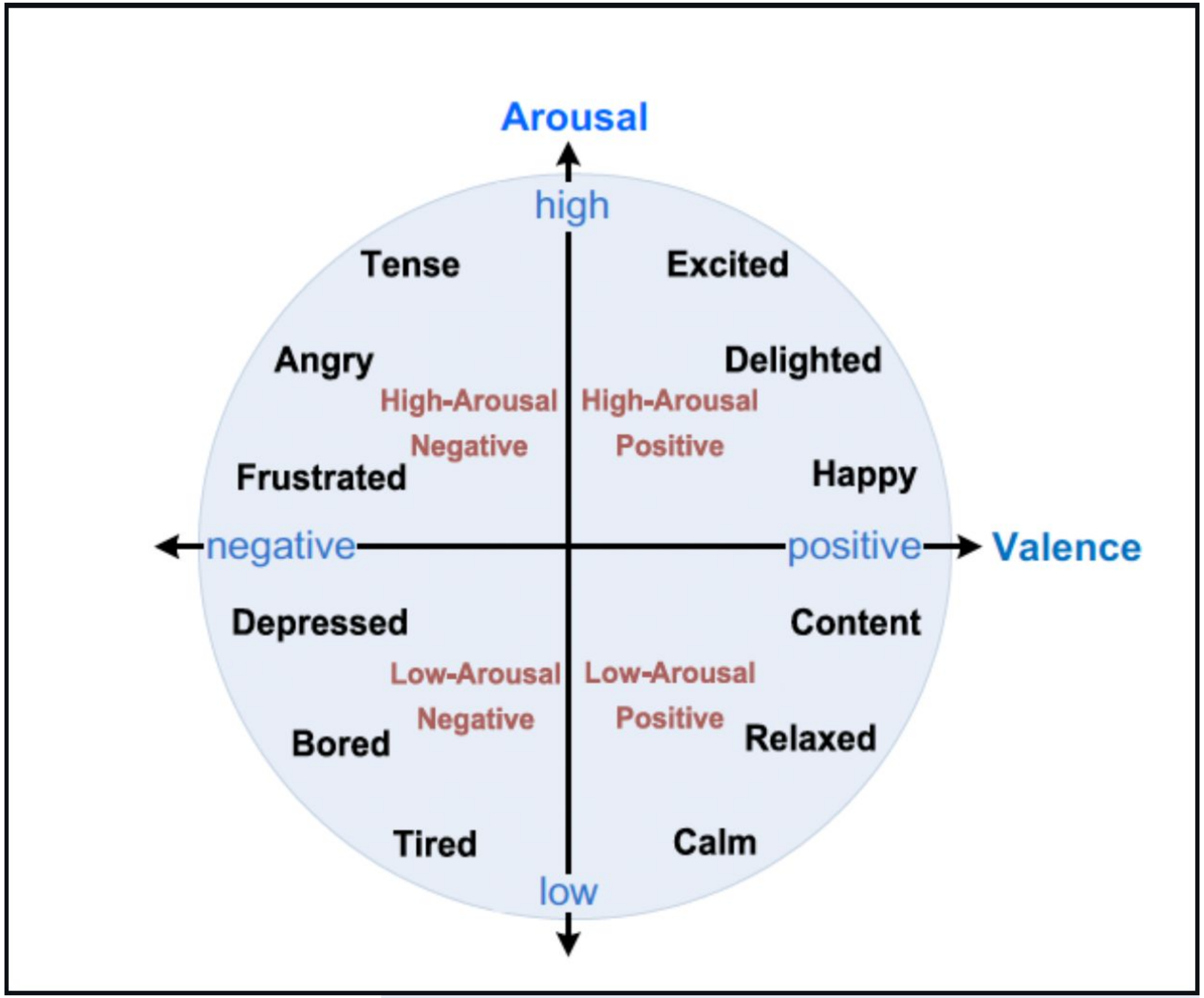
Sentiment and Emotion analysis within the 4D Picture Project

Objective: Sentiment and emotional analysis of Reddit forum data on stages and treatment of cancer.



[Analysing Emotions in Cancer Narratives: A Corpus-Driven Approach]() (Lal et al., CL4Health 2024)

https://github.com/DimABSA/DimABSA2026



QuadAI at SemEval-2026 Task 3: Ensemble Learning of Hybrid RoBERTa and LLMs for Dimensional Aspect-Based Sentiment Analysis
AJW de Vink, FK Ventirozos, N Amat-Lefort, L Han
arXiv preprint arXiv:2603.07766

Information Extraction: example project

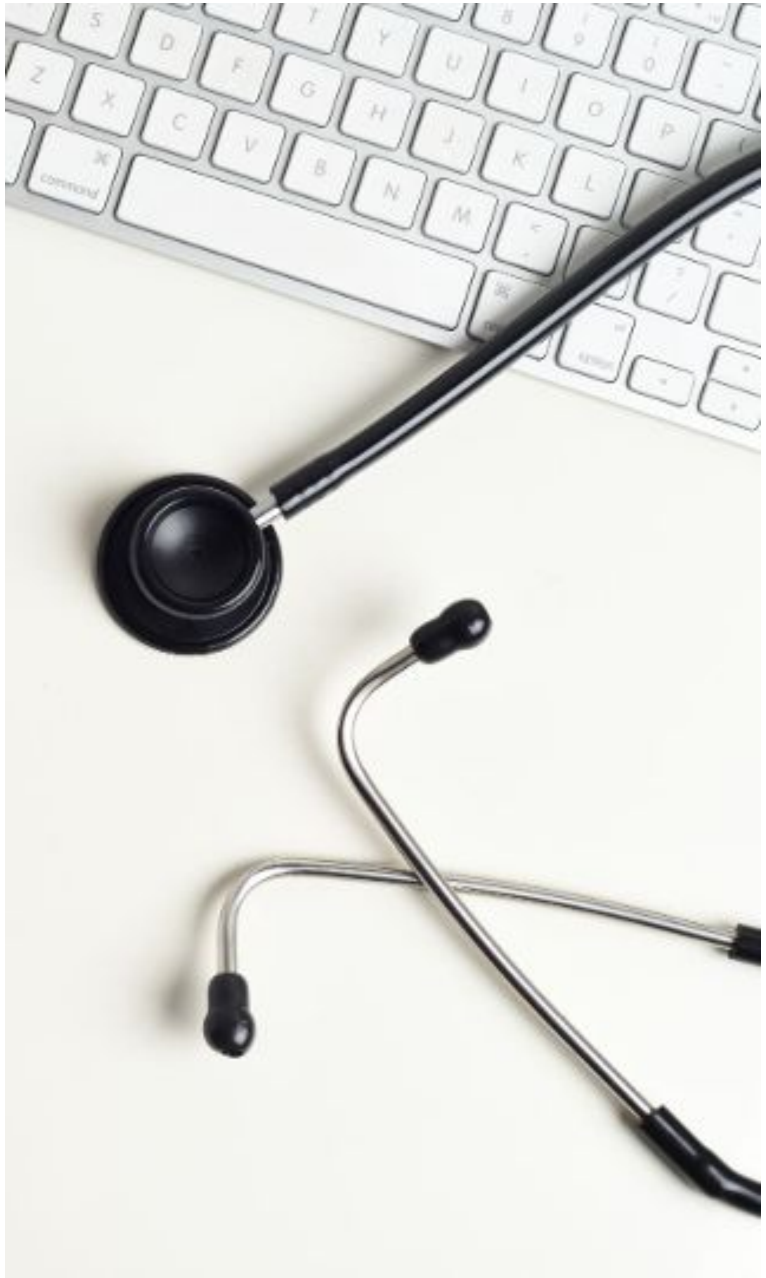
Real-world evidence from **patient forums** can complement medical perspectives

Our case: Gastrointestinal Stromal Tumor (GIST)

GIST Support International on Facebook

- The forum data was donated to us by the organization
- Data from 24 Oct 2009 until 1 Nov 2020
- 124,103 posts in 14,631 threads
- Median post length: 20 words

With LUMC. public forum but personal data. Donated to us for research.



Entity Extraction

Medical entity extraction is a machine learning task based on sequence labelling

- Word order matters
- One entity can span multiple words
- There are multiple ways to refer to the same concept

The extracted entities need to be linked to a standard form (in an ontology)

ADE = adverse drug events = side effects

Arjun Magge, Elena Tutubalina, Zulfat Miftahutdinov, Ilseyar Alimova, Anne Dirkson, Suzan Verberne, Davy Weissenbacher, Graciela Gonzalez-Hernandez, DeepADEMiner: a deep learning pharmacovigilance pipeline for extraction and normalization of adverse drug event mentions on Twitter. *Journal of the American Medical Informatics Association (JAMIA)*, Volume 28, Issue 10, October 2021, pp 2184–2192

Entity extraction approach

Named entity recognition is typically approached as a sequence labelling task: Each token gets a label

Label spans (word sequences) in the text that refer to entities

Store the data with a label for each word

- Label 'B': this word is the beginning of an entity
- Label 'I': this word is inside an entity
- Label 'O': this word is not in an entity

I can't fall asleep at all.

Token	I	can't	fall	asleep	at	all
Label	O	B-ADE	I-ADE	I-ADE	O	O

Entity extraction approaches

- Sequence labelling is a supervised learning task, typically approached with transfer learning:
- A pretrained (biomedical) base model is fine-tuned for entity recognition
- This requires labelled data

Jane Villenueva of United Airlines Holding discussed

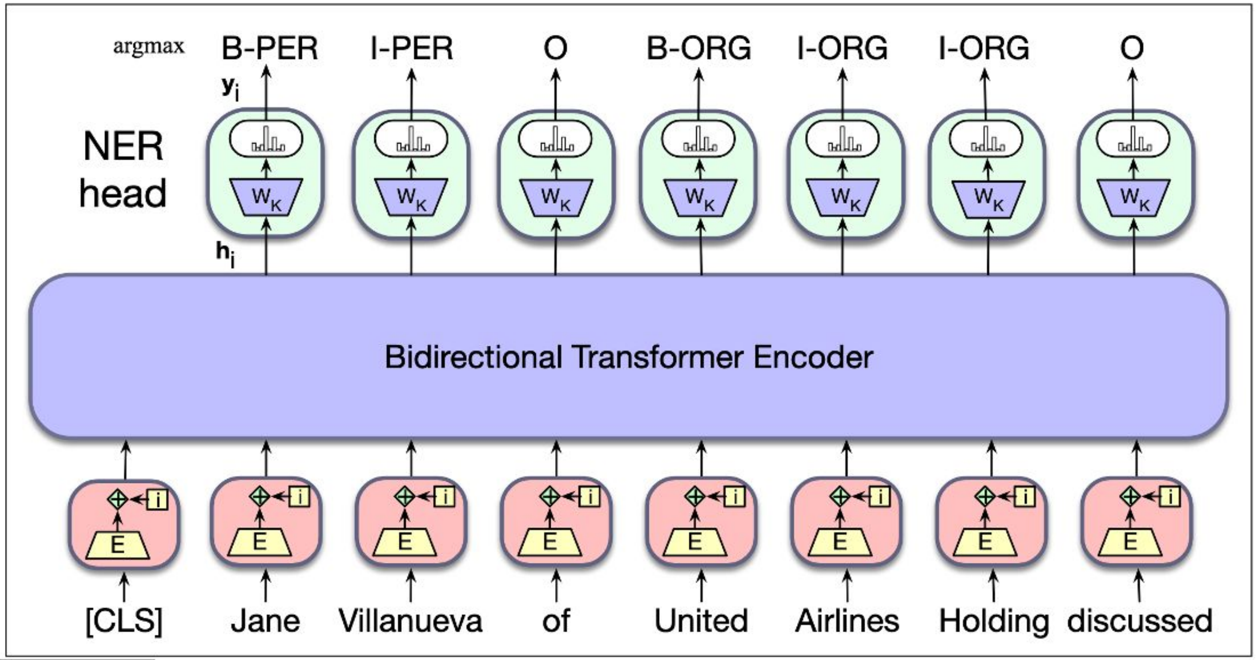
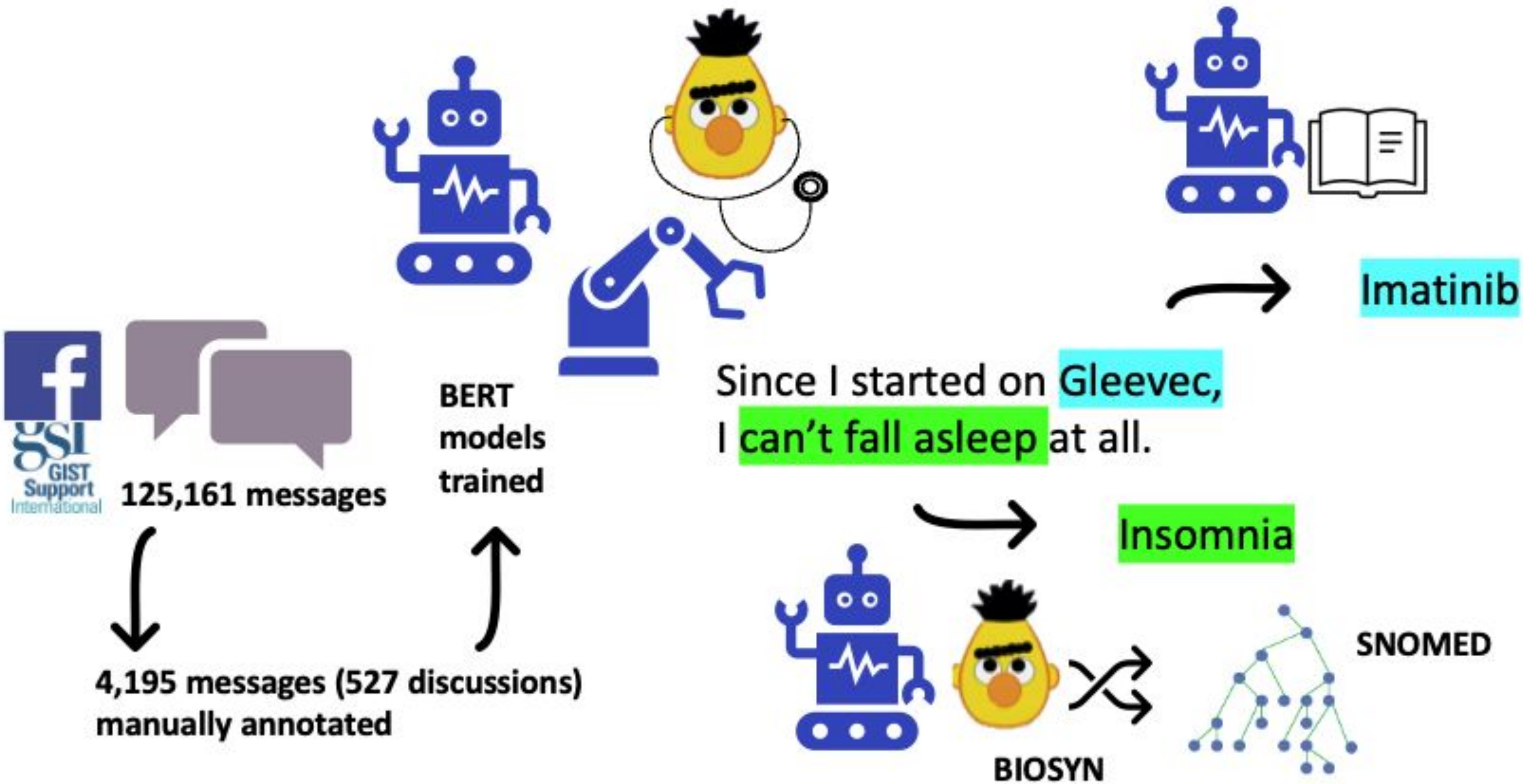


Figure 11.13 Sequence labeling for named entity recognition with a bidirectional transformer encoder. The output vector for each input token is passed to a simple k-way classifier.

Dan Jurafsky and James H. Martin: Speech and Language Processing (3rd ed. draft, 2026) https://web.stanford.edu/~jurafsky/slp3/

Medical entity extraction



Anne Dirkson, Suzan Verberne, Wessel Kraaij, Gerard van Oortmerssen and Hans Gelderblom. Automated gathering of real-world data from online patient forums can complement pharmacovigilance for rare cancers. *Nature Scientific Reports* 12, 10317 (2022)

Medical entity extraction

Evaluation of ADE extraction:

Recall	74%
Precision	70%
F1	**0.72**
Human pairwise F1	**0.80**

Model applied to the whole forum:

Drug	# of ADE mentions found
Imatinib	13,376
Sunitinib	2,335
Regorafenib	319
Ripretinib	319
Avapritinib	297

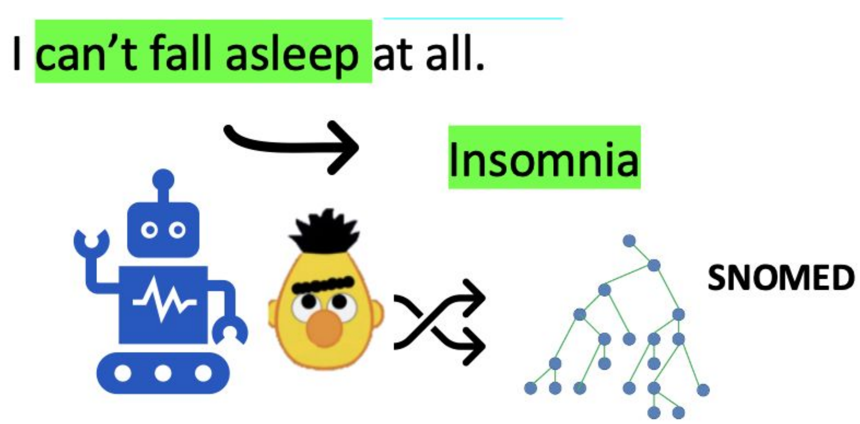
From extraction to linking (normalisation)

Identifying the spans of text that refer to an entity is not enough...

We also need to normalize the entities to the concepts they refer to.

We use medical knowledge bases/ontologies

- Most common: UMLS/Snomed



Biomedical entity linking

Why is biomedical entity linking so difficult?

1. **High diversity** in surface forms of the same concept
    - MI and heart attack both are surface forms of concept Myocardial Infarction (C0027051)
2. **Similarity** in surface form of different concepts
    - Candida and Cardia (Levenshtein distance=2) refer to a yeast and the heart respectively
3. Free text by patients and medical professionals is often **noisy**
4. **Number of concepts** in the medical domain is huge
    - The UMLS has > 3.3 million unique concepts

Biomedical entity linking

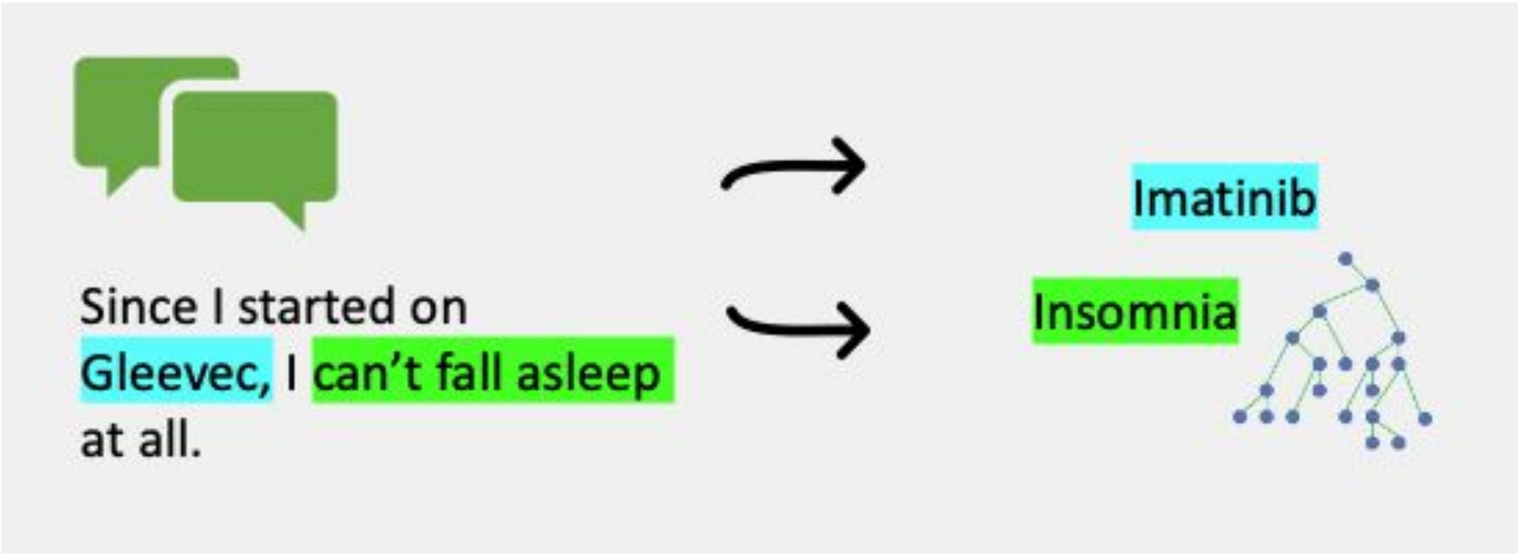
Extra challenges in languages other than English:

- UMLS consists of mostly English terms
- Large labelled datasets are sparse

Name Count by Language:

Language	Name Count	% of Metathesaurus
ENG	10161789	63.96%
SPA	1879224	11.83%
POR	452723	2.85%
FRE	446929	2.81%
JPN	345246	2.17%
RUS	306887	1.93%
DUT	301049	1.89%
GER	270780	1.7%

Biomedical entity linking



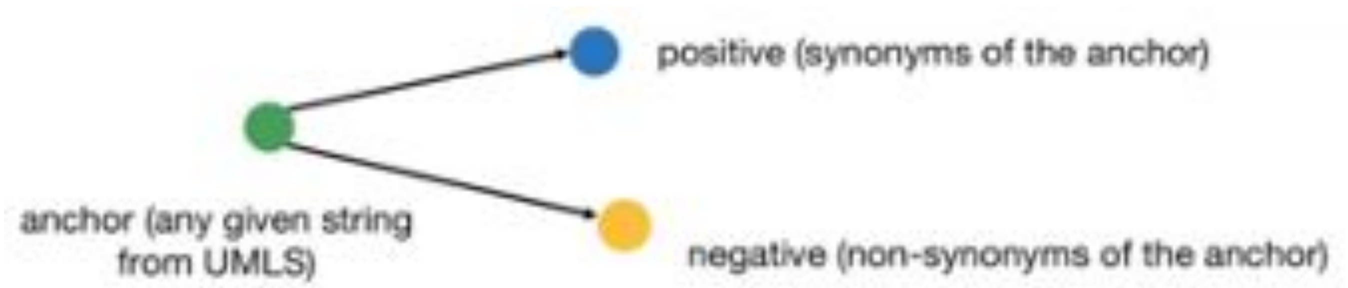
Approaches to biomedical entity linking:

1. Define it as **text classification task** with the knowledge base items as labels.
   Problems:
   - the label space is huge (Snomed: 311,000 concepts)
   - we don't have training data for all items

Biomedical entity linking

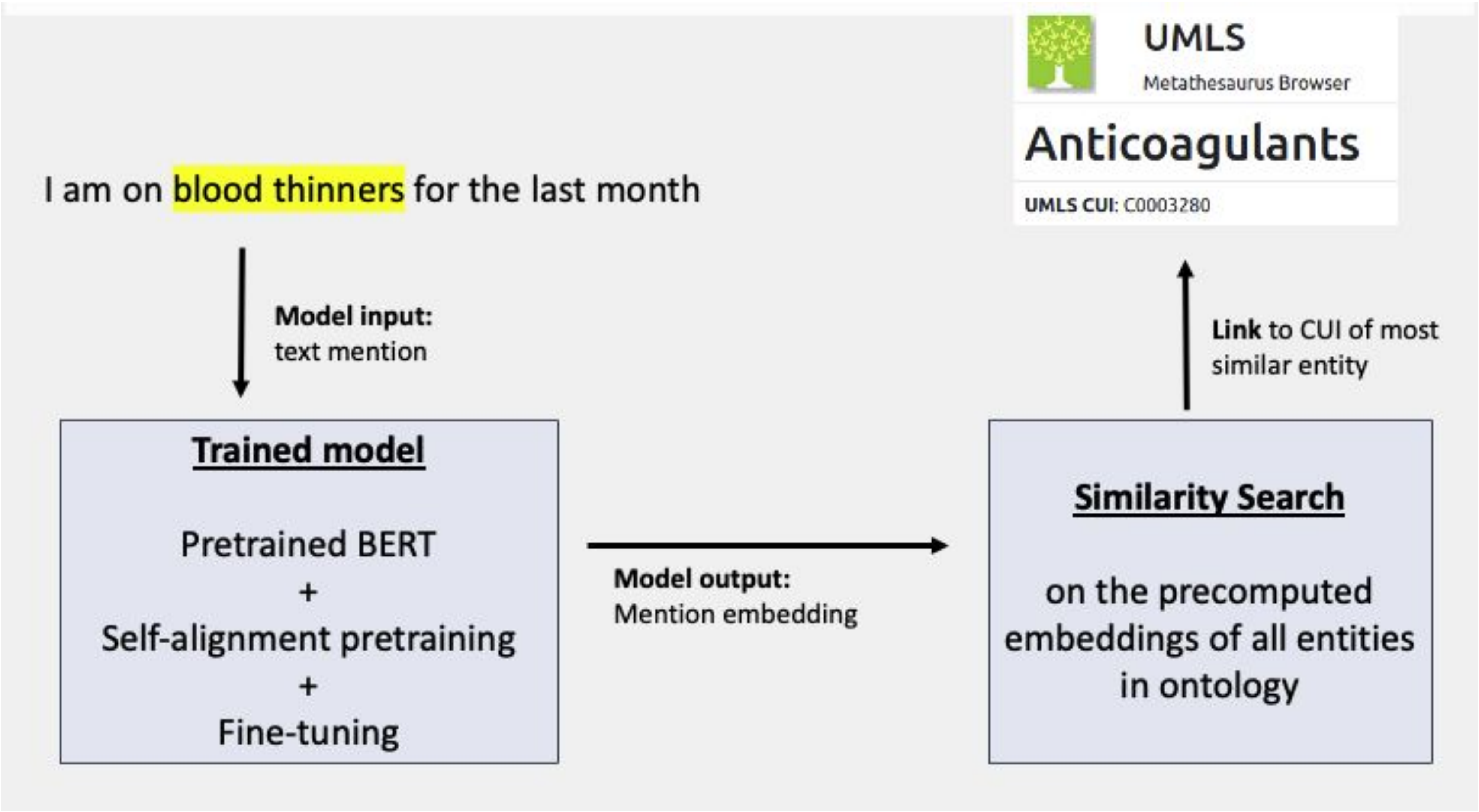
2. Define it as **term similarity task**: Self-Alignment Pretraining (SapBERT):

- Train embeddings for synonym detection
- Take pretrained embeddings (e.g. from a medical BERT model)
- Collect training data from a large ontology (e.g. ULMS)
- With **contrastive** learning, move synonyms closer together than non-synonyms



Liu et al. (2021). Self-Alignment Pretraining for Biomedical Entity Representations.
https://github.com/cambridgeltl/sapbert

Biomedical entity linking: example project



Fons Hartendorp, Tom Seinen, Erik van Mulligen & Suzan Verberne (2024, May). Biomedical Entity Linking for Dutch: Fine-tuning a Self-alignment BERT Model on an Automatically Generated Wikipedia Corpus. In *Proceedings of the First Workshop on Patient-Oriented Language Processing (CL4Health)@ LREC-COLING 2024* (pp. 253-263).

Biomedical entity linking: example project

Forum: kanker.nl (Dutch)

- 120k sentences; 2.1M tokens; 368,840 medical named entities extracted with existing **MedRoBERTa.nl** finetuned for NER
- Linked to our cleaned ULMS with our Dutch SapBERT model

Top 5 most found entities and linked concepts:

#	Entity	Linked concept
4356	*cancer*	primary malignant neoplasm
4240	*chemo*	chemo-immunotherapy
3043	*surgery*	operative surgical procedures
3034	*therapy*	milieu therapy
2287	*tumor*	neoplasms

100 random mentions manually evaluated:

- **42 mentions: error in NER**
- 20 mentions: linked to a wrong concept
- 20 mentions: linked to a related concept
- 18 mentions: are linked correctly

Fons Hartendorp, Tom Seinen, Erik van Mulligen & Suzan Verberne (2024, May). Biomedical Entity Linking for Dutch: Fine-tuning a Self-alignment BERT Model on an Automatically Generated Wikipedia Corpus. In *Proceedings of the First Workshop on Patient-Oriented Language Processing (CL4Health)@ LREC-COLING 2024* (pp. 253-263).

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NLP models with components - error accumulation issue => end to end?

Fons Hartendorp, Tom Seinen, Erik van Mulligen & Suzan Verberne (2024, May). Biomedical Entity Linking for Dutch: Fine-tuning a Self-alignment BERT Model on an Automatically Generated Wikipedia Corpus. In *Proceedings of the First Workshop on Patient-Oriented Language Processing (CL4Health)@ LREC-COLING 2024* (pp. 253-263).

Biomedical entity extraction and linking

- Biomedical Entity Linking for non-English remains challenging
- Contextual methods needed for ambiguous surface forms
- For real world cases, **entity recognition (NER) can be a bottleneck**
- Can we use few-shot learning with generative models instead?

Entity extraction with generative LLMs?

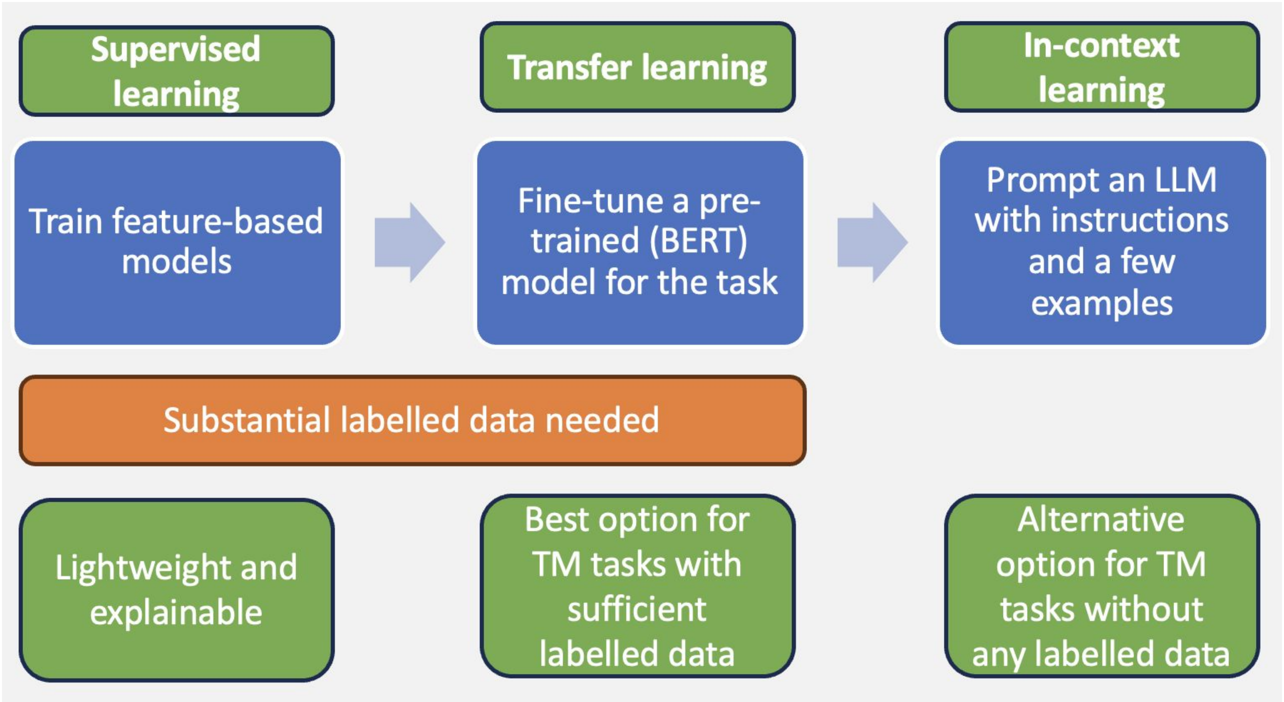
Sequence labelling is a **discriminative** task (a fixed set of labels)

But fine-tuning encoder models like BERT requires substantial labelled data

In addition, the generalizability of these models to unseen entities is limited

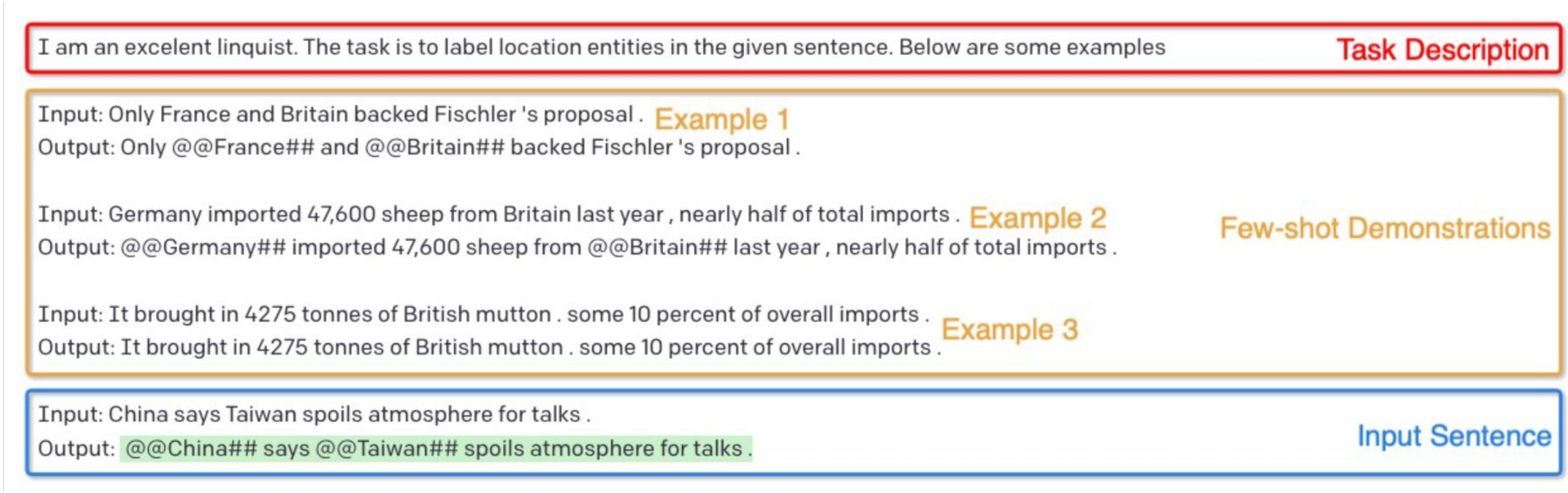
Can we use **generative LLMs** instead?

- Yes, but using generative models for discriminative tasks can be a bit challenging



Entity extraction with generative LLMs

GPT-NER: Few-shot named entity recognition



Challenge 1: it matters what the examples are. The paper evaluates several ways to select them
Challenge 2: hallucination. LLMs over-confidently label inputs as entities.

Wang et al. (2023) GPT-NER: Named Entity Recognition via Large Language Models

Entity extraction with generative LLMs

GPT-NER (Few-shot named entity recognition) cannot beat a fine-tuned model:

English OntoNotes5.0 (FULL)			
Model	**Precision**	**Recall**	**F1**
Baselines (Supervised Model)			
BERT-Tagger (Devlin et al., 2018)	90.01	88.35	89.16
BERT-MRC (Li et al., 2019a)	92.98	89.95	91.11
GNN-SL (Wang et al., 2022)	91.48	91.29	91.39
BERT-MRC+DSC (Li et al., 2019b)	91.59	92.56	**92.07** (SOTA)
GPT-NER			
GPT-3 + *random retrieval*	58.8	64.36	61.58
GPT-3 + *sentence-level embedding*	71.87	78.77	75.32
GPT-3 + *entity-level embedding*	79.17	84.29	81.73

But: “when the amount of training data is extremely scarce, the results of GPT-NER are significantly better than that of the supervised model”

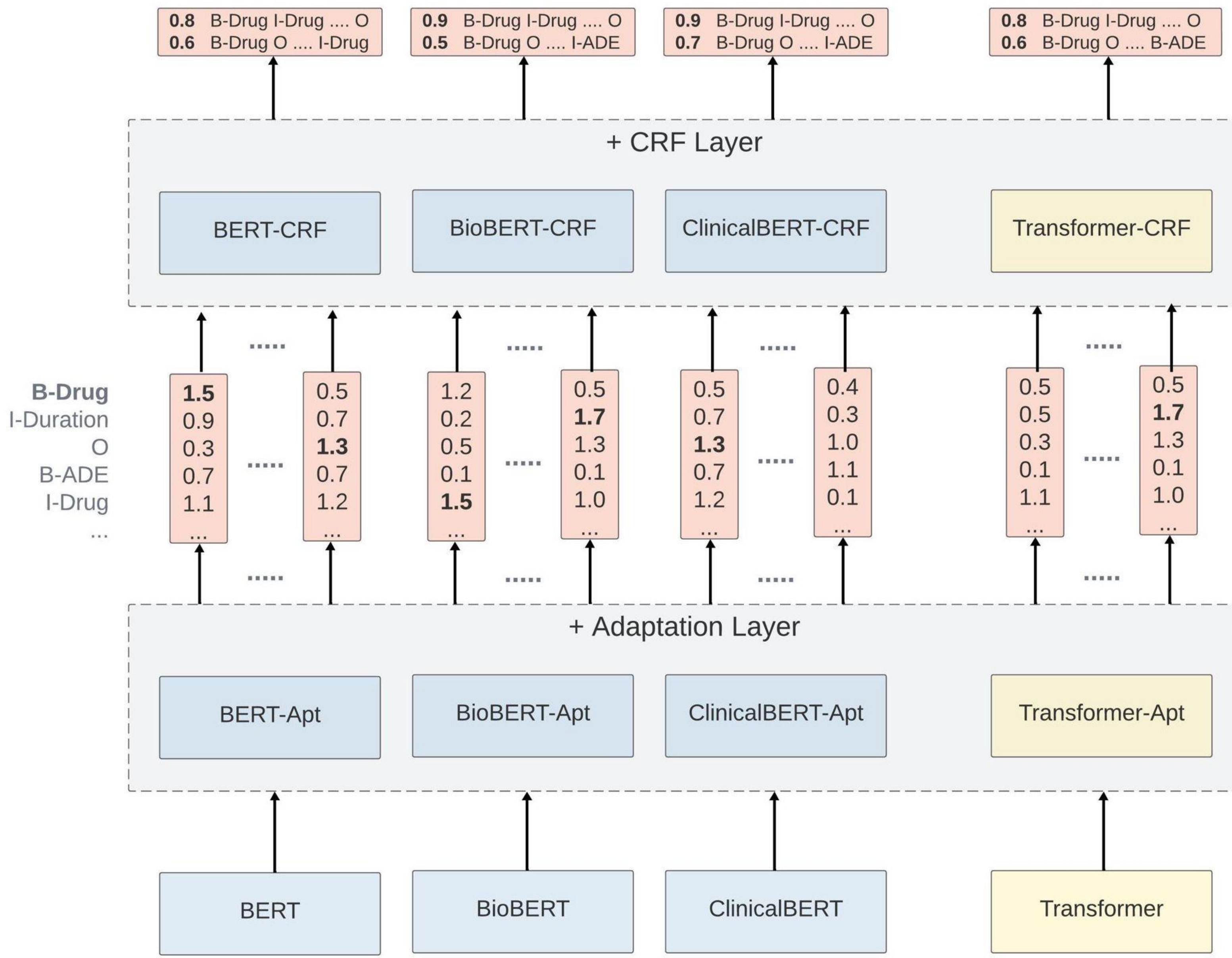
Wang et al. (2023) GPT-NER: Named Entity Recognition via Large Language Models

Entity extraction with generative LLMs

Trade-off between training data and prompt engineering / in-context learning

Supervised fine-tuning	**In-context learning**
When we have a lot of labelled data	When we have no or very limited labelled data
When the label set is restricted	When the label set is huge or open
When the entities are short	When the entities are long and fuzzily defined
When many entities are common	When there are mainly long-tail entities

Example NER project: scratch-learned



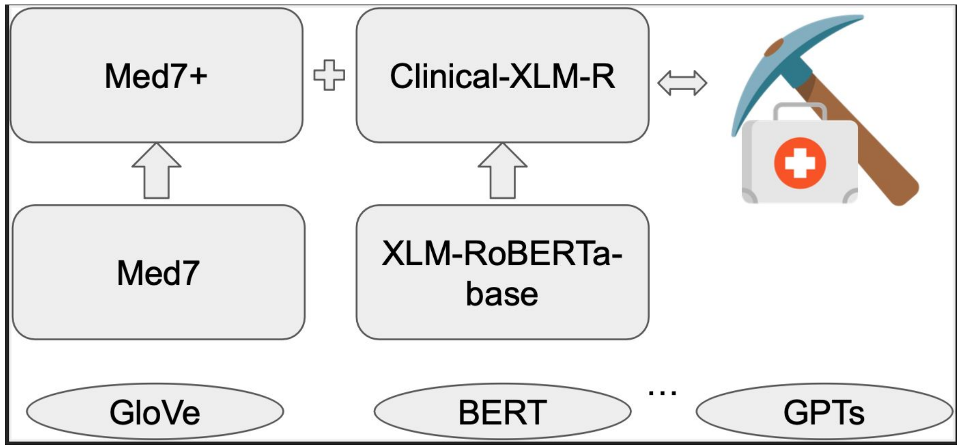
Exploring the Value of Pre-trained Language Models for Clinical Named Entity Recognition

S Belkadi, L Han, Y Wu, G Nenadic

Published in: 2023 IEEE International Conference on Big Data (BigData)

https://ieeexplore.ieee.org/abstract/document/10386154

Example NER project: finetune medicalLM

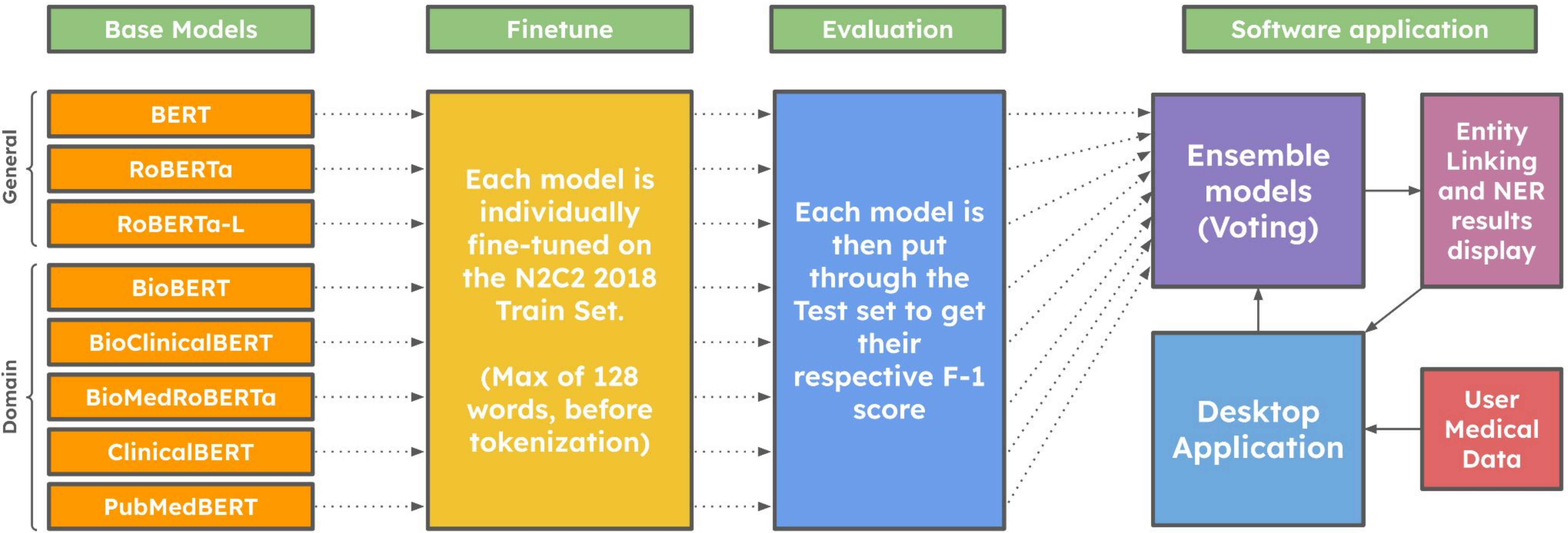


MedMine: Examining Pre-trained Language Models on Medication Mining

H Alrdahi, L Han, H Šuvalov, G Nenadic
arXiv preprint arXiv:2308.03629

true_label	true_text	Med7	Med7+	XLM-R	char_start	char_end
drug	Sulfa Sulfonamide Antibiotics	O	drug	drug	120	149
drug	Lamictal	O	drug	drug	150	158
ade	toxicity	O	ade	ade	1354	1362
ade	toxicity	O	ade	ade	2558	2566
ade	toxicity	O	ade	ade	3097	3105
reason	hypotension	O	O	reason	8717	8728
reason	BP	O	O	O	1853	1855
strength	0 3	O	O	O	1910	1913
strength	0 8	O	O	O	1942	1945
reason	resuscitation	O	O	O	1960	1973
dosage	2 8L	O	O	O	1974	1978
ade	rash	O	O	ade	2482	2486

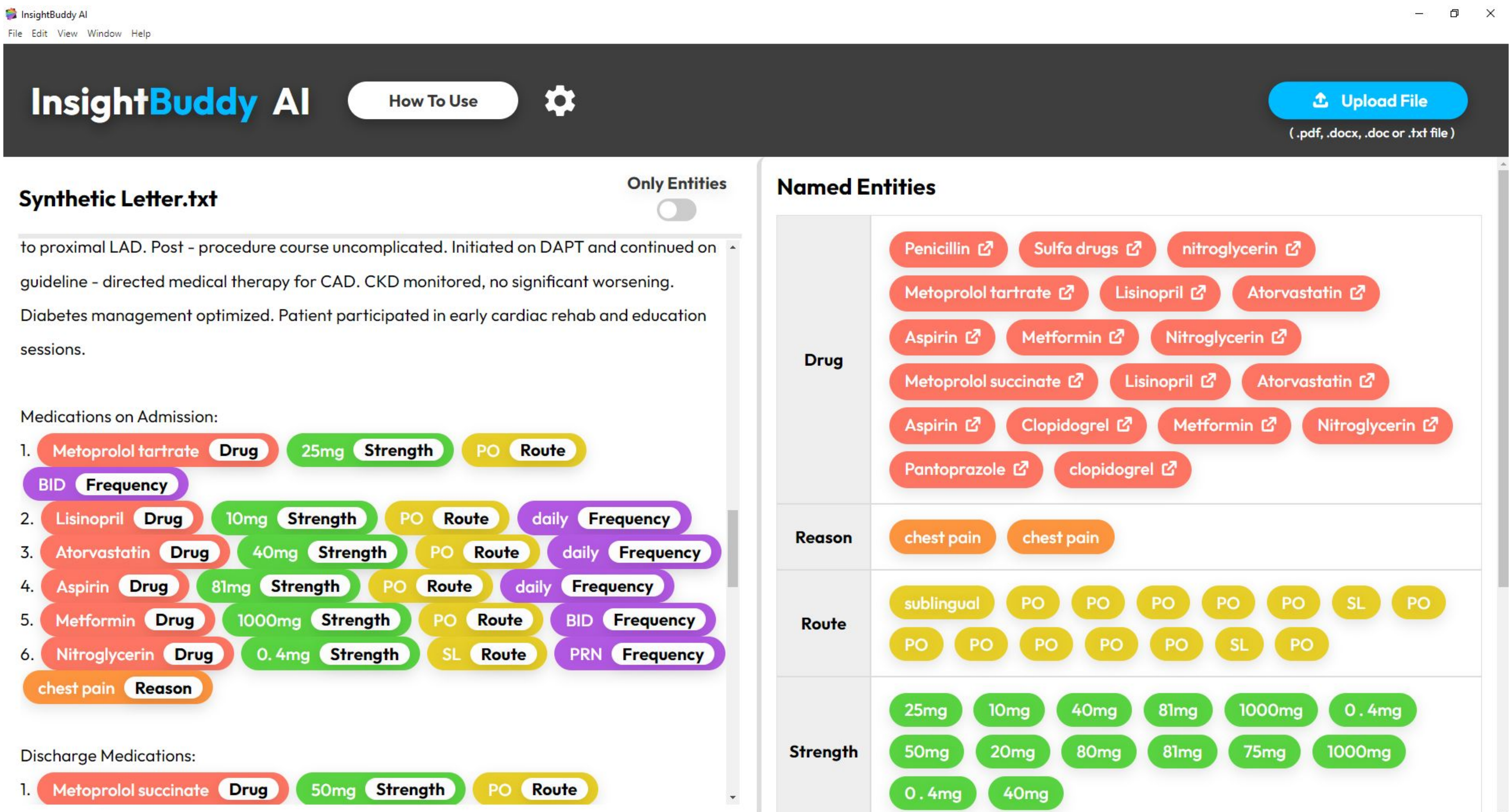
Example NER project: ensemble learning



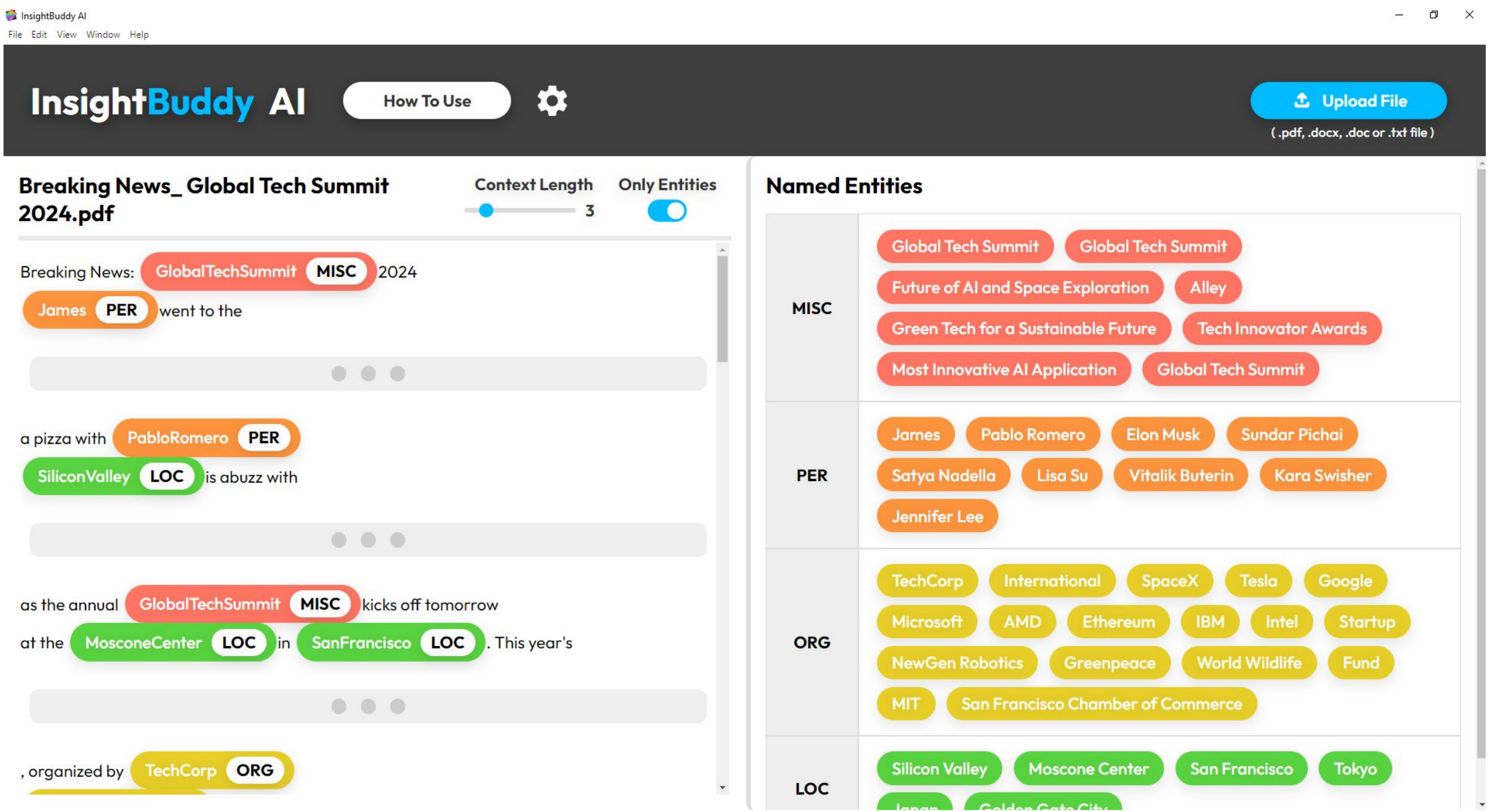
Insightbuddy-ai: Medication extraction and entity linking using pre-trained language models and ensemble learning
P Romero, L Han, G Nenadic
Proceedings of the 2025 Conference of the Nations of the Americas Chapter of ACL

Software/App: https://github.com/HECTA-UoM/ensemble-NER

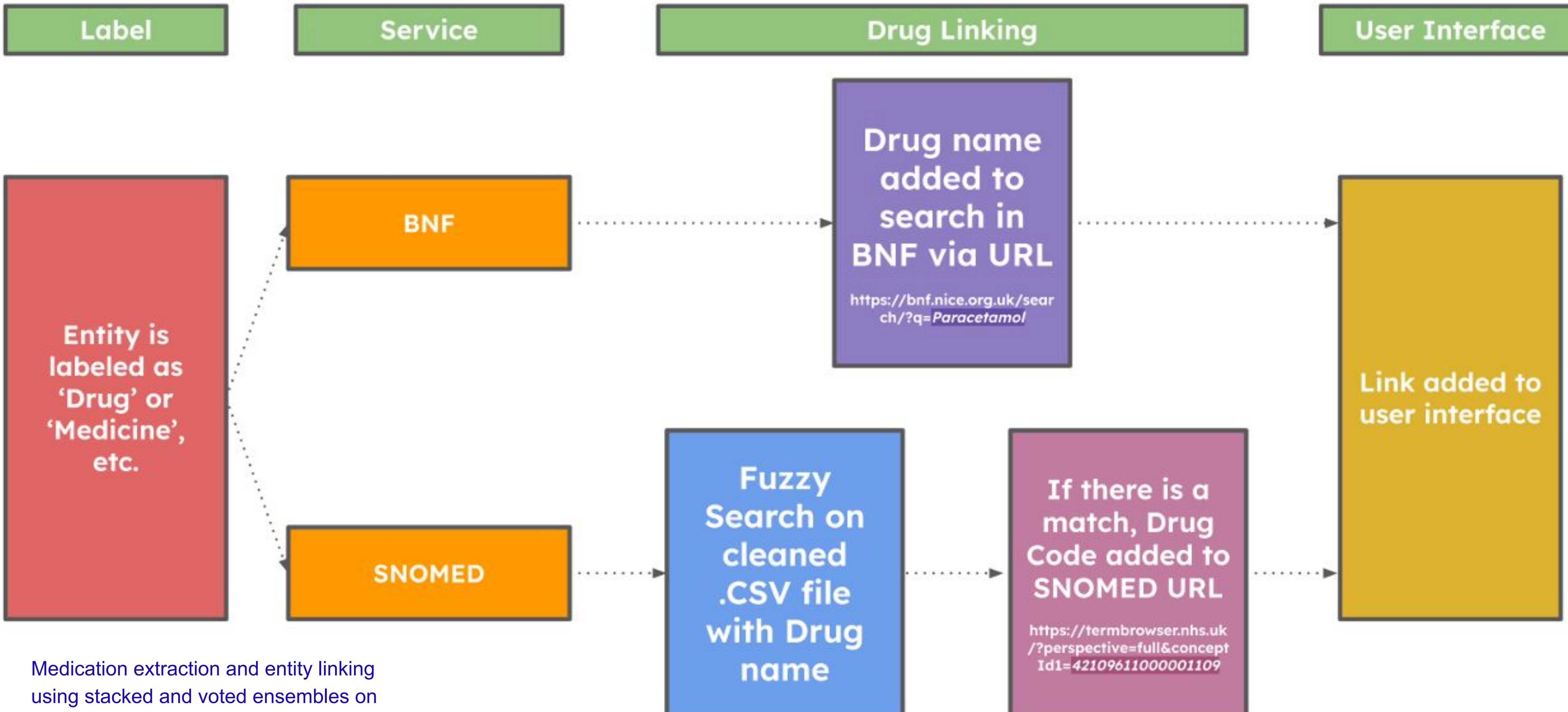
Example NER project: Insightbuddy-AI



Example NER project: Insightbuddy-AI



Example NER project: Insightbuddy-AI



Medication extraction and entity linking using stacked and voted ensembles on LLMs

P Romero, L Han, G Nenadic - CL4Health workshop, 2025 - aclanthology.org

Entity extraction and linking - references

DeepADEMiner: a deep learning pharmacovigilance pipeline for extraction and normalization of adverse drug event mentions on Twitter
Arjun Magge, Elena Tutubalina, Zulfat Miftahutdinov, Ilseyar Alimova, Anne Dirkson, Suzan Verberne, Davy Weissenbacher, Graciela Gonzalez-Hernandez. Journal of the American Medical Informatics Association (JAMIA), Volume 28, Issue 10, October 2021, pp 2184–2192

Automated gathering of real-world data from online patient forums can complement pharmacovigilance for rare cancers.
Anne Dirkson, Suzan Verberne, Wessel Kraaij, Gerard van Oortmerssen and Hans Gelderblom. Nature Scientific Reports 12, 10317 (2022)

GPT-NER: Named entity recognition via large language models
S Wang, X Sun, X Li, R Ouyang, F Wu, T Zhang, J Li, G Wang, C Guo. Findings of the association for computational linguistics: NAACL 2025

Biomedical Entity Linking for Dutch: Fine-tuning a Self-alignment BERT Model on an Automatically Generated Wikipedia Corpus.
Fons Hartendorp, Tom Seinen, Erik van Mulligen, & Suzan Verberne (2024). CL4Health@ LREC-COLING 2024

How about discontinuous entities?

Alhassan, A., Schlegel, V., Cabral, R. C., Batista-Navarro, R., Han, C., Poon, J., & Nenadic, G. Recognition and Linking of Discontinuous Named Entities in Healthcare: A Comparative Performance Analysis. *Frontiers in Digital Health*, *8*, 1758921.

Language specific - Italian

Teresa Paccosi and Alessio Palmero Aprosio. 2021. REDIT: A Tool and Dataset for Extraction of Personal Data in Documents of the Public Administration Domain. In Proceedings of the Eighth Italian Conference on Computational Linguistics (CLiC-it 2021), pages 252–259, Milan, Italy. CEUR Workshop Proceedings.

NER -> De-identification: Example Project

DeID-Clinic: A Risk-Aware Pseudonymization Framework for Clinical Text De-identification and Re-identification Risk Assessment

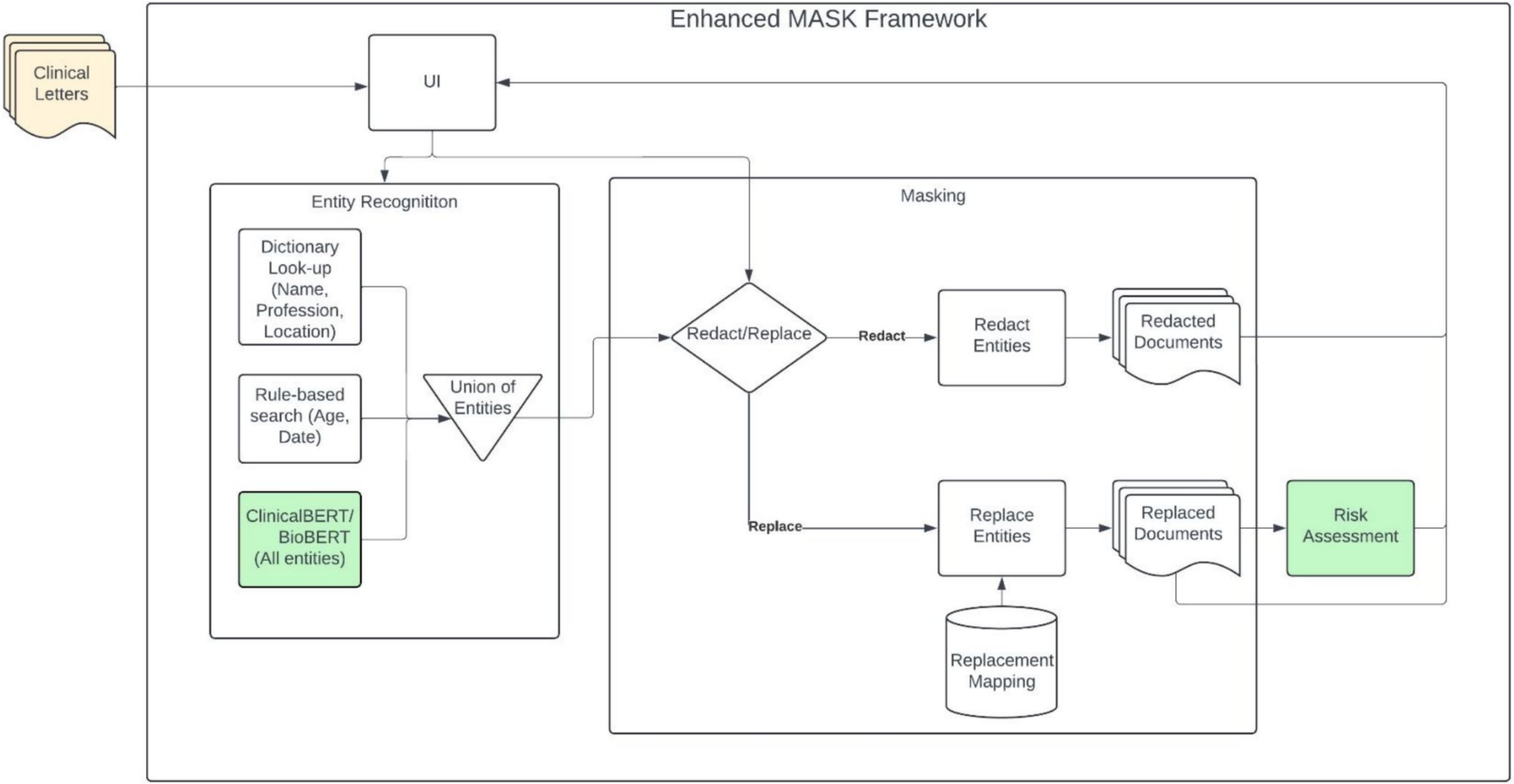
Angel Paul[1,†], **Dhivin Shaji**[1,†], **Lifeng Han**[2,3,*]
Warren Del-Pinto[1], **Goran Nenadic**[1], **Suzan Verberne**[2]
[1]University of Manchester, Manchester, UK
[2]Leiden Institute of Advanced Computer Science (LIACS), Leiden University, NL
[3]Biomedical Data Sciences, Leiden University Medical Center, Leiden, NL
[†]co-first [*]*corresponding*: {l.han, s.verberne}@liacs.leidenuniv.nl

Abstract

The increasing availability of sensitive textual data has created an urgent need for robust de-identification methods that enable compliant data sharing while preserving downstream utility. This paper presents DeID-Clinic, a multi-layered framework for automated pseudonymization and re-identification risk assessment of clinical free-text data. Our approach integrates domain-adapted transformer models, including BioBERT and ClinicalBERT, into the MASK de-identification framework to improve the detection and masking of protected health information (PHI). Beyond entity recognition, we introduce a novel document-level risk assessment module that quantifies residual re-identification risk using a combination of k-anonymity, l-diversity, t-closeness, contextual similarity, and entity co-occurrence analysis. Experiments conducted on the i2b2 2014 de-identification dataset demonstrate strong performance, achieving macro-level F1 scores above 0.96 for several entity categories, while enabling quantitative prioritization of high-risk documents for further review. Our results highlight the effectiveness of combining neural de-identification with explicit risk modeling, supporting privacy-preserving data sharing in sensitive domains. Although evaluated on clinical text, the proposed framework is generalizable to other privacy-critical domains such as legal and administrative documents, where reliable pseudonymization and risk-aware anonymization are essential.

Keywords: Automated De-Identification, Risk Assessment, Patient Privacy, Pseudonymization, Personal Health Information

This afternoon
https://mormor-karl.github.io/events/LEGAL-CALD-pseudo/


Enhanced MASK Framework
Clinical Letters
UI
Entity Recognititon
Dictionary Look-up (Name, Profession, Location)
Rule-based search (Age, Date)
ClinicalBERT/ BioBERT (All entities)
Union of Entities
Masking
Redact/Replace
Redact
Replace
Redact Entities
Redacted Documents
Replace Entities
Replaced Documents
Risk Assessment
Replacement Mapping

https://legal2026.mobileds.de/#program

legal2026.mobileds.de

Begins with | lifeng | 1 match | Done

GRAV

Home | About | Invited Talks | Submission | Topics | Dates | Program | Organizers

13:00 - 14:00	**Lunch Break**
14:00 - 14:55	**Invited Talk**
14:00 - 14:55	Privacy and anonymization in NLP: Are we barking up the wrong tree? *Prof. Dr. Ivan Habernal*
14:55 - 16:00	**Oral Presentations III: CALD-pseudo**
14:55 - 15:15	**DeID-Clinic: A Risk-Aware Pseudonymization Framework for Clinical Text De-identification and Re-identification Risk Assessment** *Angel Paul[1], Dhivin Shaji[1], Lifeng Han[2], Warren Del-Pinto[1], Goran Nenadic[1], Suzan Verberne[3]* [1]University of Manchester, [2]The University of Manchester, [3]LIACS, Leiden University
15:15 - 15:35	**Distilling Human-Aligned Privacy Sensitivity Assessment from Large Language Models** *Gabriel Loiseau[1], Damien Sileo[1], Damien Riquet[2], Maxime Meyer[2], Marc Tommasi[3]* [1]Inria, [2]Hornetsecurity, [3]Lille University
15:35 - 15:55	**Birds of a Feather: Do Embedding Representations of Personal Information Flock Together?** *Maria Irena Szawerna and Simon Dobnik* University of Gothenburg
16:00 - 16:30	**Coffee Break**

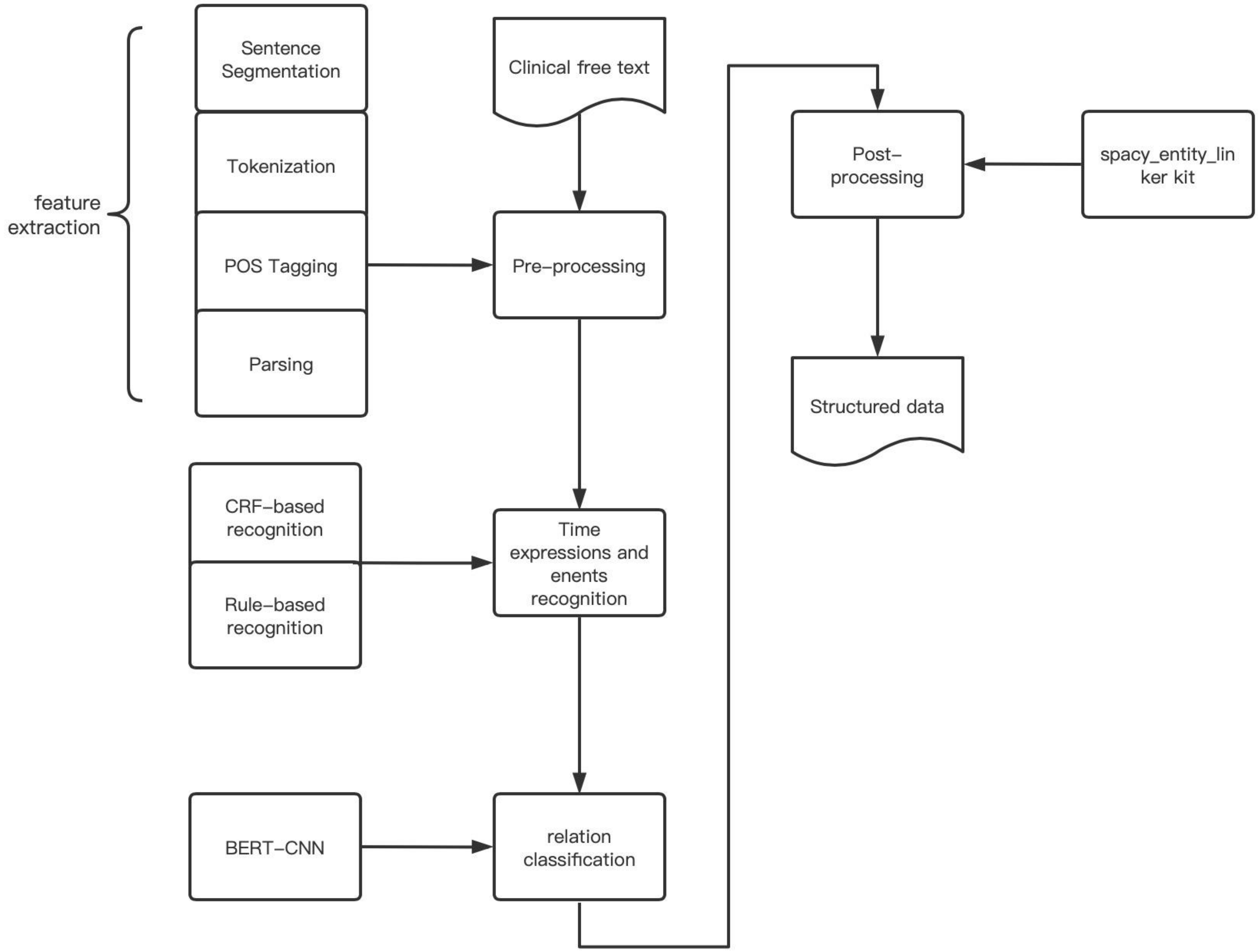
Temporal relation modelling: example

Depends on time/running,
if to go directly to exercise

clinical free text		
Admission Date	**Discharge Date**	**Doctor's Note**
02/22/92	*03/08/92*	*She was, therefore, cleared for the operating room, and on 3/3/92, she underwent* ***a left carotid endarterectomy****, with continuous electroencephalogram monitoring and* ***vein patch angioplasty****, which was uneventful .*

What treatment/s was/were used during visit

Temporal relation modelling (BERT, NNs)



MedTem1.0 (Tu, Han, Nenadic, 2022)

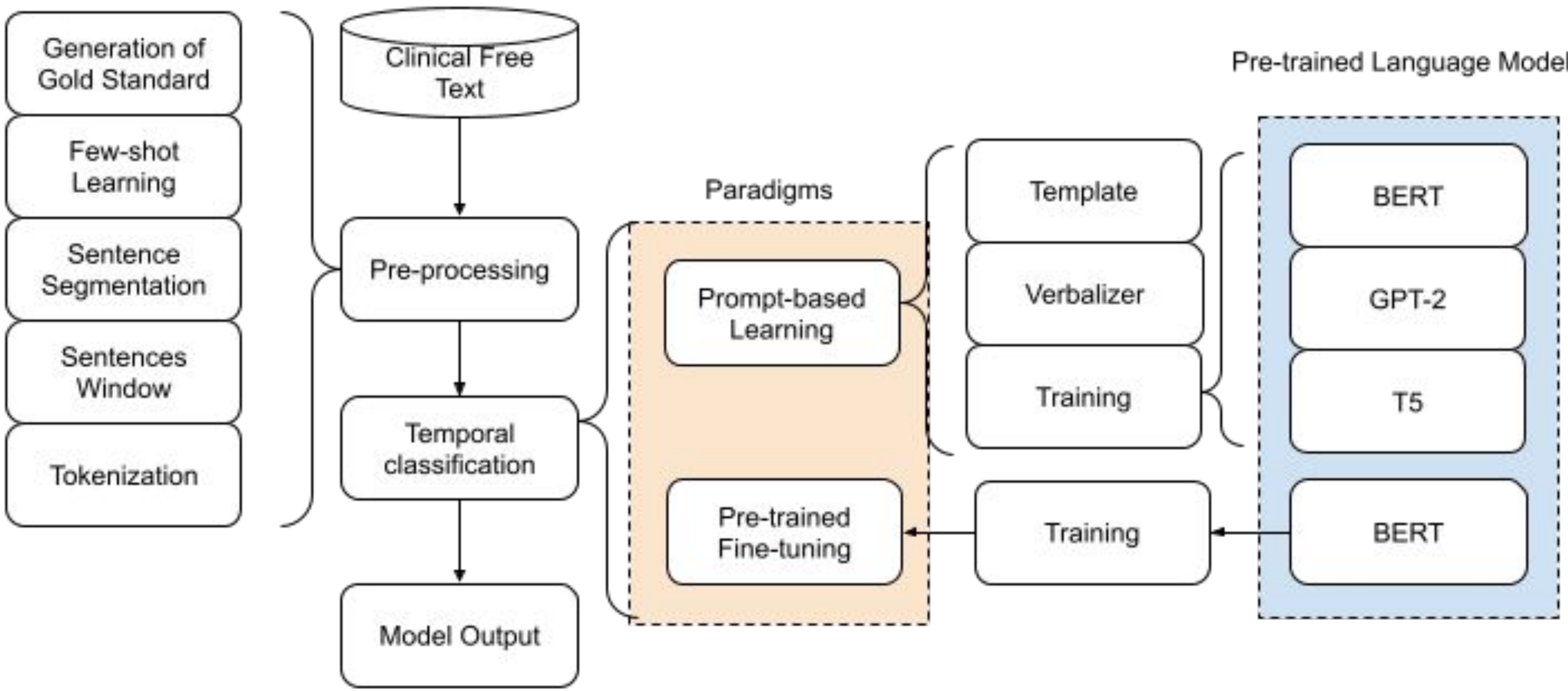
NER: BiLSTM-CRF, CNN-BiLSTM

(i2b2 2009/2012 entities)

RE: BERT-CNN

(after, overlap, before)

Temporal relation modelling (vs prompting



MedTem2. 0: Prompt-based temporal classification of treatment events from discharge summaries

Y Cui, L Han, G Nenadic. ACL 2023 Student Research Workshop

Temporal relation modelling refs

Extraction of Medication and Temporal Relation from Clinical Text using Neural Language Models

H Tu, L Han, G Nenadic

Published in: 2023 IEEE International Conference on Big Data (BigData)

MedTem2. 0: Prompt-based temporal classification of treatment events from discharge summaries

Y Cui, L Han, G Nenadic. ACL 2023 Student Research Workshop

Exercise: entity extraction

In **settings without in-domain labelled data** data we have two options:

- use of a BERT checkpoint that someone else fine-tuned for a similar task
- prompting a generative LLM to do the task

In this exercise you will compare both on a medical entity recognition task, using the same small sample of forum messages that you manually annotated before.

1. For running the BERT model, we have prepared this colab notebook: https://colab.research.google.com/drive/14XJouUPjHE9scJcK9J4pz_qEf-GwbMj0?usp=sharing

2. For running the generative LLM, you can use this privacy-preserving interface, where you can select your own favourite model: https://duck.ai/chat?ia=chat&duckai=1 (prompts on the next slide)

Exercise: entity extraction

1. For running the BERT model, we have prepared this colab notebook: https://colab.research.google.com/drive/14XJouUPjHE9scJcK9J4pz_qEf-GwbMjo?usp=sharing

2. For running the generative LLM, you can use this privacy-preserving interface, where you can select your own favourite model: https://duck.ai/chat?ia=chat&duckai=1

Prompt suggestion:

> Please extract medical entities from the following texts. In the output, first repeat the sentence, then list the extracted entity spans (one per line), and label them with one of the following categories: MEDICATION BIOLOGICAL_STRUCTURE SIGN_SYMPTOM CLINICAL_EVENT SEVERITY DATE DURATION DISEASE_DISORDER DIAGNOSTIC_PROCEDURE
>
> + All 8 texts

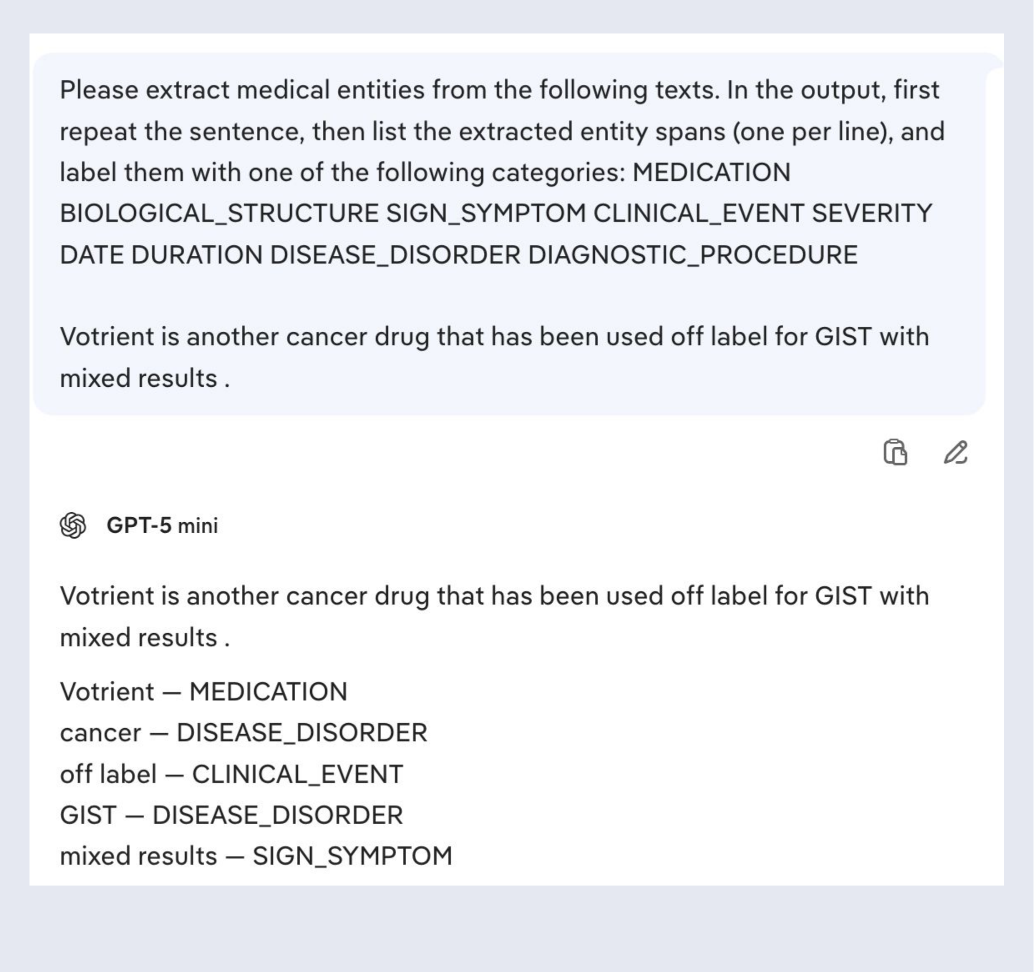
Exercise: entity extraction

3. Compare both the BERT model output and the LLM output to your own annotations.
Which modelling approach is better?
What are problems for both models?

4. Try out another model from the same interface (e.g. replace GPT-5 by GPT-OSS or Mistral)

5. Do multiple runs with the same model to investigate consistency.

Discuss with your neighbour



2nd hour done
if more time - next slides

3. Patients

Universiteit Leiden

Patient-related tasks

- Patient and Public Involvement and Engagement (PPIE) in the 4D Picture project
- Barriers in patient-doctor communication
- Communication support: the role of metaphors
- Shared decision making (SDM)
- Health literacy: translation, simplification, summary - challenges and future

4D Picture (2022-27)

- The 4D PICTURE project aims to help cancer patients, their families, and healthcare providers better understand their options. It supports their treatment and care choices, at each stage of disease, by drawing on large amounts of evidence from different types of European data. The project involves experts from many different specialist areas who are based in nine European countries. The overall aim is to improve the cancer patient journey and ensure personal preferences are respected.

- Data collection and cleaning of corpora; NLP & ML prototypes; novel entity linking (citizen's vocabulary to professional discourse); aspect-based sentiment analysis; new version of Metaphor Menu

- Funding: HORIZON Europe 2021 grant agreement 101057332, UKRI Reference Number 10041120
- https://4dpicture.eu/

Health text mining and citizen science in 4D

- NLP team + PPI members in each country + ethics work package
  - Leiden University, Erasmus Medical Centre, Zaragoza, Maastro, Rsyd, Lancaster

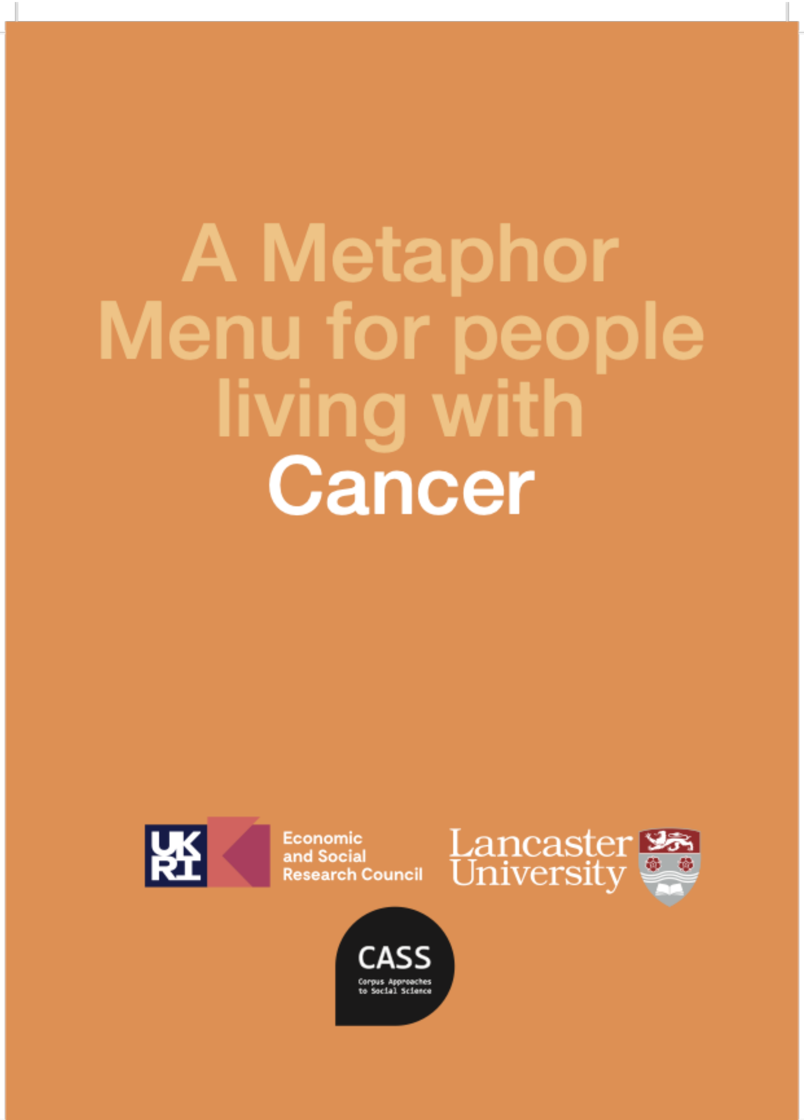
- Key objectives
  - Uncover stories and experiences of cancer patients & family and convert to usable actionable knowledge about how they experience their care paths
  - Multilingual techniques (Danish, Dutch, English, Spanish) to bridge the gap between patients informal vocabulary to formal health professional terminology
  - Further develop the “Metaphor Menu” as a multilingual conversation tool

- Main techniques
  - Data collection, NER & entity linking, Semantic analysis, Sentiment & emotion analysis, corpus linguistics

Understand patient experiences through text analysis of clinical narratives

I am a 20 year old AGE guy that just got done with immunotherapy TREATMENT for stage 3 STAGE melanoma. CANCER_TYPE I 'm more or less fine now but dealing with cancer DISEASE has caused me a lot more problems mentally than physically. Even as a little kid I would be absolutely terrified EMOTION ANALYSIS of dying which is probably very common. I used to wake up having constant nightmares EMOTION ANALYSIS of dying and they have stuck with me through life. I used to comfort myself EMOTION ANALYSIS saying that when I was a lot older I would have made peace EMOTION ANALYSIS with it and more or less accepted it. EMOTION ANALYSIS But during the heat of my cancer DISEASE experience I had to get a MRI INVESTIGATIONS on my head PART_OF_BODY since a decently sized SIZE melanoma CANCER_TYPE was found in my lymph node. PART_OF_BODY I had made the mistake of looking up what it ment if the melanoma CANCER_TYPE had reached my brain PART_OF_BODY and I totally broke down. EMOTION ANALYSIS I'm pretty healthy now but the idea of dying young became really real to me. My mental has totally went downhill and I have no idea

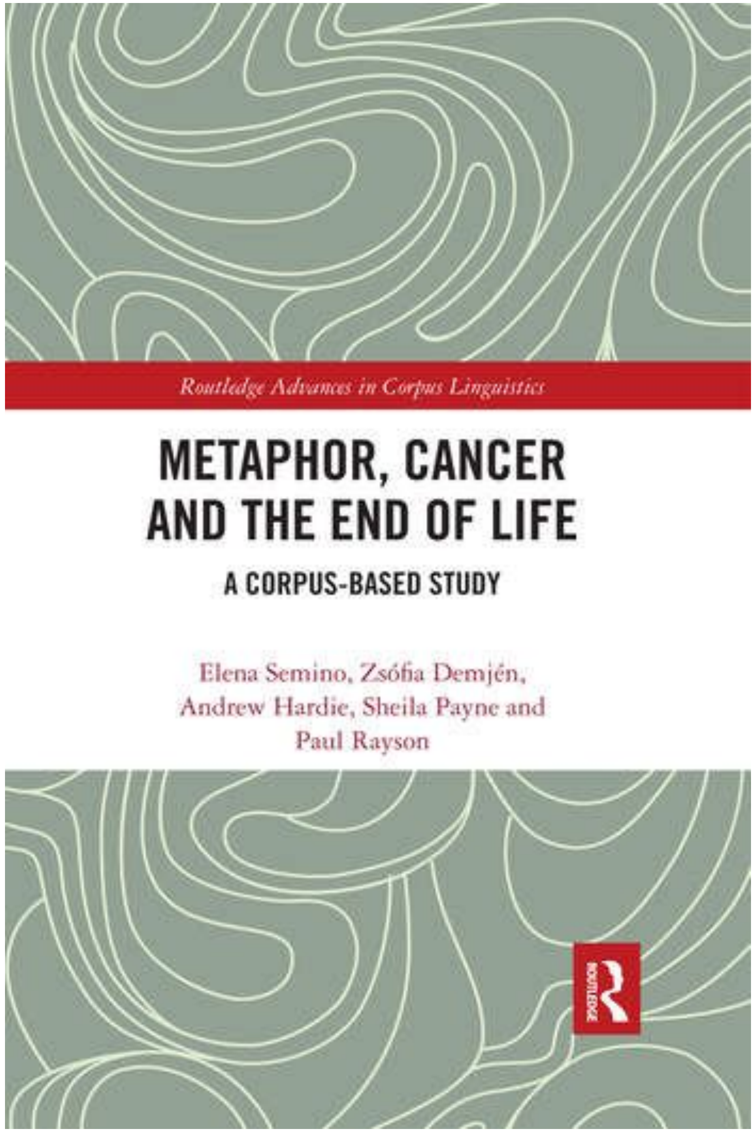
Metaphors in 4D



Previous research on metaphors for cancer has shown that metaphors can be a helpful resource for **talking and thinking** about the experience of illness.

However, different metaphors suit different people, or the same person at different times.

https://wp.lancs.ac.uk/melc/



Metaphors in 4D

1 Imagine it a bit like a scary fairground ride – it might be scary in places, but it will eventually stop and you can get off. Be strong, be brave and we will be here to hold your hand if you need it.

2 My journey with cancer may not be smooth but it certainly makes me look up and take notice of the scenery.

3 Some journeys with cancer will be longer and others short, but what matters most is how we walk that journey.

7 I don't intend to give up; I don't intend to give in. No I want to fight. I don't want it to beat me, I want to beat it.

8 I respect Cancer and never underestimate the power it has, but, if you can face up to it and hit it back head on, you stand a good chance of beating it for a bit longer.

9 Having cancer is not a fight but a relationship where I am forced to live with my disease day in, day out. Some days cancer has the upper hand, other days I do.

10 'Battle' suggests either I win or cancer does. I think of it more as 'working with cancer'. For me, seen in this more everyday way the cancer becomes easier to cope with.

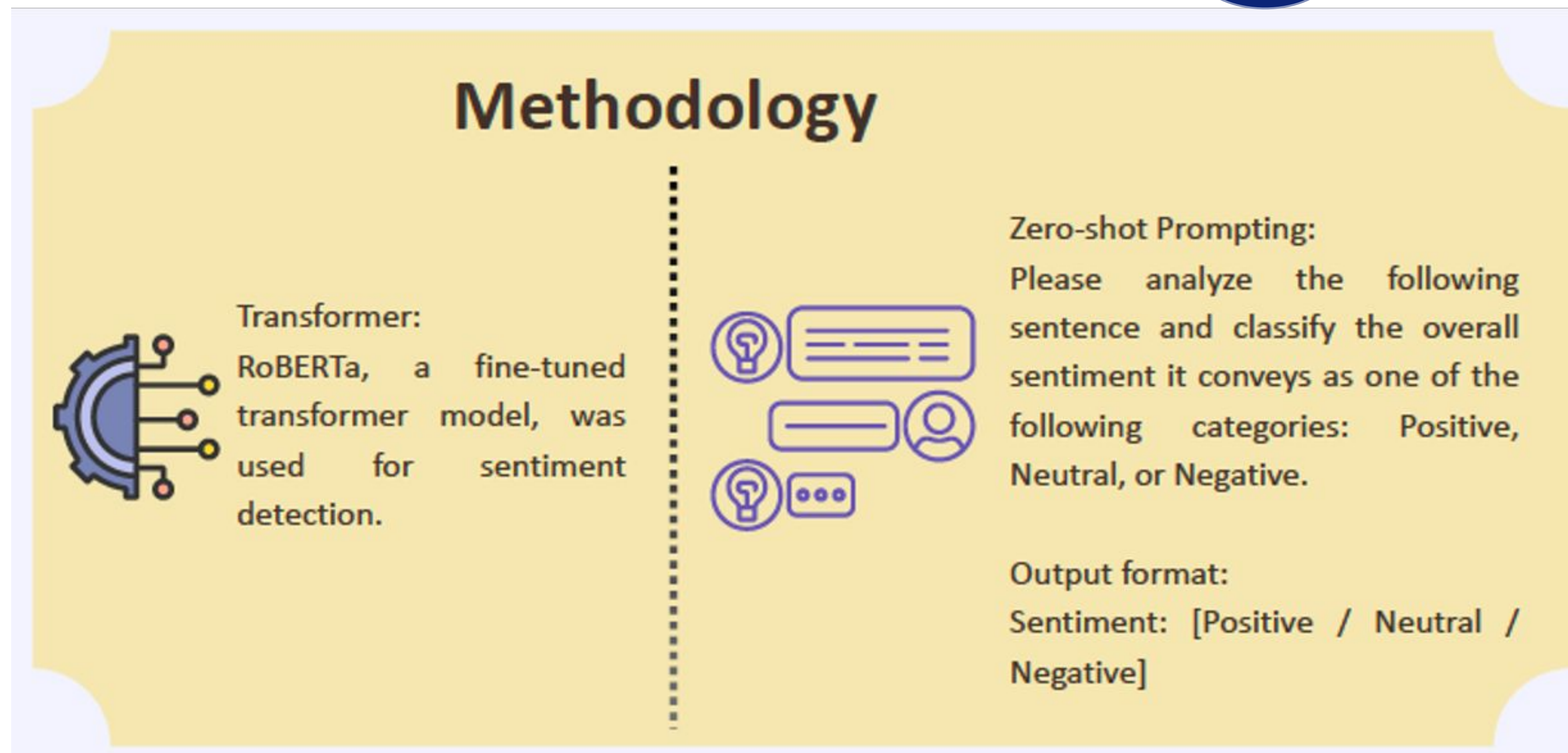
11 You fight this terrible, faceless illness that has invaded every part of you. You stand and say, 'No, this is not all there is to me'. You have fought back, taken control of the things left to you. And even when you felt crap, you forced yourself to get out of bed and keep going.

Metaphor sentiment analysis

Refer back:
NLP-sentiment
(earlier)

Data:
Narratives were extracted from the HealthUnlocked platform, specifically focusing on ovarian and prostate cancer communities. These posts were filtered using metaphor-specific keywords to create post-sets with:

- 2,624 sentences with battle references,
- 336 with enemy,
- 7,250 with fight,
- 7,274 with journey,
- 3,354 with roller coaster, and
- 714 with war.



Daisy Lal, Paul Rayson, Lucia Pitarch, Sander Puts (2025) Sentiment Analysis of Cancer Metaphors: Comparing Human and Machine Interpretation. HealTAC2025, Glasgow, UK

Metaphor sentiment analysis

Daisy Lal, Paul Rayson, Lucia Pitarch, Sander Puts (2025) Sentiment Analysis of Cancer Metaphors: Comparing Human and Machine Interpretation. HealTAC2025, Glasgow, UK

Metaphor sentiment analysis

Results

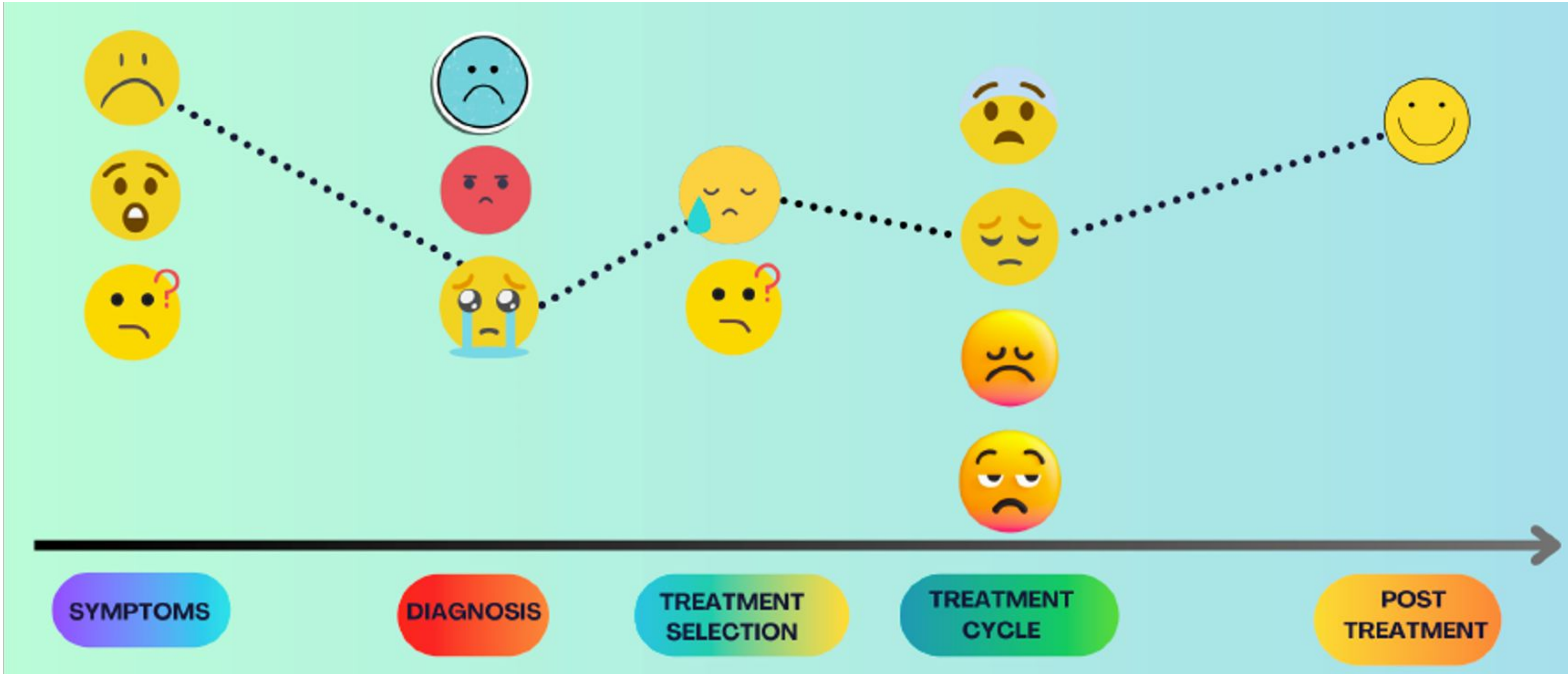
When compared to human sentiment annotations, ChatGPT showed moderate to strong correlations (0.41–0.96), suggesting its robust contextual understanding and ability to interpret metaphors with nuance. In contrast, RoBERTa's performance on metaphor-laden inputs was less promising. For battle metaphors, it showed negative or zero correlation (–0.24–0.00), while for other metaphor types, the correlation ranged from fair to moderate, with the best performance observed for war and roller coaster metaphors (0.30–0.69). These findings indicate that transformer models, which are pre-trained on literal data, struggle to process affective metaphors effectively. The results also underscore the potential of large generative models like ChatGPT in overcoming these limitations.

Conclusion

Human annotators demonstrated varying levels of agreement across different metaphors. For metaphors like fight, enemy, and roller coaster, substantial to strong correlations were observed (0.47–0.87). In contrast, for metaphors such as journey, war, and battle, the agreement was moderate to fair (0.39–0.70). Importantly, the agreement was metaphor-dependent, with no consistent annotator pair performing best across all metaphor types. This variability highlights the interpretive subjectivity involved in emotional inference from metaphor-driven language, which is influenced by personal context and familiarity with specific metaphors.

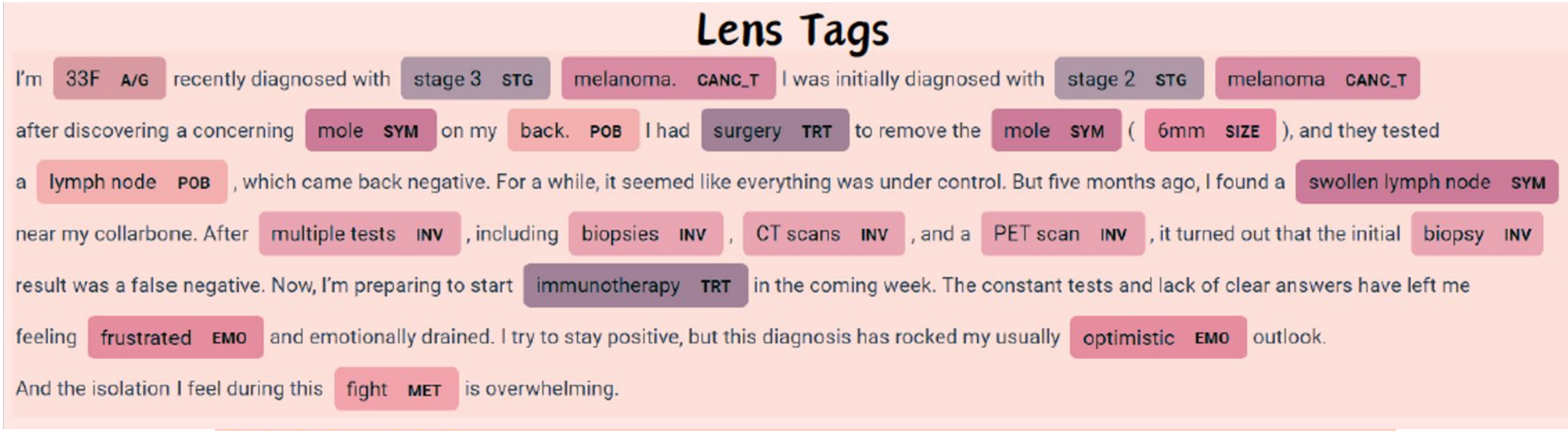
Daisy Lal, Paul Rayson, Lucia Pitarch, Sander Puts (2025) Sentiment Analysis of Cancer Metaphors: Comparing Human and Machine Interpretation. HealTAC2025, Glasgow, UK

Emotions in cancer narratives



Daisy Monika Lal, Paul Rayson, Sheila A. Payne, and Yufeng Liu. 2024. Analysing Emotions in Cancer Narratives: A Corpus-Driven Approach. In Proceedings of the First Workshop on Patient-Oriented Language Processing (CL4Health) @ LREC-COLING 2024, pages 73–83, Torino, Italia. ELRA and ICCL. https://aclanthology.org/2024.cl4health-1.9/

Entities in cancer narratives

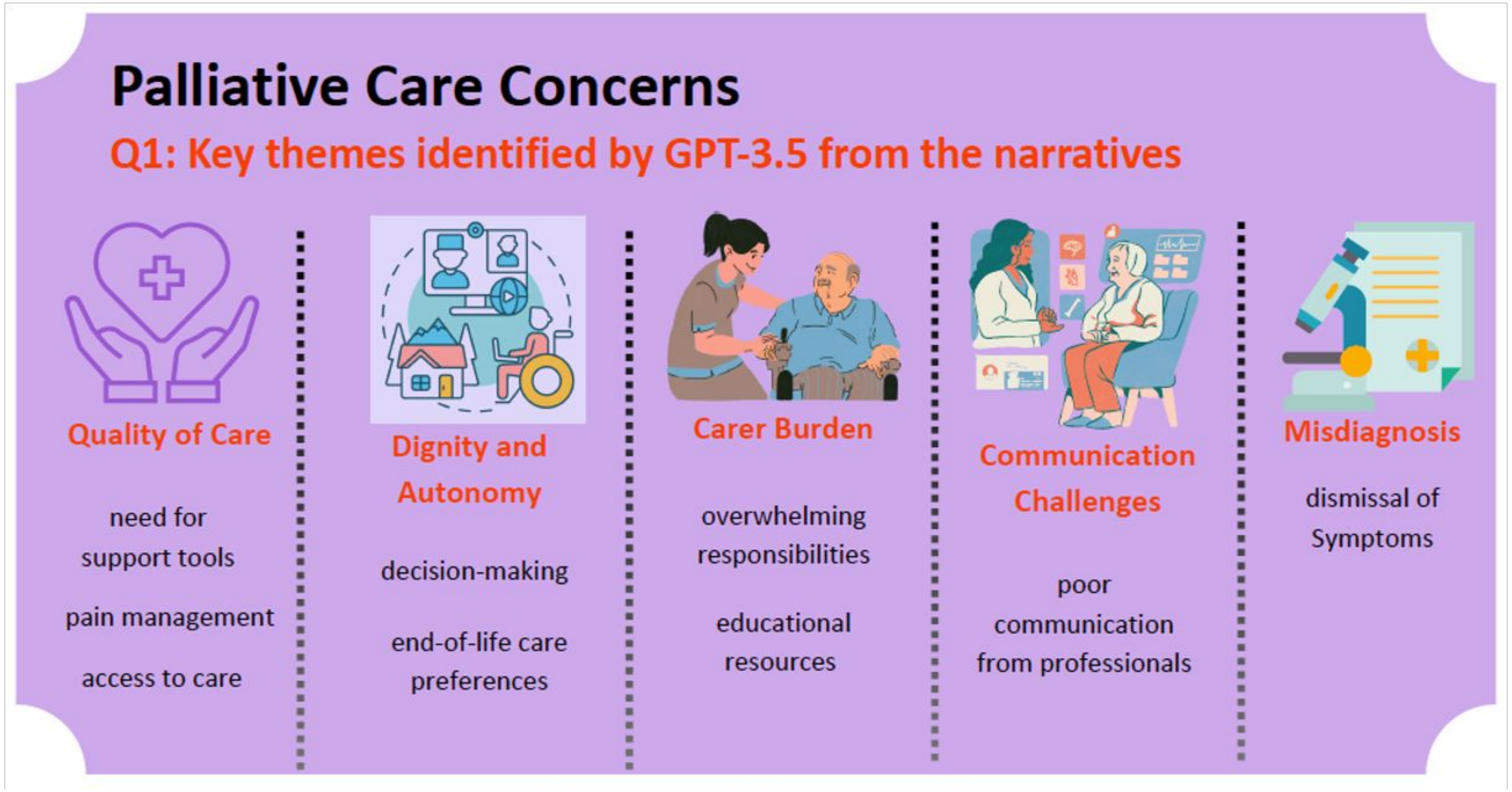


Entity Type	Labels
Clinical	Cancer Type (CANC_T), Staging and grading (STG), Treatment (TRT), Part of Body (POB), Result (RES), Symptom (SYM), Investigation (INV), Medication (MED), Adverse Effect (ADV_EFF), Etiology (EGY), Tumor size and shape (SIZE), Number (NUM), Duration (DUR), Mental Health Diagnosis (MHD), Diagnosis of other diseases (DIAG), Organization (ORG), People or Cancer care team (PPL)
Demographic	Age (AGE), Gender (GENDER), Age/Gender (A/G), Geopolitical Entity (GPE)
Psychosocial	Emotion (EMO), Metaphorical Expression (MET), Other Expressions (EXP)

Daisy Monika Lal, Paul Rayson, Christopher Peter, Ignatius Ezeani, Mo El-Haj, Yafei Zhu, and Yufeng Liu. 2025. LENS: Learning Entities from Narratives of Skin Cancer. In Proceedings of the 31st International Conference on Computational Linguistics: System Demonstrations, pages 20–27, Abu Dhabi, UAE. Association for Computational Linguistics. https://aclanthology.org/2025.coling-demos.3/

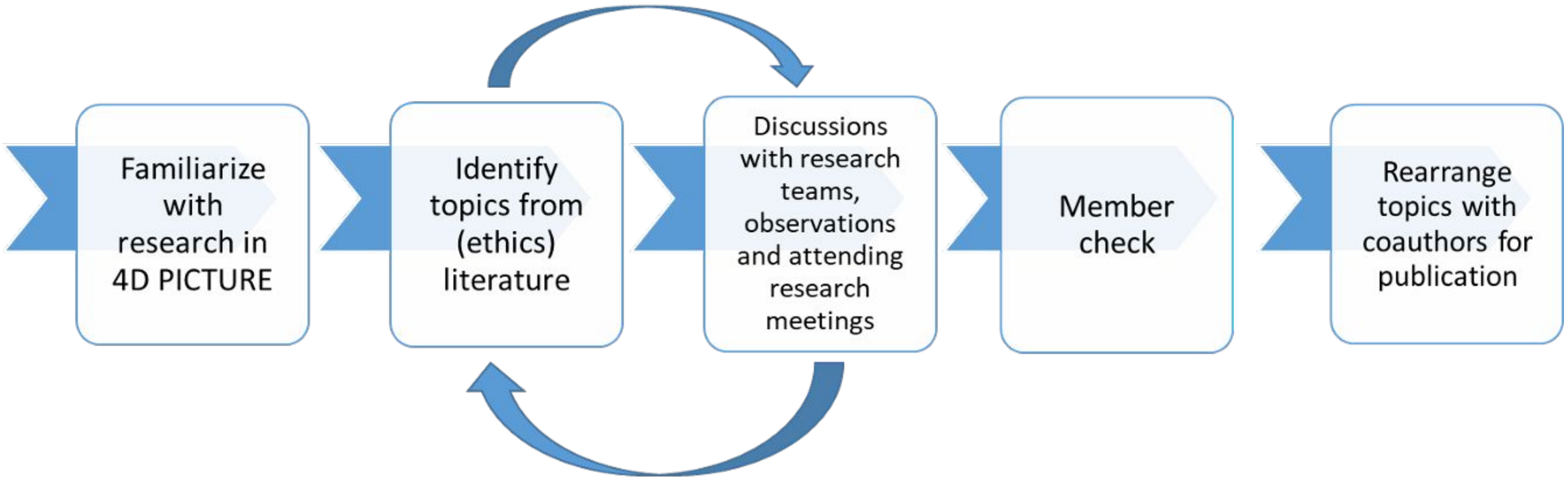
Analysis of patient and carer narratives

Identify the concerns expressed by the author in alignment with the concept of patient concerns in healthcare.
Narrative: {article}
Return the results as a JSON object in the following format:
- Label: patient concern
- Evidence: [a comma-separated list of all supporting sentences from the narrative.]



Daisy Lal, Paul Rayson, Sheila Payne, Yufeng Liu, Ida Korfage, Judith Rietjens (2025) Analysing Palliative Care Concerns: Exploring Patient and Carer Narratives with GPT-3.5. European Association for Palliative Care conference.

Embedded ethics review in 4D

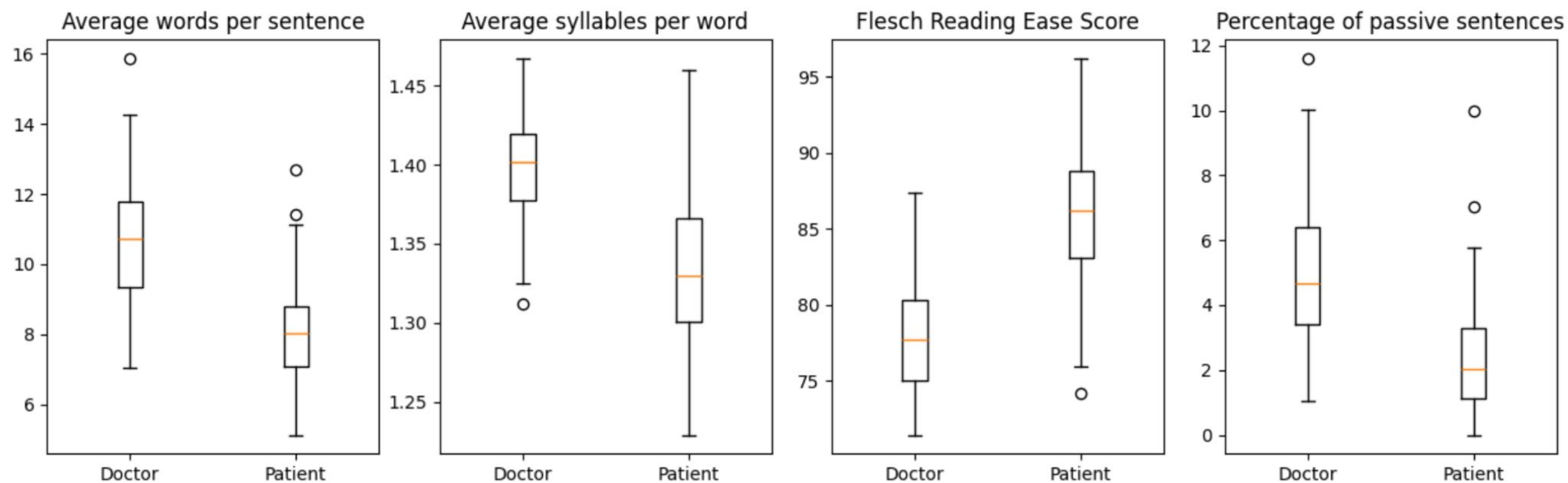


- NLP challenges
  - Bias and fairness in summarising patients' experiences
  - Preserving meaning in the data and including under-represented groups
  - Protecting privacy of cancer patients' data in the public domain
  - Growth of LLM approaches since the project started
- Embedding the ethics team from Erasmus Medical Centre, Technical University of Munich, University of Amsterdam

Barriers in patient-doctor communication

Findings on three datasets with manually transcribed consults with 191 breast cancer patients and 161 rectal cancer patients:

- The language complexity of doctors is higher than of patients.



- The language **complexity** of patients is correlated to their **education** level
- Doctors consistently used more medical terminology than patients (as % of all words)
- Doctors <u>do not adapt</u> their language throughout the consult

Baarslag, A. (Annelotte), Analyzing language complexity and medical terminology use in Dutch medical consults. Thesis Master Computer Science, LIACS, Leiden University, 2026.

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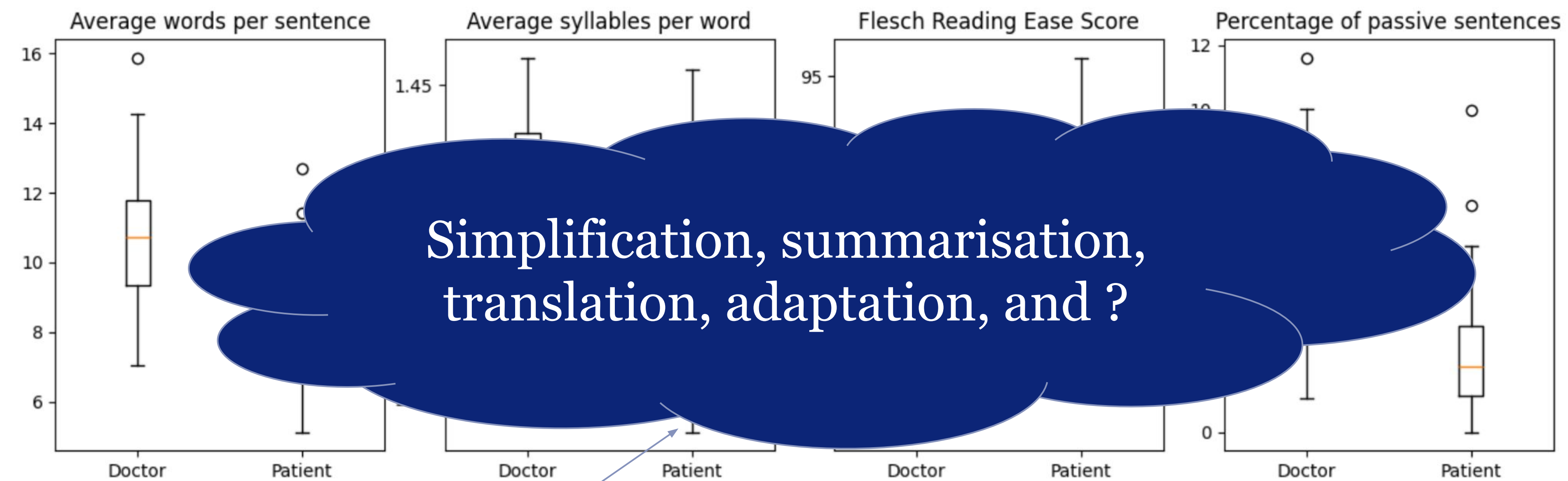

Simplification, summarisation, translation, adaptation, and ?

- The language **complexity** of patients is correlated to their **education** level
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- Doctors do not adapt their language throughout the consult

Baarslag, A. (Annelotte), Analyzing language complexity and medical terminology use in Dutch medical consults. Thesis Master Computer Science, LIACS, Leiden University, 2026.

Example project on communication:

Metaphorical language extraction and topic model - Dutch

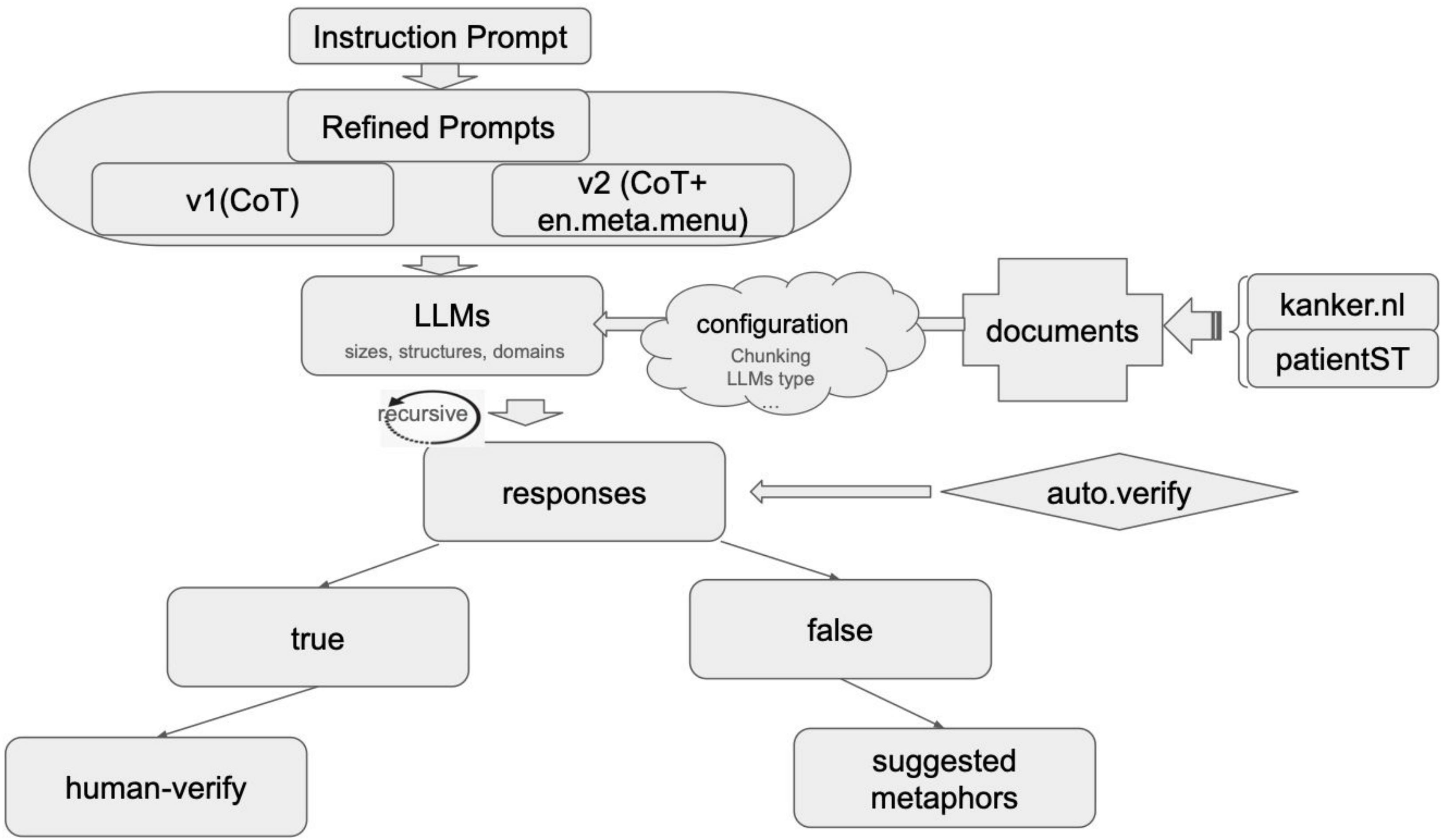

Figure 1: HEALTHQUOTE.NL: Extraction Framework using LLMs and Human in the Loop.

how to support cancer patient using NLP

1. what language patients use?
2. health literacy issue?
3. patient-doctor communication?
4. shared-decision making and care-path redesign?

5. [redacted] to

metaphor extraction from patient data: interviews and forum.
alignment to English Metaphor Menu

Data: storytelling interviews and forum

Data1: 13 interview / Patient Storytelling (patientST)

an oncology setting obtained from (griffioen2021).

There are three roles for each interview: cancer patient, a family member or friend of the patient (significant other), and interviewer (researcher).

- specified as P (Patiënt), N (Naaste), and O (Onderzoeker) in the free text.

Data2: cancer patient forum

15,653 blogs,

17,290 comments,

2,246 group discussions,

5,777 “ask a professional" items

- experts answer questions from patients, and

10,134 reactions

- on “ask a professional" from three cancer types: breast cancer (3,524), prostate cancer (2,613), and melanoma (614).

100 post used for test/pilot-study

RefinedPrompt Structure

- Persona: an expert linguist
- Strict Extraction Protocol
  - Document Scanning
  - Metaphor Identification, CoTs with exam-ples
  - Verification Requirements, different levels
  - Data Extraction Format
  - Structured Output (Population)
- Quality Control Rules
- Final Instructions

https://github.com/aaronlifenghan/HealthQuote.NL
for detailed prompts

Instruction prompt output: LLM issues

phenomena using llama3.1:8b using initial instruction prompt(I.InP) without CoT:

1) abstractive or paraphrased metaphor
2) fail to translation Dutch to English
3) Suggesting metaphors
4) Figurative language rather than metaphor
5) over-interpreting metaphors
6) Idioms instead of metaphors
7) Hallucinated metaphors

categorised issues
from LLMs

Examples

The Party Here, cancer is personified as an uninvited celebrant. The party metaphor conveys irony and defiance, transforming fear into dark humor. This creative reversal allows the writer to express anger and regain narrative control by reframing the illness as something absurd rather than purely tragic.

> *"I was told not to use worrying titles anymore, so this time I've decided to make it a celebration instead – my cancer is having a party of its own."""*

The Train The train functions as a metaphor for relentless motion and loss of control. These images capture the unstoppable pace of medical treatment and the sense that life continues regardless of one's readiness, balancing fatigue with perseverance.

> *"Just when you think you've stepped off one speeding train, you realize you're already on the next one."""*

> *"Life keeps pounding along like a fast-moving train – whether I'm ready to get on or not."*

The Final Chord The musical metaphor of the *final chord* transforms death into a form of completion rather than cessation. It provides emotional and narrative closure, allowing grief to be expressed through harmony and resolution.

> *"This feels like the final note in the song of our journey together – the moment of closure that sends everything gently into silence."*

paraphrased examples / GPT5

Blog text (paraphrased)	Metaphor (in Dutch)	map	Confidence Score	Explanation
Daar zit je dan… in de wachtkamer bij dokter De Leeuw. Goede naam, denk ik nog, hopelijk vecht ze ook als een leeuw. Mijn man zit zenuwachtig zijn handen samen te knijpen. O, wat had ik hem dit graag bespaard. ... De dokter vertelt waar ik een groot litteken zal krijgen. Ik maak een stom grapje: ik ben toch al gelukkig getrouwd. … Mijn gedachten dwalen af. Lymfeklieren, overal in je lichaam… Ineens raakt het me volledig, heel kort maar glashelder. De brok in mijn keel voel ik, maar ik druk de tranen weg. Nee, dat sta ik mezelf niet toe. De kinderen… denk aan de kinderen. We gaan gewoon voor goede uitslagen. Ik zal vechten als een leeuw. Als je wilt, kan ik er ook een meer literaire, bijna poëtische versie van maken, die de spanning en emoties nóg sterker benadrukt, zonder dat het te zwaar wordt. Wil je dat?	Hopelijk vecht ze als een leeuw.	7	0.95	Explicit simile of fighting like a lion — directly expresses a combat/fight frame, matching item 7 (I don't intend to give up; I want to fight).
	een groot litteken	13	0.72	A scar functions as a lasting trace/after-effect of illness/treatment; thematically closest to item 13 (cancer as an unwelcome lodger that leaves a mess / lasting memory/residue).
	Ineens raakt het me volledig, heel kort maar glashelder.	2	0.85	This describes sudden clarity and heightened perception; maps best to item 2 (the illness makes you look up and take notice of the scenery / increased appreciation/awareness)
	Ik zal vechten als een leeuw.	7	0.99	Straightforward, first-person battle metaphor — very strong direct fit with item 7 (determined combat).
Na een week wachten mochten we eindelijk de uitslag van het biopt van een vlekje in mijn mond horen. Volgens Dokter Google zou het zomaar een onschuldige "amalgaamtattoo" kunnen zijn, maar stiekem bleef in mijn achterhoofd het idee rondspoken dat het ook een melanoom kon zijn. .... Dankzij hem konden we snel weer	Volgens Dokter Google kon het een onschuldige 'amalgaamtattoo' zijn, … melanoom kon zijn.	1	0.65	The 'haunting in the back of my mind' gives a feeling of fear and uncertainty that is temporary, which fits item 1 (scary fairground ride)
	konden we snel weer naar huis	5	0.25	Returning home after reassurance suggests social/support element;

Dutch Metaphor Extraction from Cancer Patients' Interviews and Forum Data using LLMs and Human in the Loop

check this out?
CL4health WS
afternoon

Lifeng Han[1,2], **David Lindevelt**[2], **Sander Puts**[3], **Erik van Mulligen**[4], **Suzan Verberne**[2]
on behalf of 4D PICTURE
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Abstract

Metaphors and Metaphorical Languages (MLs) play an important role in healthcare for the information communication between clinicians, patients, and patients' family members. In this work, we focus on the Dutch language and cancer patients' data. We extract the metaphors used by patients using two data resources: 1) cancer patient storytelling interview data, 2) online forum, data including cancer patients' posts, comments, and questions to professionals. We investigate how current state of the art LLMs and perform on this task by exploring different prompting strategies such as Chain of Thought, few-shot learning, and self-prompting. With human in the loop, we verify the extracted metaphors and collect the output as a corpus, named "**HealthQuote.NL**". We believe the extracted metaphors can be useful for supporting better patient care, e.g. shared decision making, helping communication between patients and clinicians, patient health literacy, etc. It can also be integrated into the design of a care path. We share our prompts and related resources at https://github.com/aaronlifenghan/HealthQuote.NL.

how to support cancer patient using NLP

[redacted]

5. What are patients' concerns?
6. Are there patterns or common issues?
7. how to use data engineering and NLP to discover new / potential issues?

topic modelling
Bert-topic vs Topic2Vec
clinical embedding models
what themes do we have?

StoryTelling data: working on translations

H0-1

O: Ik heb hier de opname gestart. Het gaat eigenlijk om jullie ervaringen, dus het gaat eigenlijk niet zozeer om goede of foute antwoorden, dat maakt helemaal niet uit. Ik ben gewoon benieuwd naar jullie ervaringen, want daarvan willen we leren. En dat zijn de ervaringen die je allemaal hebt opgedaan tijdens het hele behandeltraject, vanaf het moment dat je dacht er is iets fout en dat is het begin van deze lijn, stip, tot eigenlijk vandaag. Het eerste dat ik jullie zou willen vragen is: zet maar eens gewoon wat voor jullie belangrijke momenten op die lijn waarvan je denkt 'nou dat waren wel gewoon voor ons momenten in het hele behandeltraject die meteen bij me opkomen waar er iets gebeurde' en die iets te maken hadden met hoe jullie het ervaren hebben. Je mag dikke pakken, dunne pennen pakken, wat je wil. Je mag ook allebei schrijven, je mag het ook samen overleggen.

N: Maart 2016, dat jij rechtop in je bed hebt gezeten 's nachts.

P: Toen de pijn zich manifesteerde.

N: Wil je het dan ook eronder schrijven?

O: Ja, schrijf het er maar bij.

N: Waar was dat, in je zij meer, of in je rug?

P: In de zij, ja. Alleen 's nachts was het toen, die pijn.

O: Dat was het begin?

P: Ja. Dat hebben we een paar dagen aangezien en toen ben ik naar de dokter gegaan, naar de huisarts. Toen moesten we naar het ziekenhuis, voor een echo.

N: Een buikfoto en een echo heb je toen gekregen in het ziekenhuis. Toen zagen ze ontlasting vast zitten op de plek waar jij pijn had.

P: Ja, dat was ook zo, ja.

N: Dus toen zeiden ze, ah verklaard.

P: Ik wees niet eens dat het kon.

H0-1

O: I started the recording here. It's actually about your experiences, so it's really not so much about right or wrong answers, that doesn't matter at all. I'm just curious about your experiences, because that's what we want to learn from. And those are the experiences that you all have had throughout the treatment process, from the moment you thought there's something wrong and that's the beginning of this line, dot, to actually today. The first thing I'd like to ask you is: just put some moments that are important to you on that line that you think 'well those were just moments for us throughout the treatment process that immediately come to mind where something happened' and that had something to do with how you experienced it. You may take thick suits, thin pens, whatever you want. You may also both write, you may also discuss it together.

N: March 2016, that you sat upright in your bed at night.

P: When the pain manifested.

N: So do you want to write it underneath?

O: Yes, write it in.

N: Where was that, in your side more, or in your back?

P: In the side, yes. Only at night it was then, that pain.

O: That was the beginning?

P: Yes. We watched that for a few days and then I went to the doctor, to the family doctor. Then we had to go to the hospital, for an ultrasound.

N: You then got an abdominal x-ray and ultrasound at the hospital. Then they saw stool stuck where you had pain.

P: Yes, that was also true, yes.

N: So then they said, ah explained.

P: I didn't even point out that it was possible.

Global Analysis

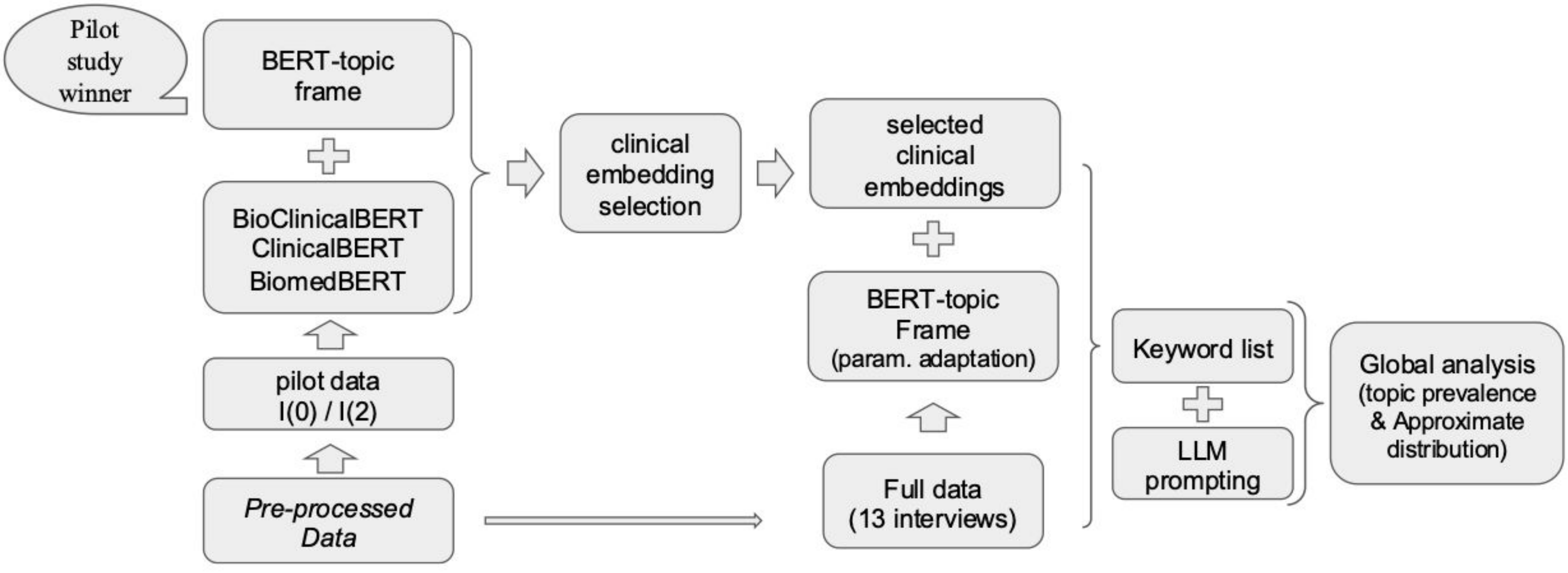
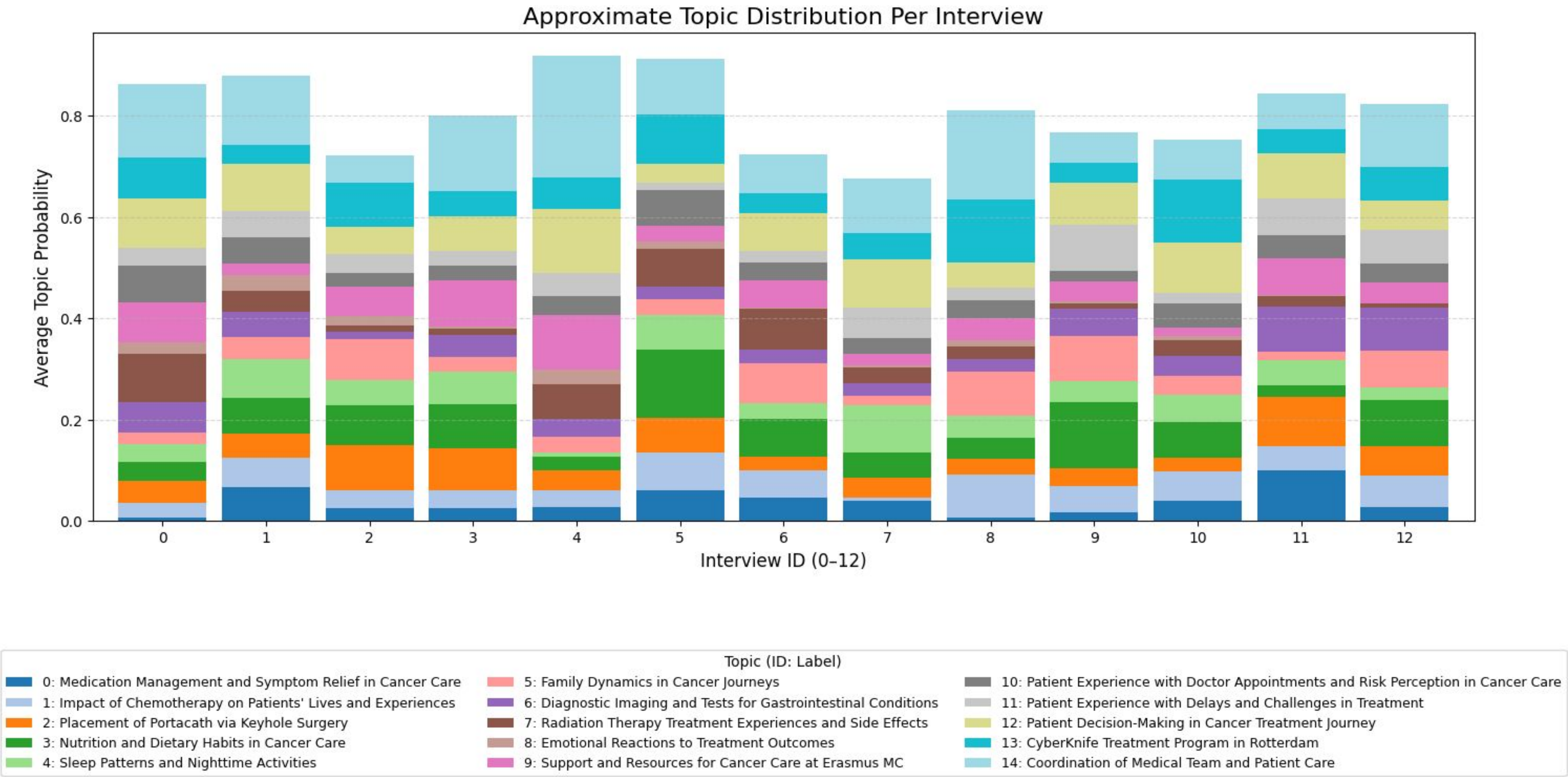


Figure 5: BERT-topic Deployment on all Interviews with Clinical LM Embeddings

topic distribution



Analyzing Cancer Patients' Experiences with Embedding-based Topic Modeling and LLMs

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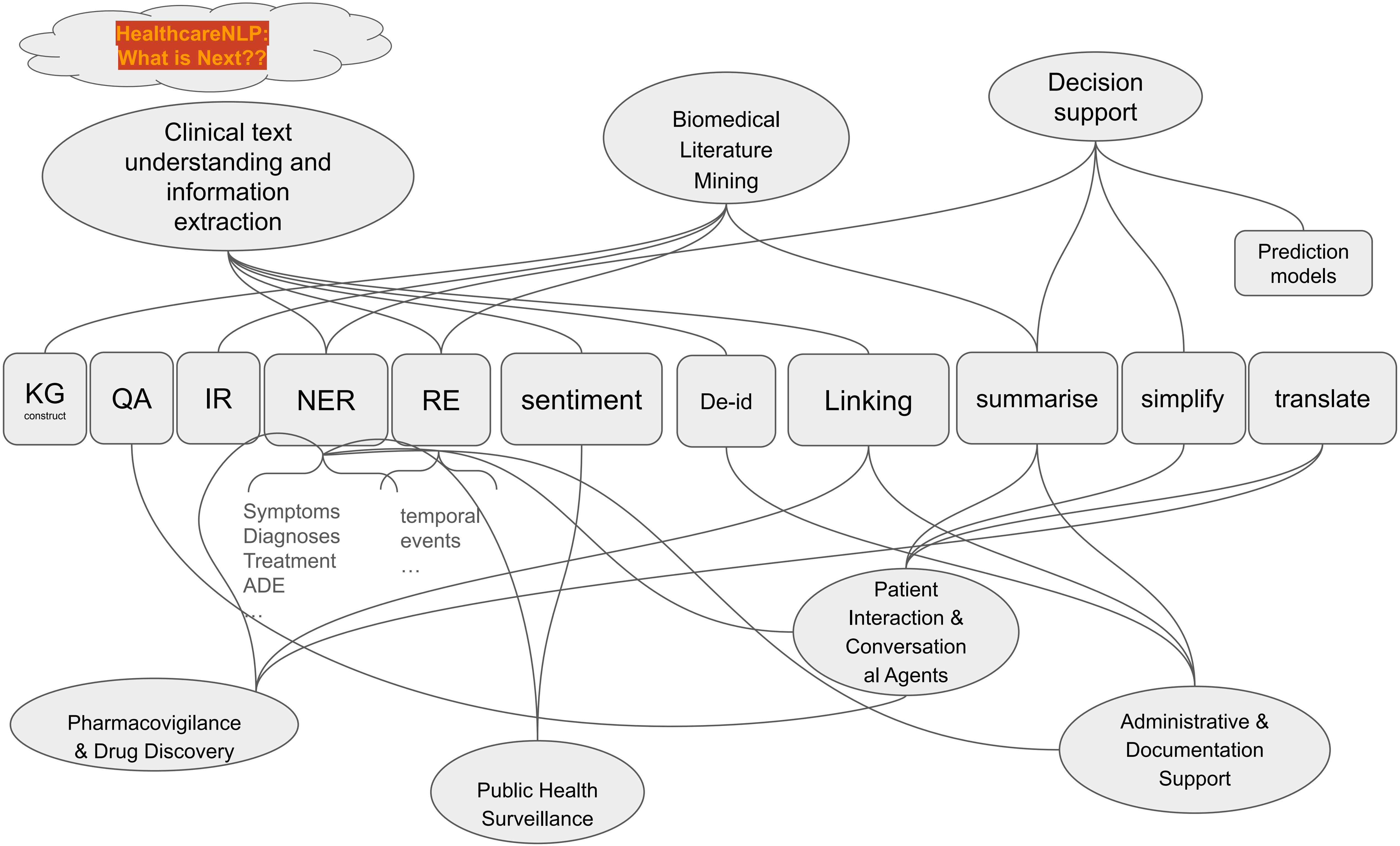
Accepted: in publishing process with CLIN journal

https://arxiv.org/pdf/2601.12154

Abstract

This study investigates the use of neural topic modeling and LLMs to uncover meaningful themes from patient storytelling data, with the goal of offering insights that could contribute to more patient-oriented healthcare practices. We analyze a collection of transcribed interviews with cancer patients (132,722 words in 13 interviews). We first evaluate BERTopic and Top2Vec for the purpose of individual interview summarization, by using similar preprocessing, chunking, and clustering configurations to ensure a fair comparison on Keyword Extraction. LLMs (GPT4) are then used for next step topic labeling. Their outputs for a single interview (I0) are rated through a small-scale human evaluation, focusing on *coherence*, *clarity*, and *relevance*. Based on the preliminary results and evaluation, BERTopic shows stronger performance and is selected for further experimentation using three *clinically oriented embedding* models. We then analyzed the full interview collection with the best model setting. Results show that domain-specific embeddings improved topic *precision* and *interpretability*, with BioClinicalBERT producing the most consistent results across transcripts. The global analysis of the full dataset of 13 interviews, using the BioClinicalBERT embedding model, reveals the most dominant *topics* throughout all 13 interviews, namely "Coordination and Communication in Cancer Care Management" and "Patient Decision-Making in Cancer Treatment Journey". Although the interviews are machine translations from Dutch to English, and clinical professionals are not involved in this evaluation, the findings suggest that neural topic modeling, particularly BERTopic, can help provide useful *feedback* to clinicians from patient interviews. This pipeline could support more efficient document navigation and strengthen the role of patients' voices in healthcare workflows. Affiliated resources created in this work will be shared publicly at `https://github.com/4dpicture/TM4health` including codes on preprocessing and stopword list prepared.

HealthcareNLP: What is Next??


HealthcareNLP:
What is Next??
Clinical text understanding and information extraction
Biomedical Literature Mining
Decision support
Prediction models
KG
construct
QA
IR
NER
RE
sentiment
De-id
Linking
summarise
simplify
translate
Symptoms
Diagnoses
Treatment
ADE
…
temporal
events
…
Patient Interaction & Conversation al Agents
Pharmacovigilance & Drug Discovery
Public Health Surveillance
Administrative & Documentation Support

Discussion-1: AI for Reflexive QR

Jowsey, T., Braun, V., Clarke, V., Lupton, D., & Fine, M. (2025). We reject the use of generative artificial intelligence for reflexive qualitative research. *Qualitative Inquiry*, 10778004251401851.

(Position/opinion paper: 1.5 pages)

(participants download this paper pre-Tutorial?)

=> how do you think of this paper's point?

=> do/will you do "reflexive QR", if or not to use AI?

Discussion-1: AI for Reflexive QR

1) What is Reflexive Qualitative Research?

Reflexive Qualitative Research (RQR) is an approach where:

> Knowledge is *co-constructed* by the researcher and the data through interpretation and reflection.

It is **not** about extracting "objective facts" from text.
Instead, it asks:

- What meanings do participants construct?
- How do I (as researcher) interpret those meanings?
- How does my background influence what I see?

So the researcher is **part of the analytic instrument**, not a neutral observer.

Discussion-1: AI for Reflexive QR

The paper outlines three primary objections:

1. **GenAI Cannot Make Meaning**
   - The authors argue that simulated intelligence *does not and cannot engage in authentic meaning-making*, a central task in qualitative analysis.
2. **Qualitative Analysis Should Be a Human Practice**
   - They assert that reflexive qualitative work is inherently human — it involves interpretation, positioning, and ethical engagement with data — attributes they believe AI cannot possess or perform.
3. **Ethical & Justice Concerns**
   - The paper cites *broader ethical issues*: environmental impact of GenAI technologies, labor and extraction concerns (especially relating to Global South contributors), and methodological justice.

how to do address these using AI/NLP?

Discussion-1: AI for Reflexive QR

The paper outlines three primary objections:

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Hint: Agent-based learning, Persona, Meaning, interpretable, Smarter-instead-of-larger?

Hands-on Options

INSIGHTBUDDY-AI: Medication Extraction and Entity Linking using Large Language Models and Ensemble Learning

https://github.com/HECTA-UoM/InsightBuddy-AI

Learning Entities from Narratives of Skin Cancer (LENS)

https://github.com/4dpicture/LENS

MedCAT Tutorials

https://github.com/CogStack/MedCATtutorials

Dank u voor uw aandacht.

thank you for you attention

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Universiteit Leiden

Quick feedback survey

https://www.menti.com/alyixj47x2s1

Join at menti.com with code 1563 1233

Appendices

Universiteit Leiden

Summarisation and Simplification

We move this and Translation tasks to appendix and open-source Github pages for further reading.

Simplification

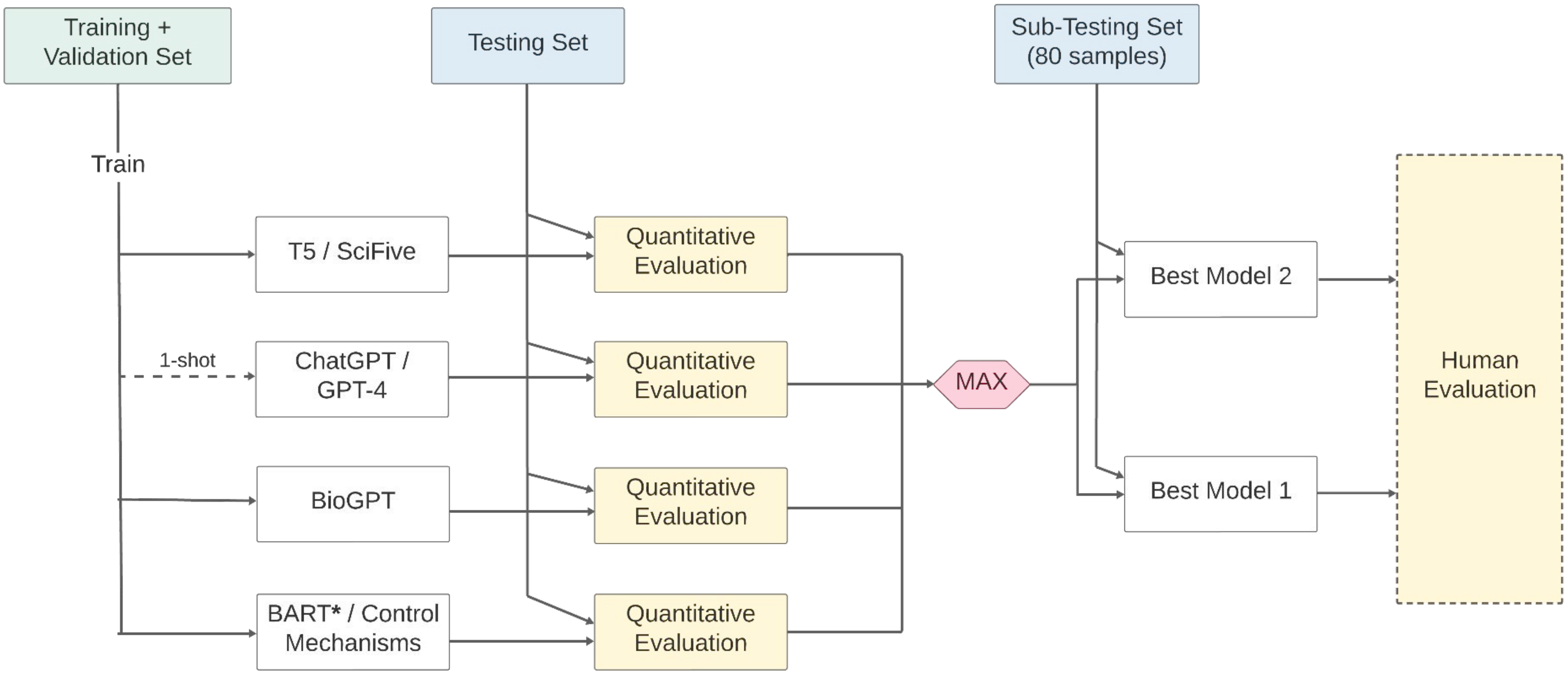
Investigating large language models and control mechanisms to improve text readability of biomedical abstracts
Z Li, S Belkadi, N Micheletti, L Han*, M Shardlow… - 2024 IEEE 12th International Conference on Health Informatics

MaLei at the PLABA Track of TREC 2024: RoBERTa for Term Replacement--LLaMA3. 1 and GPT-4o for Complete Abstract Adaptation

INPUT (ABSTRACT)	OUTPUT (ADAPTED)
Acute pharyngitis/tonsillitis, which is characterized by inflammation of the posterior pharynx and tonsils, is a common disease.	Sore throat/tonsillitis, or when the back of the throat or tonsils is inflamed, is common.
Several viruses and bacteria can cause acute pharyngitis; however, Streptococcus pyogenes (also known as Lancefield group A β-hemolytic streptococci) is the only agent that requires an etiologic diagnosis and specific treatment.	Many viruses and bacteria can cause short-term sore throat. However, group A strep, caused by Group A strep bacteria, is the only cause that must be identified based on signs and symptoms and treated.

Sentence structure simplification | Term simplification | Paraphrase / synonyms

"language gap used by patients and specialists"



Investigating large language models and control mechanisms to improve text readability of biomedical abstracts

Z Li, S Belkadi, N Micheletti, L Han*, M Shardlow… - 2024 IEEE 12th International Conference on Health Informatics

Finetuning LLM / domain-LLMs + Control Tokens

Examples from https://bionlp.nlm.nih.gov/plaba2024/

PLABA-2024

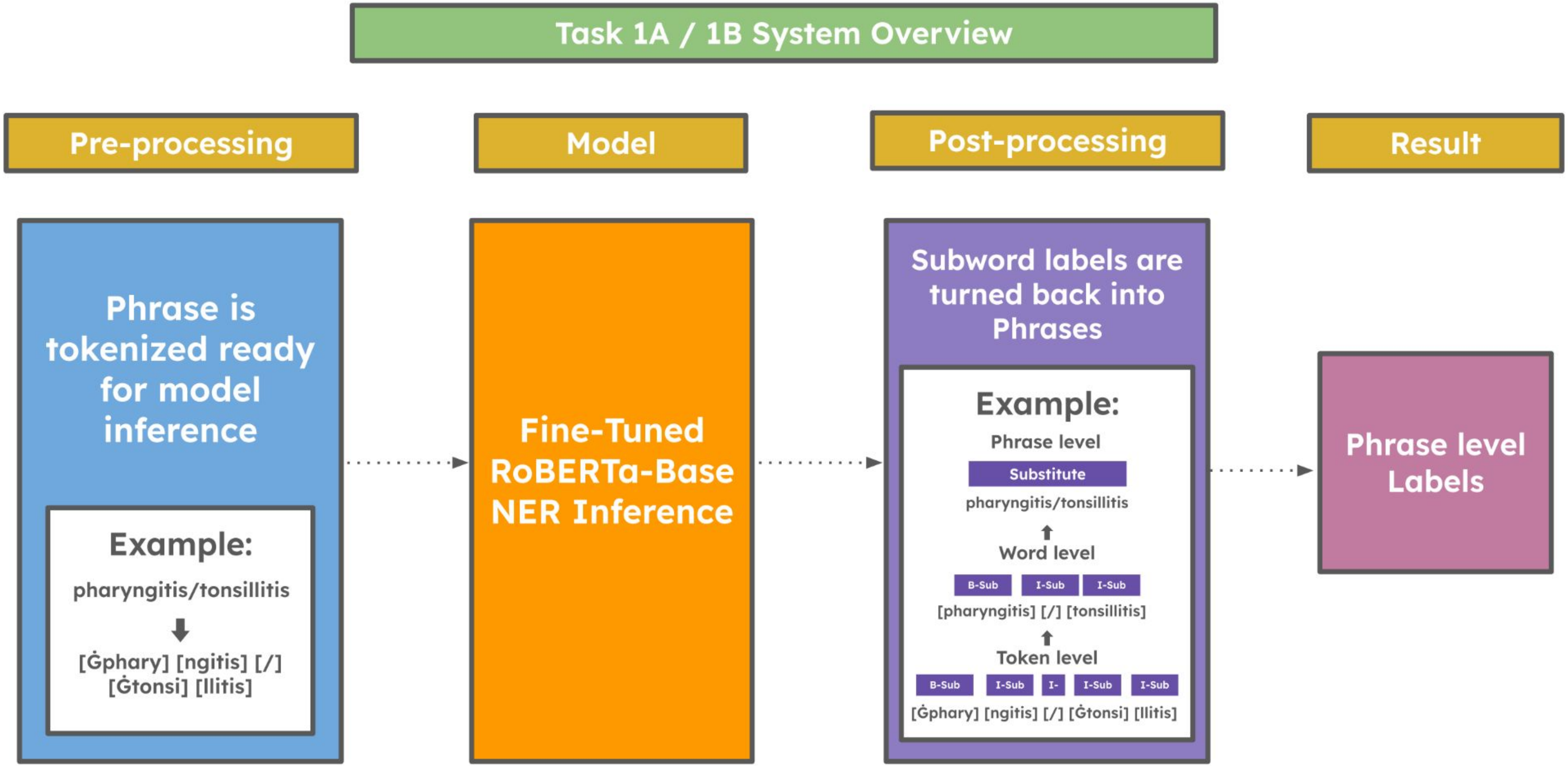
1A: Identifying non-consumer terms
1B: Classifying replacement
1C: Generation

1. **Substituted:** the term is jargon with a common alternative (e.g. "myocardial infarction" can be "heart attack").
2. **Explained:** there is no alternative, or the term is important to the topic, and it should be explained. For example: "This study looked at treatments for sleep apnoea (*when you stop breathing while sleeping*)."
3. **Generalized:** the term can be replaced with a more general category without losing its significance. For example: "Clearing of the infection is confirmed with a ~~Nucleic Acid Amplification Test (NAAT)~~ common lab test."
4. **Exemplified:** the term has a specific example that would give a general audience an idea of what it is. For example: "Depression is common in people with neurodegenerative diseases (*like Parkinson's*)."
5. **Omitted:** The term is not relevant to understanding the sentence or too technical to explain, and does not need to appear in a consumer version.

This is a *multilabel* task; systems should identify any and all types of replacements that would be appropriate for each term.

MaLei at the PLABA Track of TREC 2024:
RoBERTa for Term Replacement--LLaMA3. 1
and GPT-4o for Complete Abstract Adaptation

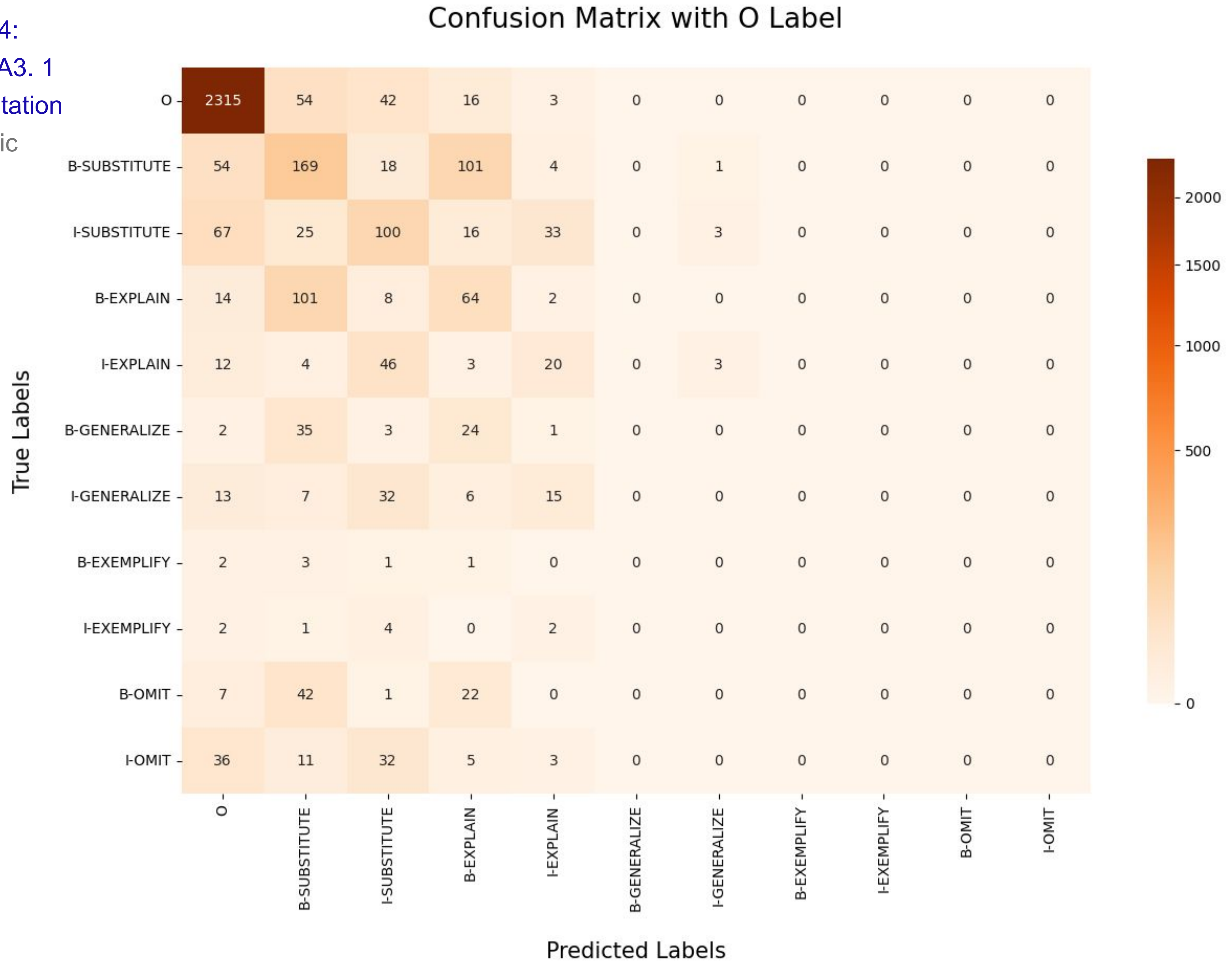
Z Ling, Z Li, P Romero, L Han, G Nenadic



MaLei at the PLABA Track of TREC 2024: RoBERTa for Term Replacement--LLaMA3. 1 and GPT-4o for Complete Abstract Adaptation

Z Ling, Z Li, P Romero, L Han, G Nenadic

Confusion Matrix Across Labels For Insights



Submission to the PLABA-2024

Team	Run	Rank	Task 1A (F1)	Task 1B (F1)	Average Score
BU	MLPClassifier-identify-classify-replace-v1	1	0.0459	**0.7788**	0.4124
CLAC	mistral	1	0.4410	0.6663	0.5537
CLAC	gpt	2	0.3767	0.3795	0.3781
IITH	First	1	0.1956	0.7014	0.4485
UM	Roberta-base	1	*0.4787*	*0.7765*	**0.6276**
ntu_nlp	gemini-1.5-pro_demon5_replace-demon5	1	*0.4885*	0.6335	0.5610
ntu_nlp	gemini-1.5-flash_demon5_replace-demon5	2	0.4431	0.6544	0.5488
ntu_nlp	gpt-4o-mini_demon5_replace-demon5	3	0.4518	0.6197	0.5358
Yseop	roberta-gbc	1	**0.5036**	0.6838	0.5937

Task 1A and Task 1B (F1) Results: Roberta-base highest Averaged-Score

Section	Content
Objective	You are tasked with simplifying a complex medical text to make it easily understandable by non-medical professionals, such as patients or caregivers.
Guidelines	• **Maintain Accuracy:** Ensure that the simplified text accurately reflects the original meaning. Do not omit critical details or introduce incorrect information. • **Use Plain Language:** Replace medical jargon and complex terms with simple, everyday language. If a technical term is necessary, provide a brief explanation or analogy. • **Shorter Sentences:** Break down long, complicated sentences into shorter, more manageable ones. Multiple sentences are allowed but must fit within a single line per numbered point. • **Use Active Voice:** Where possible, use the active voice to make the text more direct and easier to read. • **Provide Examples:** Use examples or analogies to help explain complex concepts, making them relatable to everyday experiences. • **Simplify Structure:** Organize the information in a logical, easy-to-follow structure. Use bullet points or lists to highlight key points when applicable.
Output Guidelines	1. Align the output strictly with the numbered points from the original text. 2. Maintain the numbering format exactly as it appears in the input. 3. Each numbered point must be a single line but can include multiple sentences if necessary for clarity. 4. Ensure that each simplified point corresponds directly to its numbered counterpart in the original text.

Continued on next page

Task2: end to end:

GPT4o Prompts

Section	Content
Original Text Example	1. The dystonias are a group of disorders characterized by excessive involuntary muscle contractions leading to abnormal postures and/or repetitive movements. 2. A careful assessment of the clinical manifestations is helpful for identifying syndromic patterns that focus diagnostic testing on potential causes. 3. If a cause is identified, specific etiology-based treatments may be available. 4. In most cases, a specific cause cannot be identified, and treatments are based on symptoms. 5. Treatment options include counseling, education, oral medications, botulinum toxin injections, and several surgical procedures. 6. A substantial reduction in symptoms and improved quality of life is achieved in most patients by combining these options.
Simplified Text Example	1. Dystonias are disorders with a lot of uncontrollable muscle contractions. This causes awkward postures and/or repetitive movements. 2. Checking the symptoms helps spot patterns and guides testing to find possible causes. 3. If a cause is found, treatments specific to that cause may be available. 4. When no specific cause is found, treatments focus on managing symptoms. 5. Treatment options include counseling, education, oral medications, botox (a muscle relaxant), and surgeries. 6. Combining these treatments usually leads to a noticeable decrease in symptoms and a better quality of life for most patients.

GPT4o Prompts

Summarisation

MaLei at MultiClinSUM: Summarisation of Clinical Documents using Perspective-Aware Iterative Self-Prompting with LLMs

L Ren, YM Ng, L Han

https://doi.org/10.48550/arXiv.2509.07622

The Manchester Bees at PerAnsSumm 2025: Iterative Self-Prompting with Claude and o1 for Perspective-aware Healthcare Answer Summarisation

P Romero, L Ren, L Han, G Nenadic

Proceedings of the Second Workshop on Patient-Oriented Language Processing ...

MaLei at MultiClinSUM: Summarisation of Clinical Documents using Perspective-Aware Iterative Self-Prompting with LLMs

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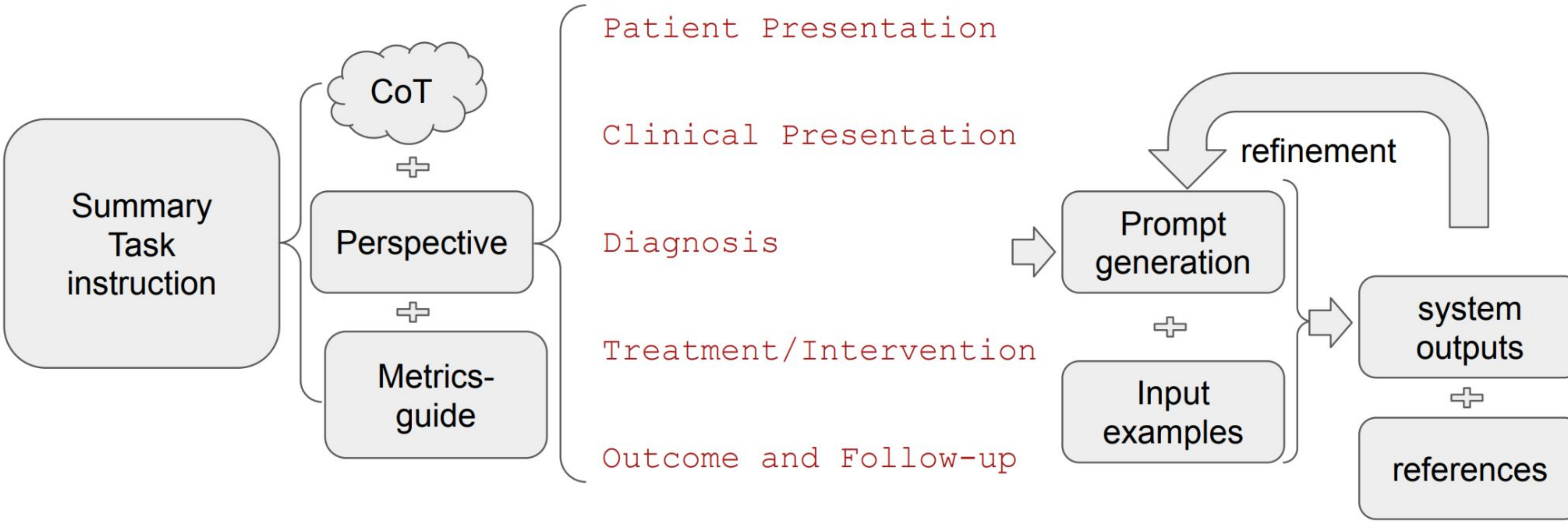


Figure 1: Perspective-aware Iterative Self-Prompting (PA-ISP) Illustration Diagrams from the MaLei team.

The Manchester Bees at PerAnsSumm 2025: Iterative Self-Prompting with Claude and o1 for Perspective-aware Healthcare Answer Summarisation

P Romero, L Ren, L Han, G Nenadic

Proceedings of the Second Workshop on Patient-Oriented Language Processing
…

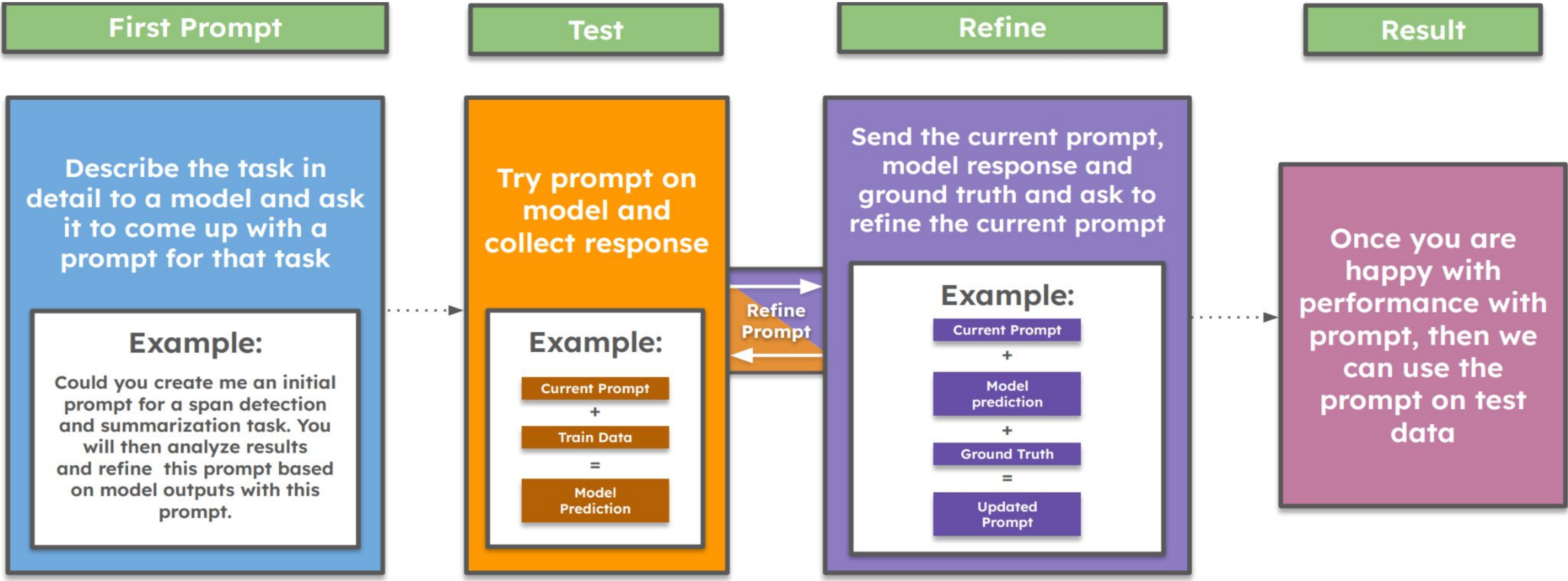


Figure 1: Iteration Cycle for ISP showing the process of prompt refinement through feedback loops.

translation

Neural Machine Translation of Clinical Text: An Empirical Investigation into Multilingual Pre-Trained Language Models and Transfer-Learning

L Han, S Gladkoff, G Erofeev, I Sorokina, B Galiano… - Frontiers in Digital Health… - frontiersin.org

Investigating Massive Multilingual Pre-Trained Machine Translation Models for Clinical Domain via Transfer Learning

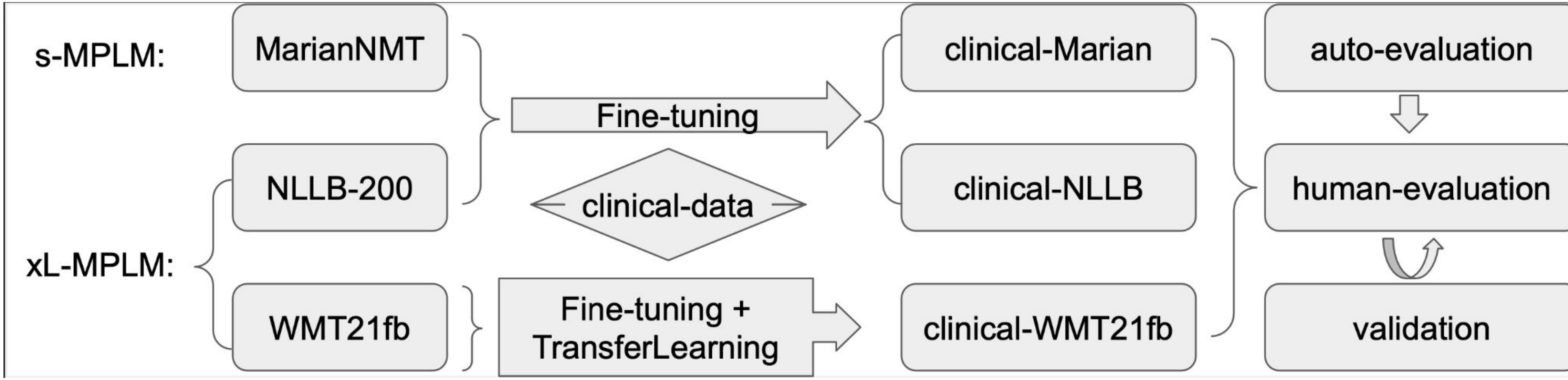
L Han, I Sorokina, S Gladkoff, G Nenadic

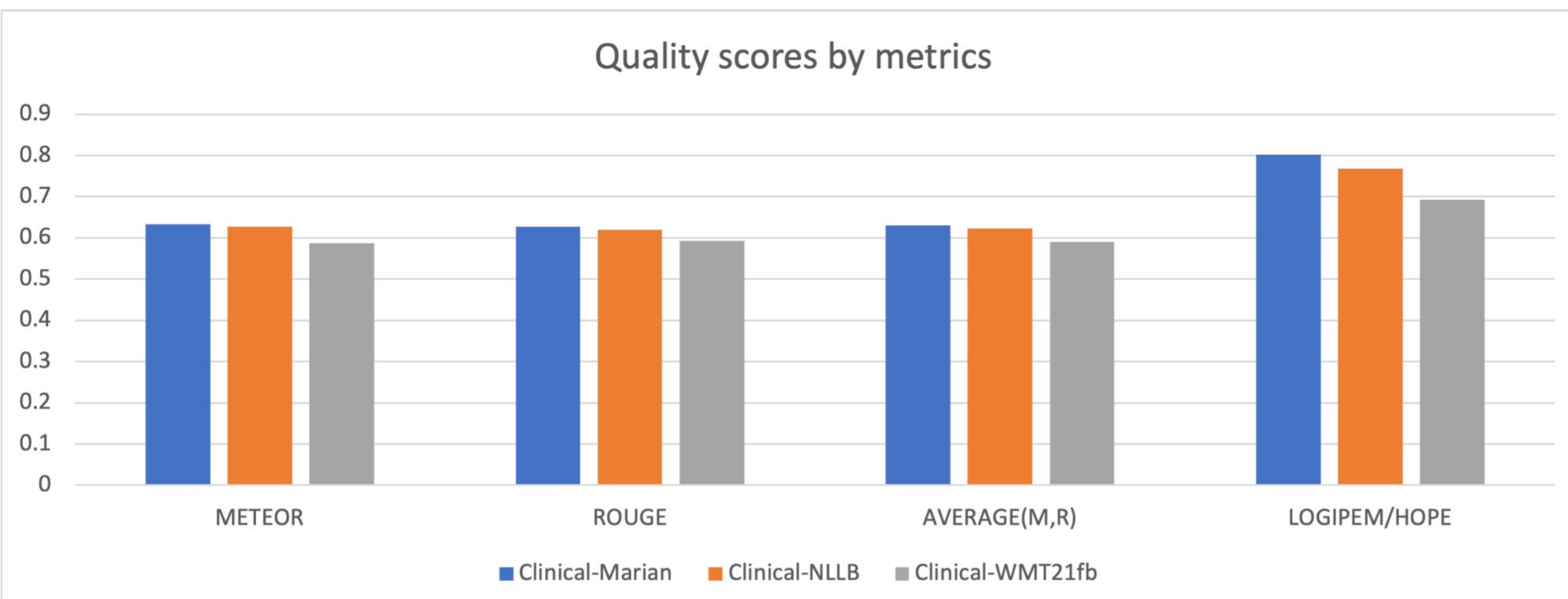
Proceedings of the 5th Clinical Natural Language Processing Workshop, 31-40

Examining Large Pre-Trained Language Models for Machine Translation: What You Don't Know About It

L Han, G Erofeev, I Sorokina, S Gladkoff, G Nenadic

Proceedings of the Seventh Conference on Machine Translation (WMT), 908–919





not: the larger the better
but: the smarter the better

(Han, Gladkoff, et al. 2024)

Clinical Knowledge Transfer

How good/bad MT in clinical domain?

xL-LM vs S-LM?

Transfer-learning

https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2024.1211564/abstract